A Cheatsheet for Algebraic Topology

Contents

Chapter 1

Basic Category Theory

1.1 Construction of Category

Definition 1.1.1. *A metagraph consists of* $\begin{cases} \textit{objects } a,b,c,\dots \\ \textit{arrows } f,g,h\dots \end{cases}$ *and two operations* $\begin{cases} \textit{domain} & a = domf \\ \textit{codomain} & b = codf \end{cases}$

→ a **metacategory** is a metagraph with two additional operations $\begin{cases} \text{identity} & id_a = 1_a : a \to a \\ \text{composition} & <g,f> \to g \circ f : domf \to codg \end{cases}$ where

$$\begin{tikzcd} & b \arrow[dr, "g"] & \\ a \arrow[ur, "f"] \arrow[rr, "g \circ f"] & & c \end{tikzcd}$$

commutes, and the following axioms are satisfied;

1. associativity

   for given objects and arrows $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{k} d$, one has the equality $k \circ (g \circ f) = (k \circ g) \circ f$

2. unit law

   for all arrows $\begin{matrix} f : a \to b \\ g : b \to c \end{matrix}$, composition with the identity arrow gives $\begin{matrix} 1_b \circ f = f \\ g \circ 1_b = g \end{matrix}$

If b is any object of a metacategory C, the corresponding identity arrow 1_b is uniquely determined by the unit law, and therefore we may identify the identity arrow with the object. This identification gives an arrow-only metacategory C, which consists of arrows, certain ordered pair $< g,f >$ called the composable pair and an operation assigning to each composable pair $< g,f >$ an arrow g,f called their composite.

Definition 1.1.2. *A directed graph is a set O of objects, a set A of arrows and two functions*

$$A \underset{dom}{\overset{cod}{\rightrightarrows}} O$$

$\rightarrow A \times_O A = \{< g, f > | g, f \in A,\ domg = codf\}$ the product over O is the set of composable pairs of arrows.

Definition 1.1.3. *A category is a directed graph with two additional functions; the identity* $\begin{array}{ccc} O & \rightarrow & A \\ c & \mapsto & id_c \end{array}$ *and the composition* $\begin{array}{ccc} A \times_O A & \rightarrow & A \\ < g, f > & \mapsto & g \circ f \end{array}$ *s.t.* $dom(id_a) = a = cod(id_a)$, $dom(g \circ f) = domf, cod(g \circ f) = codg$, $\forall a \in O$, $\forall < g, f > \in A \times_O A$ *and the associativity and the unit law axioms hold.*

$\rightarrow \hom(b, c) = \{f | f$ in $C,\ domf = b,\ codf = c\}$ is the set of arrows from b to c.

e.g.

- we can regard each ordinal number n as the linearly ordered set of all the preceding ordinals $n = \{0, 1, \ldots, n-1\}$. Then each ordinal n is linearly ordered and hence it is a preorder and therefore is a category.
  - **0** is the empty category
  - **1** is the category with one object and one identity arrow
  - **2** is the category with two objects a, b, and just one non-identity arrow $a \rightarrow b$
  - **3** is the category with three objects whose non-identity arrows are arranged as in the triangle
    $$\begin{tikzcd} & b \arrow[dr] & \\ a \arrow[ur] \arrow[rr] & & c \end{tikzcd}$$
- $\downdownarrows$ is the category with two objects a, b and two non-identity arrows $a \rightrightarrows b$.
- a category is discrete when every arrow is an identity.
- a monoid is a category with one object. Since any two arrows have a composite, a monoid may be described as a set M with a binary operation $M \times M \rightarrow M$ which is associative and has an identity.

Definition 1.1.4. *Given categories C,B, a functor $T : C \rightarrow B$ with domain C and codomain B consists of two functions; the object function* $\begin{array}{ccc} T : ObjC & \rightarrow & ObjB \\ c & \mapsto & T(c) \end{array}$

and
the arrow function $\begin{array}{rcl} T : ArrowC & \to & ArrowB \\ f & \mapsto & T(f) \end{array}$ *where* $f : c \to c'$ *is an arrow of* C *and* $T(f) : Tc \to Tc'$ *is an arrow of* B*, in such a way that* $T(1_c) = 1_{T_c}$,
$T(g \circ f) = Tg \circ Tf$.

$\Leftrightarrow$ as an arrow-only fashion, a functor is a function T from arrows f of C to arrows Tf of B, carrying each identity of C to an identity of B and each composable pair $< g, f >$ of C to a composable pair $< Tg, Tf >$ in B with $Tg \circ Tf = T(g \circ f)$.

e.g.

- **a forgetful functor** : a functor which forgets some or all of the structure of an algebraic object is called a forgetful functor.

Functors

- may be composed.
  given functors $C \xrightarrow{T} B \xrightarrow{S} A$ between categories A, B, C, the composite functions $c \mapsto S(Tc)$, $f \mapsto S(Tf)$ on objects c and arrows f of C define a functor $S \circ T : C \to A$, called the composite of S with T. This is associative.

- satisfies the unit law
  for each category B, there exists an identity functor $I_B : B \to B$, which acts as an identity for the composition.

$\to$**we may consider the metacategory of all categories**; $\left\{ \begin{array}{ll} \text{objects :} & \text{categories} \\ \text{arrows :} & \text{functors} \end{array} \right.$
with the compositions defined above.

Definition 1.1.5. *A functor* $T : C \to B$ *is full when* $\forall c, c' \in ObjC$ *and to every arrow* $g : Tc \to Tc'$ *of* B*, there is an arrow* $f : c \to c'$ *of* C *s.t.* $g = Tf$*. A functor* $T : C \to B$ *is faithful when to every pair* $c, c' \in ObjC$ *and to every pair* $f_1, f_2 : c \to c'$ *of parallel arrows of* C*, the equality* $Tf_1 = Tf_2 : Tc \to Tc'$ *implies* $f_1 = f_2$.

$\Leftrightarrow \forall c, c' \in C$, $\forall f : c \to c'$ arrow of C, define a function
$\begin{array}{rcl} T_{c,c'} : \hom(c, c') & \to & \hom(Tc, Tc') \\ f & \mapsto & Tf \end{array}$. Then T is full when every such function is surjective and faithful when every such function is injective.

Definition 1.1.6. *A subcategory* S *of a category* C *is a collection of some of the objects and some of the arrows of* C*, which includes with each arrow* f *both the object* $domf$*,* $codf$ *with each object* s *its identity arrow* 1_s *and with each pair of composable arrows their composite.*

S is a full subcategory of C when the inclusion functor $S \to C$ is full.

Definition 1.1.7. *Given two functors $S, T : C \to B$, a natural transformation is a function* $\begin{array}{ccc} \tau : S & \to & T \\ c & \mapsto & \tau_c \end{array}$ *where $\tau_c = \tau c : Sc \to Tc$ of B in such a way that every arrow $f : c \to c'$ in C yields a commutative diagram*

$$\begin{tikzcd} Sc \arrow[r, "\tau c"] \arrow[d, "Sf"] & Tc \arrow[d, "Tf"] \\ Sc' \arrow[r, "\tau c'"] & Tc' \end{tikzcd}.$$

e.g. Consider an ordinal category **3** with two functors S, T. The natural transformation between S, T can be depicted as

$$\begin{tikzcd} Sa \arrow[rr, "\tau a"] \arrow[dr, "Sf"] \arrow[dd, "Sh"'] & & Ta \arrow[dr, "Tf"] \arrow[dd] & \\ & Sb \arrow[rr, "\tau b"] \arrow[dl, "Sg"] & & Tb \arrow[dl, "Tg"] \\ Sc \arrow[rr, "\tau c"] & & Tc & \end{tikzcd}$$

where $\tau a, \tau b, \tau c$ are called the components of the natural transformation τ.

A natural transformation τ with every component τc invertible in B is called a natural isomorphism, written as $\tau : S \cong T$.

Definition 1.1.8. *An equivalence between categories C and D is defined to be a pair of functors $S : C \to D$, $T : D \to C$ together with natural isomorphisms $I_C \cong T \circ S$, $I_D \cong S \circ T$.*

1.2 Derivations of Categories

1.2.1 Opposite Category

A statement Σ is defined to be any phrase built up from the types of atomic statements by means of the ordinary propositional connectives and usual quantifiers ($\forall, \exists, \exists!, \ldots,$). A sentence is a statement with all variables quantified.

Definition 1.2.1. *The dual of any statement Σ, written as Σ^*, is formed by making the replacements throughout inΣ :*

- *codomain$\leftrightarrow$domain*
- $h = g \circ f \leftrightarrow h = f \circ g$
- $\to \leftrightarrow \leftarrow$

- *monic*$\leftrightarrow$*epi*
- *terminal* $\leftrightarrow$ *initial*

$\rightarrow \Sigma^{**} = \Sigma$

the Duality Principle : if a statement Σ is a consequence of the axioms, so is the dual statement Σ^*.

Definition 1.2.2. *The opposite category C^{op} of any category C consists of objects the objects of C, and arrows f^{op} in one to one correspondence $f \mapsto f^{op}$ with arrows $f : a \to b$ of C, where $f^{op} : b \to a$. The composition $f^{op} g^{op} = (gf)^{op}$ is defined in C^{op} exactly when the composite gf is defined in C.*

$\rightarrow$ if Σ is any statement with free variables $f, g, \ldots$, then Σ is true for arrows $f, g, \ldots$, of a category C iff the dual statement Σ^* is true for the arrows $f^{op}, g^{op}, \ldots$ of the opposite category C^{op}.

If $T : B \to C$ is a functor, its object function $c \mapsto Tc$ and its mapping function $f \mapsto Tf$ can define a functor from C^{op} to B^{op}, which denote as $T^{op} : C^{op} \to B^{op}$ with the same object function and rewritten mapping function $f^{op} \mapsto (Tf)^{op}$.

1.2.2 Products of Categories

Definition 1.2.3. *Given two categories B, C, the product of B and C is a new category, written as $B \times C$; object is a pair $< b, c >$, $b \in ObjB$, $c \in ObjC$ and arrow $< b, c > \to < b', c' >$ is a pair $< f, g >$ of arrows $\begin{cases} f : b \to b' \\ g : c \to c' \end{cases}$ and the composite of two such arrows is defined in terms of the composites in B and C by*

$$< f', g' > \circ < f, g > = < f' \circ f, g' \circ g >$$

Functors $B \xleftarrow[P]{} B \times C \xrightarrow{Q} C$, called the projections of the product, are defined on arrows by $\begin{cases} P < f, g > = f \\ Q < f, g > = g \end{cases}$ with the property called **the universality**; given any category D and two functors $\begin{cases} R : D \to B \\ T : D \to C \end{cases}$, there is a unique functor $F : D \to B \times C$ with $\begin{cases} PF = R \\ QF = T \end{cases}$.

Two functors $\begin{cases} U : B \to B' \\ V : C \to C' \end{cases}$ have a product $U \times V : B \times C \to B' \times C'$ defined explicitly on $\begin{cases} \text{objects} & (U \times V) < b, c > = < Ub, Vc > \\ \text{arrows} & (U \times V) < f, g > = < Uf, Vg > \end{cases}$.
$\to$ to each pair $< B, C >$ of categories, a new category $B \times C$ to each pair of functors $< U, v >$ a new functor $U \times V$ s.t. the composites $\begin{cases} U' \circ U \\ V' \circ V \end{cases}$ are defined, $(U' \times V') \circ (U \times V) = U'U \times V'V$, i.e. $\times : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$ is a functor.

Proposition 1.2.1. *Let B, C, D be categories. For all objects $c \in C$, $b \in B$, let $L_c : B \to D$, $M_b : C \to D$ be functors s.t. $M_b(c) = L_c(b)$ for all $b \in B$, $c \in C$. Then there exists a bifunctor $S : B \times C \to D$ with $\begin{cases} S(-, c) = L_c \\ S(b, -) = M_b \end{cases}$ iff for every pair of arrows $\begin{cases} f : b \to b' \\ g : c \to c' \end{cases}$, one has*

$$M_{b'} g \circ L_c f = L_{c'} f \circ M_b g$$

pf.

- if we write b, c for the corresponding identity arrows, $< b', c > \circ < f, c > = < b'f, gc > = < f, g > = < fb, c'g > = < f, c' > \circ < b, g >$ by the definition of the composition. Applying the functor S to the above equality gives $S(b', g)S(f, c) = S(f, c')S(b, g)$ i.e. the condition in the proposition is satisfied.
- given all L_c and M_b, the condition in the proposition defines $S(f, g)$ for every pair f, g. This yields a bifunctor S with the required properties.

Consider two bifunctors $S, S' : B \times C \to D$ and let α be a function which assigns to each pair of objects $b \in B$, $c \in C$, an arrow $\alpha(b, c) : S(b, c) \to S'(b, c)$ in D. Then α is called natural in b if $\forall c \in C$, $\alpha(b, c)$ for all b define $\alpha(-, c) : S(-c) \to S'(-, c)$, a natural transformation of functors $B \to D$.

Proposition 1.2.2. *For bifunctors S, S', the function α defined above is a natural transformation $\alpha : S \to S'$ iff $\alpha(b, c)$ is natural in b, $\forall c \in C$ and natural in c, $\forall b \in B$.*

e.g. Given two ordinal categories $\mathbf{3}$, $\mathbf{2}$, we can construct a product $\mathbf{3} \times \mathbf{2}$

$$\begin{tikzcd}
& \cdot \arrow[dr] & \\
\cdot \arrow[ur] \arrow[rr] & & \cdot \\
& \cdot \arrow[uu] \arrow[dr] & \\
\cdot \arrow[uu] \arrow[ur] \arrow[rr] & & \cdot \arrow[uu]
\end{tikzcd}$$

. More generally, instead of $\mathbf{3}$, consider any random category C

and the functors $T_0.T_1 : C \to C\times\mathbf{2}$ are defined for each arrow f of C by $\begin{cases} T_0 f = <f,0> \\ T_1 f = <f,1> \end{cases}$.
If $\downarrow$ denotes the unique non-identity arrow $0 \to 1$ of $\mathbf{2}$, we may define a transformation between $T_0, T_1 : C \to C \times \mathbf{2}$ by $\mu : T_0 \to T_1$, $\mu c = <c, \downarrow>$ for any object c. It maps bottom to top and is natural. Here μ is called the universal natural transformation from C.

1.2.3 Higher Categories

Given categories B, C, consider all functors $R, S, T, \cdots : C \to B$. If $\begin{cases} \sigma : R \to S \\ \tau : S \to T \end{cases}$ are two natural transformations, their components for each $c \in C$, define composite arrows $(\tau \cdot \sigma)c = \tau c \circ \sigma c$ which are components of a transformation $\tau \cdot \sigma : R \to T$.

For any arrow $f : c \to c''$ in C, $\begin{array}{ccc} Rc & \xrightarrow{Rf} & Rc' \\ \Big\downarrow{\sigma c} & & \Big\downarrow{\sigma c'} \\ Sc & \xrightarrow{Sf} & Sc' \\ \Big\downarrow{\tau c} & & \Big\downarrow{\tau c'} \\ Tc & \xrightarrow{Tf} & Tc' \end{array}$, the small squares in the diagram commute since σ, τ are natural. Hence the outer rectangle commutes, i.e. $\tau \cdot \sigma$ is natural. The composition of natural transformations is associative and $\exists 1_T : T \to T$ with components $1_T c = 1_{T_c}$.
$\to$ a functor category $B^C = Funct(C, B)$ with
$\begin{cases} \text{objects} & \text{the functors } T : C \to B \\ \text{arrow} & \text{natural transformations between functors} \end{cases}$. Here, $Nat(S,T) = B^C(S,T) = \{\tau | \tau : S \to T \text{ natural } \}$ is the hom-set for this category.

e.g.

- $B^{\mathbf{1}}$ isomorphic to B, $B^{\mathbf{2}}$ is the category of arrows of B

- if K is a commutative ring and G a group, $(K-\mathbf{Mod})^G$ is the category of K-linear representations of G.
  each functor $T : G \to K-\mathbf{Mod}$ is determined by a K-module V and a morphism $T : G \to Aut(V)$ of groups. If T' is a second such representation, a natural transformation $\sigma : T \to T'$ is given by a single arrow $\sigma : V \to V'$ s.t.
  $\begin{array}{ccc} V & \xrightarrow{\sigma} & V' \\ \Big\downarrow{Tg} & & \Big\downarrow{T'g} \\ V & \xrightarrow{\sigma} & V' \end{array}$ commutes $\forall g \in G$. Here σ is an interwining operator.

Given functors $C \underset{T}{\overset{S}{\rightrightarrows}} B \underset{T'}{\overset{S'}{\rightrightarrows}} A$ and natural transformation $\begin{cases} \tau : S \to T \\ \tau' : S' \to T' \end{cases}$, we can define the composite $(\tau' \circ \tau)$ for an object c, $(\tau' \circ \tau)c = T'\tau c \circ \tau' Sc = \tau' Tc \circ S'\tau c$ and for an arrow $f : c \to b$, $(\tau' \circ \tau)f : S'Sf \to T'Tf$ which is natural. The naturality comes from the commutative diagram $\begin{array}{ccccc} S'Sc & \xrightarrow{S'\tau c} & S'Tc & \xrightarrow{\tau' Tc} & T'Tc \\ \downarrow{S'Sf} & & \downarrow{S'Tf} & & \downarrow{T'Tf} \\ S'Sb & \xrightarrow{S'\tau b} & S'Tb & \xrightarrow{\tau' Tb} & T'Tb \end{array}$ where the left square commutes, given τ natural and S' a functor. Also the right square commutes given τ' natural and Tf an arrow. Thus the outer square also commutes and therefore $\tau' \circ \tau$ is natural. $\circ$ satisfies the associativity and the unit law axioms. $\to \begin{cases} S' \circ \tau : S' \circ S \to S' \circ T \\ \tau' \circ T : S' \circ T \to T' \circ T \end{cases}$ gives the composition $\tau' \circ \tau = (T' \circ \tau) \cdot (\tau' \circ S) = (\tau' \circ T) \cdot (S' \circ \tau)$.

Theorem 1.2.1. *The collection of all natural transformations is the set of arrows of two different categories under two different operations of composition $\cdot$ and $\circ$, which satisfy the interchange law*

$$(\tau' \cdot \sigma') \circ (\tau \cdot \sigma) = (\tau' \circ \tau) \cdot (\sigma' \circ \sigma)$$

Moreover any transformation which is an identity for the composition $\circ$ is also an identity for the composition $\cdot$.

Definition 1.2.4. *A double category is the set of arrows for two different compositions which together satisfy the interchange law. A 2-category is a double category in which every identity arrow for the first composition is also an identity for the second composition.*

The functor category B^C is itself a functor of the categories B, C. The arrow function sends a pair of functors $\begin{cases} F : B \to B' \\ G : C \to C' \end{cases}$ to the functor $F^G : B^C \to B'^{C'}$ defined on objects $S \in B^C$ as $F^G S = F \circ S \circ G$ and on arrows $\tau : S \to T$ in B^C as $F^G \tau = F \circ \tau \circ G$.

1.2.4 Comma Categories

- for $b \in ObjC$, the category of objects under b is the category $(b \downarrow C)$ with objects all pairs $< f, c >$ where $\begin{cases} c \in ObjC \\ f : b \to c \text{ an arrow of } C \end{cases}$ $\begin{array}{c} b \\ \downarrow{f} \\ c \end{array}$, $\begin{tikzcd} & b \arrow[dl, "f"'] \arrow[dr, "\bar{f}"] & \\ c \arrow[rr, "h"] & & \bar{c} \end{tikzcd}$

  and with arrows $h :< f, c > \to < \bar{f}, \bar{c} >$ those arrows $h : c \to \bar{c}$ of C for which

$h \circ f = \bar{f}$.

e.g. X any set, $*$one-point set. Then each function $* \to X$ is a selection of a point in the set X. Thus $(* \downarrow \mathbf{Set})$ is the category of pointed set.

- for $a \in ObjC$, the category $(C \downarrow a)$ of objects over a consists of objects all pairs $< f, c >$, where $c \in ObjC$ and $f : c \to a$ an arrow of C, and arrows $h :< f, c > \to < \bar{f}, \bar{c} >$ those arrows $h : c \to \bar{c}$ of C for which $\bar{f} \circ h = f$. Then $\begin{array}{c} c \\ \downarrow f \\ c \end{array}$,

$$\begin{tikzcd} c \arrow[rr, "h"] \arrow[dr, "f"'] & & \bar{c} \arrow[dl, "\bar{f}"] \\ & a & \end{tikzcd}$$

e.g. $*$ is terminal in $\mathbf{Set}$, so there is a unique $X \to *$. Thus $(\mathbf{Set} \downarrow *) \cong \mathbf{Set}$.

- if $b \in ObjC$ and $S : D \to C$ a functor, the category $(b \downarrow S)$ of objects S-under b consists of objects $< f, d >$ where $\begin{cases} d \in ObjD \\ f : b \to Sd \end{cases}$ and arrow $h :< f, d > \to < \bar{f}, \bar{d} >$ all those arrows $h : d \to d'$ in D for which $\bar{f} = Sh \circ f$ $\begin{array}{c} b \\ \downarrow f \\ Sd \end{array}$,

$$\begin{tikzcd} & d \arrow[dl, "f"'] \arrow[dr, "\bar{f}"] & \\ Sd \arrow[rr, "Sh"] & & S\bar{d} \end{tikzcd}.$$

e.g. For a forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$, $(x \downarrow U)$ is a function $x \to Ug$ from x into the underlying set of some group g.

- if $a \in ObjC$ and $T : E \to C$ a functor, a category $(T \downarrow a)$ of objects T-over a consists of objects $< f, d >$ where $\begin{cases} d \in ObjE \\ f : Ed \to a \end{cases}$ and arrows $h :< f, d > \to < \bar{f}, \bar{d} >$ all those arrows $h : d \to \bar{d}$ in E for which $\bar{f} \circ h = f$.

Definition 1.2.5. *Given categories and functors* $E \xrightarrow{T} C \xleftarrow[S]{} D$ *, the comma category* $(T \downarrow S)$ *has as objects all triples* $< e, d, f >$ *where* $\begin{cases} d \in ObjD, e \in ObjE \\ f : Te \to Sd \end{cases}$ *and as arrows* $< e, d, f > \to < e', d', \bar{f} >$ *all pairs* $< k, h >$ *of arrows* $\begin{cases} k : e \to e' \\ h : d \to d' \end{cases}$ *s.t.*

$\bar{f} \circ Tk = Sh \circ f$.

$$\to \begin{array}{c} Te \\ \big\downarrow{\scriptstyle f} \\ Sd \end{array}, \quad \begin{tikzcd} Te \arrow[r, "Tk"] \arrow[d, "f"] & Te' \arrow[d, "\bar{f}"] \\ Sd \arrow[r, "Sh"] & Sd' \end{tikzcd}$$

The construction of the comma category is visualised by the commutative diagram

$$\begin{tikzcd} & & (T \downarrow S) \arrow[dll, "P"'] \arrow[d, "R"] \arrow[drr, "Q"] & & \\ E \arrow[r, "T"] & C & C^{\mathbf{2}} \arrow[l, "C^{d_0}"] \arrow[r, "C^{d_1}"] & C & D \arrow[l, "S"] \end{tikzcd}$$

$$\Leftrightarrow \begin{tikzcd} & & <e, d, f : Te \to Sd> \arrow[dll] \arrow[d] \arrow[drr] & & \\ e \arrow[r] & Te & f : Te \to Sd \arrow[l] \arrow[r] & Sd & d \arrow[l] \end{tikzcd}$$

where d_0, d_1 are the two functors $\mathbf{1} \to \mathbf{2}$ so that C^{d_0}, C^{d_1} are the functors which send each arrow f of C to its domain and codomain respectively.

1.2.5 Free Categories

Let FX denote a free monoid generated by a set X. Then it consists of all finite strings $x_1x_2 \ldots x_n$, $\forall x_i \in X$. The multiplication of these strings is given by juxtaposition.

For any monoid M, let UM denote the set of elements of M. Then any function $f : X \to UM$ extends to a unique morphism of monoids $g : FX \to M$.

Let G be a directed graph with set of objects O, set of arrows A, and a pair of functions $A \rightrightarrows O$. A morphism $D : G \to G'$ of graphs is a pair of functions

$$\begin{cases} D_O : O \to O' \\ D_A : A \to A' \end{cases} \text{s.t.} \begin{cases} D_O \partial_O f = \partial_O D_A f \\ D_O \partial_1 f = \partial_1 D_A f \end{cases} \text{for all arrow } f \in A.$$

Every category C determines a graph UC with the same objects and arrows, forgetting the structure of the composition of arrows. Every functor $F : C \to C'$ is also a morphism $UF : UC \to UC'$ between the corresponding graphs. Thus we may define the forgetful functor $U : \mathbf{Cat} \to \mathbf{Grph}$.

Let O be a fixed set. An O-graph is one with O as its set of objects. A morphism D of O-graphs will be one with $D_O : O \to O$ the identity. If A, B are two O-graphs, the product over O is $A \times_O B = \{< g, f > | \partial_0 g = \partial_1 f,\ g \in A, f \in B\}$, called the set of composable pairs of arrows. Then this set itself is a O-graph by the definition of the composition. This satisfies the associativity and the unit law axioms.

$\to$a category with objects O may be described as an O-graph A equipped with two morphisms $\begin{cases} c : A \times_O A \to A \\ \quad i : O \to A \end{cases}$. So **Set** can be replaced by O-**Grph** and product of sets by $\times_O$.

Theorem 1.2.2. *Let $G = \{A \rightrightarrows O\}$ be a small graph. There is a small category $C = C_G$ with O as set of objects and a morphism $P : G \to UC$ of graphs from G to the underlying graph UC of C with the universality; given any category B and any morphism $D : G \to UB$ of graphs, there is a unique functor $D' : C \to B$ with*

$(UD') \circ P = D$

$$\begin{tikzcd} G \arrow[r, "P"] \arrow[dr, "D"'] & UC \arrow[d, dotted, "UD'"] \\ & UB \end{tikzcd}$$

pf. Take the objects of C to be those of G and the arrows of C to be the finite strings $a_1 \xrightarrow{f_1} a_1 \xrightarrow{f_2} a_3 \longrightarrow \ldots \xrightarrow{f_{n-1}} a_n$ composed of n objects $a_1, \ldots, a_n$ of G connected by $n-1$ arrows $f_i : a_i \to a_{i+1}$ of G. Regard each of such string as an arrow $< a_1 f_1 f_{n-1} a_n >: a_1 \to a_n$ in C and define the composite of two strings by juxtaposition, identifying the common end. Then it is associative and strings $< a_i >$ of length $n = 1$ are its identities and any string with length $n > 1$ is a composite $< a_1 f_1 a_2 \ldots a_{n-1} f_{n-1} a_n >=< a_{n-1} f_{n-1} a_n) \circ \cdots \circ < a_1 f_1 a_2 >$.

- Consider any other morphism $D : G \to UB$ of the given graph G to the underlying graph of some category B. If there is a functor $D' : C \to B$ with $UD' \circ P = D$ as in the diagram above, D' must be $\begin{cases} D' < a >= Da & \text{on objects} \\ D' < a_1 f_1 a_2 >= Df_1 & \text{on arrows} \end{cases}$. Since any string of length $n > 1$ is a composite in C, D' must be given by $D' < a_1 f_1 \ldots a_{n-1} f_{n-1} a_n >= Df_{n-1} \circ \cdots \circ Df_1$.

- the formula does define a functor $D' : B \to C$ for which the indicated diagram commutes.

Corollary 1.2.1. *To any set X, there is a monoid M and a function $p : X \to UM$ where UM is the underlying set of M, with the following universality; for any monoid L and any function $h : X \to X \to UL$, there is a unique morphism $h' : M \to L$ of monoids with $h = Uh' \circ p$.*

1.3 Universality

Definition 1.3.1. *If $S : D \to C$ is a functor and $c \in ObjC$, a universal arrow from c to S is a pair $< r, u >$ consisting of an object r of D and an arrow $u : c \to Sr$ of C s.t. to every pair $< d, f >$ with d an object of D and $f : c \to Sd$ an arrow of C, there* *is a unique arrow $f' : r \to d$ of D with $Sf' \circ u = f$. In other words, every arrow f to S factors uniquely through the universal arrow u ;*

$$\begin{tikzcd} c \arrow[r, "u"] \arrow[dr, "f"'] & Sr \arrow[d, dotted, "Sf'"] \\ & Sd \end{tikzcd}$$

$\Leftrightarrow$ $u : c \to Sr$ is universal from c to S when the pair $< r, u >$ is an initial object in the comma category $(c \downarrow S)$, whose objects are the arrows $c \to Sd$.

e.g.

- Let $\mathbf{Vct_K}$ denote the category of all vector spaces over a fixed field K, with arrows linear transformations. Let $U : \mathbf{Vct_K} \to \mathbf{Set}$ be the forgetful functor, sending each vector space V to the set of its elements. For any set X, there is a familiar vector space V_X with X as a set of basis vectors. Then the function which sends $\forall x \in X$ into the same x regarded as a vector of V_X is an arrow $j : X \to U(V_X)$. For any other vector space W, each function $f : X \to U(W)$ can be extended to a unique linear transformation $f' : V_X \to W$ with $Uf' \circ j = f$, i.e. j is a universal arrow from X to U.

- Let **Met** be the category of all metric spaces $X, Y, \ldots$, with arrows $X \to Y$ those functions which preserves the metric. Then the complete metric spaces form a full subcategory of **Met**, and the familiar completion $\bar{X}$ of a metric space X provides an arrow $X \to \bar{X}$ which is universal for the forgetful functor.

Definition 1.3.2. *If D is a category and $H : D \to$ **Set** a functor, a universal element of the functor H is a pair $< r, e >$ consisting of an object $r \in D$ and an element $e \in Hr$ s.t. for every pair $< d, x >$ with $x \in Hd$, there is a unique arrow $f : r \to d$ of D with $(Hf)e = x$.*

The notion of "universal element" is a special case of the notion "universal arrow".
if $*$ is the set with one point, then any element $e \in Hr$ can be regarded as an arrow $e : * \to Hr$. Thus the universal element $< r, e >$ of H is exactly a universal arrow from $*$ to H.
Conversely, when C has small hom-sets, if $S : D \to C$ is a functor and $c \in C$ is an object, $< r, u : c \to Sr >$ is a universal arrow from c to S iff the pair $< r, u \in C(c, Sr) >$ is a universal element of the functor $H = C(c, S-)$.

Definition 1.3.3. *A universal arrow from S to c is a pair $< r, v >$ consisting of an object $r \in D$ and an arrow $v : Sr \to c$ with codomain c s.t. to every pair $< d, f >$*

with $f: Sd \to c$, there is a unique $f': d \to r$ with $f = v \circ Sf'$;
$$\begin{tikzcd} Sd \arrow[r, "f"] \arrow[d, dotted, "Sf'"'] & c \\ Sr \arrow[ur, "v"'] & \end{tikzcd}$$

→ the dual of an universal arrow from an object $c \in C$ to a functor $S: D \to C$

e.g. the projections $\left\{ \begin{array}{l} p: a \times b \to a \\ q: a \times b \to b \end{array} \right.$ of a product in C.

Proposition 1.3.1. *For a functor $S: D \to C$, a pair $< r, u: c \to Sr >$ is universal from c to S iff the function sending each $f': r \to d$ into $Sf' \circ u: c \to Sd$ is a bijection of hom-sets*

$$D(r,d) \cong C(c, Sd)$$

This is natural in d. Conversely, given r, c, any natural isomorphism above is determined in this way by a unique arrow $u: c \to Sr$ s.t. $< r, u >$ is universal from c to S.

pf.

- The statement that $< r, u >$ is universal is exactly the statement that $f' \mapsto Sf' \circ u = f$ is a bijection. This is natural in d since if $g': d \to d'$, $S(g'f') \circ u = Sg' \circ (Sf' \circ u)$.

- A natural isomorphism gives for each object $d \in ObjD$ a bijection $\phi_d: D(r,d) \to C(c, Sd)$. Then for any arrow $f': r \to d$,
$$\begin{tikzcd} D(r,r) \arrow[r, "\phi_r"] \arrow[d] & C(c,Sr) \arrow[d] \\ D(r,d) \arrow[r, "\phi_d"] & C(c,Sd) \end{tikzcd}$$
commutes because ϕ is natural. In this diagram, the identity $1_r \in D(r,r)$ is mapped to $\left\{ \begin{array}{l} Sf' \circ u \\ \phi_d(f') \end{array} \right.$. Since ϕ_d is a bijection, that means each $f: c \to Sd$ has the form $f = Sf' \circ u$ for a unique f', i.e. $< r, u >$ is universal.

→ the functor $C(c, S-)$ to **Set** is isomorphic to a hom-functor $D(r, -)$, which is called representations.

Definition 1.3.4. *Let D have small hom-sets. A representation of a functor K : $D \to$ **Set** is a pair $< r, \psi >$ with $r \in ObjD$ and $\psi: D(r,-) \cong K$ a natural isomorphism. The object r is called the representing object. The functor K is said to be representable when such a representation exists.*

Proposition 1.3.2. *Let $*$ denote any one-point set and let D have small hom-sets. If $< r,u : * \to Kr >$ is a universal arrow from $*$ to $K : D \to$ **Set**, then the function ψ which for each object $d \in ObjD$ sends the arrow $f' : r \to d$ to $K(f')(u*) \in Kd$ is a representation of K. Every representation of K is obtained in this way from exactly one such universal arrow.*

pf. For any set X, a function $f : * \to X$ is determined by the element $f(*) \in X$. This correspondence $f \mapsto f(*)$ is a bijection $\mathbf{Set}(*,X) \to X$, natural in $X \in \mathbf{Set}$. Composing with K yields a natural isomorphism $\mathbf{Set}(*,K-) \to K$. This with the previous proposition gives $\mathbf{Set}(*,K-) \cong K \cong D(r,-)$. Therefore a representation of K amounts to a natural isomorphism $\mathbf{Set}(*,K) \cong D(r,-)$.
$\to$ given the universal arrow u, and a representation ψ, ψ_r maps $1 : r \to r$ to an element of Kr, which is a universal element, hence also a universal arrow $* \to Kr$ and therefore $f' \mapsto K(f')(u(*))$ is a representation.

Lemma 1.3.1 (Yoneda). *If $K : D \to$ **Set** is a functor from D and $r \in ObjD$, there is a bijection*

$$y : Nat(D(r,-),K) \cong Kr$$

which sends each natural transformation $\alpha : D(r,-) \to K$ to $\alpha_r 1_r$, the image of the identity $r \to r$.

pf. $\begin{array}{ccccc} D(r,r) & \xrightarrow{\alpha_r} & K(r) & & r \\ \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle K(f)} & & \downarrow{\scriptstyle f} \\ D(r,d) & \xrightarrow{\alpha_d} & K(d) & & d \end{array}$ commutes and the lemma follows.

$\to y$ is natural in K and r.
consider K as an object in the functor category $\mathbf{Set}^D$ and regard both domain and codomain of y as functors of the pair $< K,r >$. Consider the pair $< K,r > \in Obj\mathbf{Set}^D \times D$. Then the codomain of y is the evaluation functor E, which maps each pair $< K,r > \to Kr$ and the domain of y is the functor N which maps the object $< K,r >$ to the set $Nat(D(r,-),K)$ of all natural transformations. Then $y : N \to E$ between the functors $E,N : \mathbf{Set}^D \times D \to \mathbf{Set}$, called the **Yoneda functor**. Its dual is another functor $Y' : D \to \mathbf{Set}^{D^{op}}$, which sends $f : s \to r$ to the natural transformation $D(-,f) : D(-,s) \to D(-,r) : D^{op} \to \mathbf{Set}$.

Corollary 1.3.1. *For objects $r,s \in ObjD$, each natural transformation $D(r,-) \to D(s,-)$ has the form $D(h,-)$ for a unique arrow $h : s \to r$.*

1.4 Adjoint

Definition 1.4.1. *Let A,X be categories. An adjunction from X to A is a triple $<F,G,\phi>: X \to A$ where F,G are functors*

$$X \underset{F}{\overset{G}{\rightleftarrows}} A$$

while ϕ is a function which assigns to each pair of objects $x \in ObjX, a \in ObjA$ a bijection of sets $\phi = \phi_{x,a} : A(Fx,a) \cong X(x,Ga)$ which is natural in x,a.

$\to A(Fx,a)$ is the bifunctor

$$\begin{array}{ccc} X^{op} \times A & \xrightarrow{F^{op}\times Id} A^{op} \times A \xrightarrow{hom} & \mathbf{Set} \\ <x,a> & \longrightarrow & A(Fx,a) \end{array}$$

and $X(x,Ga)$ is a similar bifunctor $X^{op} \times A \longrightarrow \mathbf{Set}$. Thus the naturality of the bijection ϕ means both of the diagrams

$$\begin{array}{ccc} A(Fx,a) & \xrightarrow{\phi} & X(x,Ga) \\ \downarrow{\scriptstyle k_*} & & \downarrow{\scriptstyle (Gk)_*} \\ A(Fx,a') & \xrightarrow{\phi} & X(x,Ga') \end{array},$$

$$\begin{array}{ccc} A(Fx,a) & \xrightarrow{\phi} & X(x,Ga) \\ \downarrow{\scriptstyle (Fh)^*} & & \downarrow{\scriptstyle h^*} \\ A(Fx',a) & \xrightarrow{\phi} & X(x',Ga) \end{array}$$

commute.

Given an adjunction, the functor F is said to be a left-adjoint for G, while G is called a right-adjoint for F.

e.g. For a fixed field K, $\mathbf{Set} \underset{V}{\overset{U}{\rightleftarrows}} \mathbf{Vct}_K$ are functors where for each vector space W, $U(W)$ is the set of all vectors in W, so that U is the forgetful functor, while for any set X, $V(X)$ is the vector space with basis X. Each function $g : X \to U(W)$ extends to a unique linear transformation $\begin{array}{ccc} f : V(X) & \to & W \\ \sum r_i x_i & \mapsto & \sum r_i (g x_i) \end{array}$. This correspondence $\psi : g \mapsto f$ has an inverse $\phi : f \mapsto f|_X$, the restriction of f to $X \to \phi$: $\mathbf{Vct_K}(V(X),W) \cong \mathbf{Set}(X,U(W))$.

Theorem 1.4.1. *An adjuction $<F,G,\phi>: X \to A$ determines*

1. *a natural transformation $\eta : I_X \to GF$ s.t. for each object x, the arrow η_x is universal to G from x while the right adjunct of each $f : Fx \to a$ is $\phi f = Gf \circ \eta_x : x \to Ga$*

2. *a natural transformation* $\epsilon : FG \to I_A$ *s.t. each arrow* ϵ_a *is universal to* a *from* F*, while each* $g : x \to Ga$ *has left adjunct* $\phi^{-1}g = \epsilon_a \circ Fg : Fx \to a$

Moreover, both the following composites are the identities

$$G \xrightarrow{\eta G} GFG \xrightarrow{G\epsilon} G \quad F \xrightarrow{F\eta} FGF \xrightarrow{\epsilon F} F$$

pf.

- Set $a = Fx$ in the natural isomorphism in the definition. The LHS hom-sets then contains the identity $1 : Fx \to Fx$, and call its ϕ-image η_x. By the Yoneda's lemma, η_x is a universal arrow $\eta_x : x \to GFx$, $\eta_x = \phi(1_{Fx})$ from $x \in X$ to G. The adjunction gives a such a universal arrow η_x, $\forall x \in ObjX$. The function$x \mapsto \eta_x$ is a natural transformation $I_X \to GF$ because every diagram $\begin{array}{ccc} y & \xrightarrow{\eta_y} & GFy \\ \downarrow h & & \downarrow GFh \\ x & \xrightarrow{\eta_x} & GFx \end{array}$ commutes for $h : y \to x$ ($GFh \circ \phi(1_{Fy}) = \phi(Fh \circ 1_{Fy}) = \phi(1_{Fx} \circ Fh) = \phi(1_{Fx}) \circ h$ by the naturality of ϕ. The bijection ϕ can be expressed in terms of the arrows η_x as $\phi(f) = G(f)\eta_x$ for $f : Fx \to a$ and therefore $\phi(f) = \phi(f \circ 1_{Fx}) = Gf \circ \phi 1_{Fx} = Gf \circ \eta_x$).

- Set $x = Ga$ in the natural isomorphism in the definition. The identity arrow $1 : Ga \to Ga$ is in the RHS hom-set. Its image under ϕ^{-1} is called $\epsilon_a : FGa \to a$, $\epsilon_a = \phi^{-1}(1_{Ga})$, $\forall a \in A$. Then similarly it is a universal arrow from F to a.

- Take $x = Ga$. Then $\epsilon_a = \phi^{-1}(1_{Ga})$ gives $1_{Ga} = \phi(\epsilon_a) = G(\epsilon_a) \circ \eta_{Ga}$, which assets that the composite natural transformation $G \xrightarrow{\eta G} GFG \xrightarrow{G\epsilon} G$ is the identity transformation.

$\to \left\{ \begin{array}{l} \eta \\ \epsilon \end{array} \right.$ is called the $\left\{ \begin{array}{c} \text{the unit} \\ \text{the counit} \end{array} \right.$ of the adjunction.

Lemma 1.4.1. *Let* $f^* : A(a,-) \to A(b,-)$ *be the natural transformation induced by an arrow* $f : b \to a$ *of* A*. Then* f^* *is monic iff* f *is epi, while* f^* *is epi iff* f *is a split monic.*

pf.

- For $h \in A(A,c)$, $f^* = hf$. Hence the first statement follows by the definition of epi. If f^* is epi, there is an $h_0 : a \to b$ s.t. $f^*h_0 = h_0 f = 1 : b \to b$, so f has a left inverse, i.e. split monic.

- the converse is trivial.

$\to f^* \mapsto f$ is the bijection $Nat(A(a,-),A(b,-)) \cong A(b,a)$ by the Yoneda lemma.

Theorem 1.4.2. *For an adjunction $<F,G,\eta,\epsilon>: X \to A$*

1. *G is faithful iff every component ϵ_a of the counit ϵ is epi.*

2. *G is full iff every ϵ_a is a split monic.*

Hence G is full and faithful iff each ϵ_a is an isomorphism $FGa \cong a$.

pf. By the Yoneda lemma to the natural transformation, $A(a,c) \xrightarrow{G_{a,c}} X(Ga,Gc) \xrightarrow{\phi^{-1}} A(FGa,c)$, determined by the image of $1: a \to a$, i.e. the counit $\epsilon_a : FGa \to a$. But ϕ^{-1} is an isomorphism, hence this natural isomorphism is monic or epi respectively when every $G_{a,c}$ is injective or surjective respectively. Then the theorem follows by the lemma.

Definition 1.4.2. *A subcategory A of B is called reflective in B when the inclusion functor $K : A \to B$ has a left adjoint $F : B \to A$. The functor F is called a reflector and the adjunction $<F,K,\phi>=<F,\phi>: B \to A$ a reflection of B in A.*

$\to A \subset B$ is reflective in B iff $\exists R : B \to B$ a functor with values in the subcategory A and a bijection of sets $A(Rb,a) \cong B(b,a)$ natural in $b \in ObjB, a \in ObjA$.

$\Leftrightarrow A \subset B$ is **coreflective** in B when the inclusion functor $A \to B$ has a right adjoint.

e.g. **Ab** is reflective in **Grp** $\to \hom(G/[G,G],A) \cong \hom(G,A)$ for all $A \in Obj$**Ab**.

Definition 1.4.3. *A functor $S : A \to C$ is an isomorphism of categories when there is a functor $T : C \to A$ s.t.* $\begin{cases} ST = I : C \to C \\ TS = I : A \to A \end{cases}$.

$\to \begin{cases} \eta : I \to ST \\ \epsilon : TS \to I \end{cases}$ make $<T,S;\eta,\epsilon> C \to A$ an adjunction.

Definition 1.4.4. *A functor $S : A \to C$ is an equivalence of categories when there is a functor $T : C \to A$ and natural isomorphisms* $\begin{cases} ST \cong I : C \to C \\ TS \cong I : A \to A \end{cases}$.

e.g. A skeleton of C is any full subcategory A s.t. each object of C is isomorphic to exactly one object of A. Then the inclusion functor $K : A \to C$ is an equivalence of categories ($\forall c \in ObjC$, $\theta_c : c \cong Tc$ with $Tc \in ObjA$ yields a functor $T : C \to A$ in exactly one way so that θ will become a natural isomorphism $\theta : I \cong KT$. Moreover $TK \cong I$, and therefore it is an equivalence),
→a category is equivalent to its skeletons.

Definition 1.4.5. *An adjoint equivalence of categories is an adjunction $< T,S;\eta,\epsilon >: C \to A$ in which both the unit $\eta : I \to ST$ and the counit $\epsilon : TS \to I$ are natural isomorphisms.*

$\to < S,T,\epsilon^{-1},\eta^{-1} >: A \to C$ is an adjunction with ϵ^{-1} as unit and η^{-1} as counit.

Theorem 1.4.3. *TFAE for a functor $S : A \to C$*

1. *S is an equivalence of categories*
2. *S is part of an adjoint equivalence $< T,S;\eta,\epsilon >: C \to A$*
3. *S is full and faithfull, and each $c \in ObjC$ is isomorphic to Sa for some object $a \in A$.*

pf.

- $2 \to 1$ is trivial
- $1 \to 3$
  $ST \cong I$ shows $\forall c \in ObjC$, $\exists a = Tc \in A$.s.t. $c \cong S(Tc)$. The natural isomorphism $\theta : TS \cong I$ gives for each $f : a \to a'$, the diagram
  $$\begin{tikzcd} TSa \arrow[r, "\theta_a"] \arrow[d, "TSf"] & a \arrow[d, "f"] \\ TSa' \arrow[r, "\theta_{a'}"] & a' \end{tikzcd}$$
  Hence $f = \theta_{a'} \circ TSf \circ \theta_a^{-1}$, i.e. S is faithful. Symmetricall $ST \cong I$ proves T is faithful.
  Consider any $h : Sa \to Sa'$ and set $f = \theta_{a'} \circ Th \circ \theta_a^{-1}$. Then the square above commutes also with Sf replaced by h, so $TSf = Th$. Since T is faithful, $Sf = h$, which means S is full.
- $3 \to 2$
  $\forall c \in ObjC$, we can choose $a_0 = T_0 c \in A$ and an isomorphism η_c :
  $$\begin{tikzcd} c \arrow[r, "\sim"] \arrow[dr, "f"'] & S(T_0c) \arrow[d, "Sg"] \\ & Sa \end{tikzcd}$$
  where $g : T_0 c \to a$. For every arrow $f : c \to Sa$, the composite $f \circ \eta_c^{-1}$ has the form Sg for some g, given S full. This g is unique, given S faithful. In other words, $f = Sg \circ \eta_c$ for a unique g, so η_c is universal from c to S. Thus T_0 can be made a functor $T : C \to A$ so that $\eta : I \to ST$ is natural. Thus T is the left adjoint of S with the unit η. As with any adjunction $S\epsilon_a \cdot \eta_{Sa} = 1$, and therefore $S\epsilon_a = (\eta_{Sa})^{-1}$ is invertible. Since S is full and faithful, the counit ϵ_a is also invertible. Thus 2 follows.

Corollary 1.4.1. *If A is a full subcategory of C and every $c \in ObjC$ is isomorphic to some object of A, then the insertion $K : A \to C$ is an equivalence and is part of an adjoint equivalence $< T, K; \eta, 1 >: C \to A$ with counit the identity. Therefore A is reflective in C.*

pf. In the previous proof, if we suppose A is a full subcategory of C and S is the insertion, $\forall a \in ObjA \subseteq ObjC$, we can choose $a_0 = a = Ka$ and η_{Ka} the identity. Then $K\epsilon_a = 1$, hence $\epsilon_a = 1$, $\forall a \in A$.

Definition 1.4.6. *A category C with all finite products is called cartesian closed when each of the following functors*

$$\begin{array}{ccc} C & \to & \mathbf{1} \\ c & \mapsto & 0 \end{array}, \quad \begin{array}{ccc} C & \to & C \times C \\ c & \mapsto & < c, c > \end{array}, \quad \begin{array}{ccc} C & \to^{\times b} & C \\ a & \mapsto & a \times b \end{array}$$

has a specified right adjoint

$$t \leftarrow\!\dashv 0, \ a \times b \leftarrow\!\dashv < a, b >, \ c^b \leftarrow\!\dashv c$$

$$\to \left\{ \begin{array}{r} \text{the first is to specify a terminal object } t \in ObjC \\ \text{the second is specifying } \forall a, b \in C \text{ a product } a \times b \text{ with projections } a \leftarrow a \times b \to b \\ \text{the third specifies } \hom(a \times b, c) \cong \hom(a, c^b) \text{ natural in } a, c \end{array} \right.$$

Definition 1.4.7. *Given two adjunctions* $\left\{ \begin{array}{c} < F, G, \phi, \eta, \epsilon >: X \to A \\ < F', G', \phi', \eta', \epsilon' >: X' \to A' \end{array} \right.$, *define a map of adjunctions to be a pair of functors* $\left\{ \begin{array}{l} K : A \to A' \\ L : X \to X' \end{array} \right.$ *s.t.*

$$\begin{array}{ccccc} A & \xrightarrow{G} & X & \xrightarrow{F} & A \\ \downarrow{\scriptstyle K} & & \downarrow{\scriptstyle L} & & \downarrow{\scriptstyle K} \\ A' & \xrightarrow{G} & X' & \xrightarrow{F} & A' \end{array}$$

of functors commute and s.t. the diagram of hom-sets and adjunctions commute

$$\begin{array}{ccc} A(Fx, a) & \xrightarrow{\phi} & X(x, Ga) \\ \downarrow{\scriptstyle K = K_{Fx,a}} & & \downarrow{\scriptstyle L = L_{x,Ga}} \\ A'(KFx.Ka) & & X'(Lx, LGa) \\ \downarrow{\scriptstyle \equiv} & & \downarrow{\scriptstyle \equiv} \\ A'(F'Lx, Ka) & \xrightarrow{\phi} & X'(Lx, G'Ka) \end{array}$$ *for all $x \in ObjX$, $a \in ObjA$.*

Proposition 1.4.1. *Given adjunctions as above and functors K, L satisfying the first diagram in the definition, the condition of the hom-set diagram is equivalent to* $\left\{ \begin{array}{l} L\eta = \eta' L \\ \epsilon' K = K\epsilon \end{array} \right.$.

pf.

- Given the hom-set diagram, set $a = Fx$ and chase the identity arrow $1 : Fx \to Fx$. Then we get the units η, η' and the equality $< L\eta : L \to LGF > = < \eta' L : L \to G'F'L >$ where $LGF = G'F'L$ by the first diagram in the definition.

- given the equality $L\eta = \eta' L$ of the natural transformations, the definition of the adjunctions ϕ, ϕ' by the their units give the hom-set diagram.

Given two adjunctions $\left\{ \begin{array}{c} < F, G, \phi, \eta, \epsilon > \\ < F', G', \phi', \eta', \epsilon' > \end{array} \right. : X \to A$, between the same two categories, two natural transformations $\left\{ \begin{array}{l} \sigma : F \to F' \\ \tau : G' \to G \end{array} \right.$ are said to be conjugate when

$$\begin{array}{ccc} A(F'x, a) & \xrightarrow{\sim} & X(x, G'a) \\ \downarrow{\scriptstyle (\sigma_x)^*} & & \downarrow{\scriptstyle (\tau_a)_*} \\ A(Fx, a) & \xrightarrow{\sim} & X(x, Ga) \end{array}$$

commutes for every pair of objects $x \in ObjX$, $a \in ObjA$

where $\left\{ \begin{array}{l} (\sigma_x)^* = A(\sigma_x, a) \\ X(x, \tau_a) = (\tau_a)_* \end{array} \right.$.

Theorem 1.4.4. *Given the two adjunctions* $\left\{ \begin{array}{c} < F, G, \phi, \eta, \epsilon > \\ < F', G', \phi', \eta', \epsilon' > \end{array} \right. : X \to A$, *the natural transformations* σ, τ *are conjugate iff any one of the four following diagrams commute;*

$$\begin{array}{ccc} G' & \xrightarrow{\tau} & G \\ \downarrow{\scriptstyle \eta G} & & \uparrow{\scriptstyle G\epsilon} \\ GFG' & \xrightarrow{G\sigma G} & GF'G' \end{array} \quad \begin{array}{ccc} F & \xrightarrow{\sigma} & F \\ \downarrow{\scriptstyle F\eta} & & \uparrow{\scriptstyle \epsilon F} \\ FG'G' & \xrightarrow{F\tau F} & FGF' \end{array} \quad \begin{array}{ccc} FG' & \xrightarrow{F\tau} & FG \\ \downarrow{\scriptstyle \sigma G} & & \downarrow{\scriptstyle \epsilon} \\ F'G' & \xrightarrow{\epsilon} & I_A \end{array} \quad \begin{array}{ccc} I'_X & \xrightarrow{\eta} & GF \\ \downarrow{\scriptstyle \eta} & & \downarrow{\scriptstyle G\sigma} \\ G'F' & \xrightarrow{\tau F} & GF' \end{array}$$

Also given the natural transformation $\sigma : F \to F'$*, there is a unique* $\tau : G' \to G$ *s.t.* $< \sigma, \tau >$ *is conjugate. Dually given* $\tau : G' \to G$*, there is a unique* σ *with* $< \sigma, \tau >$ *conjugate.*

pf.

- in the hom-set diagram, start with the identity arrow $1 : G'a \to G'a$ in the upper right and use the definition of ϕ, ϕ' by unit and counit to chase the element 1 around the diagram. Then we can get all the diagrams in the theorem.

- suppose τ is not given. Then by the Yoneda lemma to $\phi \circ (\sigma_x)^* \circ \phi'^{-1}$ there is a unique family of arrows τ'_a for which the hom-set diagram commutes and this family is a natural transformation. Since each $\epsilon_a : FGa \to a$ is universal from F to a, there is a unique family of arrows $\tau''_a : G'a \to Ga$ for which the third diagram commutes. Since the hom-set diagram implies the third diagram, $\tau'_a = \tau''_a$, i.e. if $\tau = \tau''$ makes the third diagram commute, it also make the hom-set diagram commute. Thus the third diagram implies the hom-set diagram. Given σ, there is immediately a unique natural transformation $\tau : G' \to G$ for which the first diagram commute. Since the hom-set diagram implies the first diagram, $\tau'_a = \tau_a$ and hence the solution τ'_a in the hom-set diagram is necessarily natural and the first diagram implies the hom-set diagram.

$\to$ a conjugate pair $<\sigma,\tau>$ of natural transformations is regarded as a morphism from the first to the second adjunction.

The vertical composite of two such $<F,G,\eta,\epsilon> \to_{<\sigma,\tau>} <F'G',\eta',\epsilon'> \to_{<\sigma',\tau'>} < F'',G'',\eta'',\epsilon''>$ is evidently a transformation $<\sigma',\tau'> \circ <\sigma,\tau> = <\sigma'\cdot\sigma, \tau\cdot\tau'>$ from the first to the third adjunction.
$\to$ for the two given categories X, A, thus we have a new category the category of adjunctions from X to A; $A^{(adj)X}$ with $\begin{cases} \text{objects} & \text{adjunctions } <F,G;\eta,\epsilon> \\ \text{arrows} & \text{transformations } <\sigma,\tau> \end{cases}$ with the composition defined above.

Theorem 1.4.5. *Given two adjunctions* $\begin{cases} <F,G,\eta,\epsilon>: X \to A \\ <\bar{F},\bar{G},\bar{\eta},\bar{\epsilon}>: A \to D \end{cases}$*, the composite functors yield an adjunction*

$$<\bar{F}F, G\bar{G}, G\bar{\eta}F\cdot\eta, \bar{\epsilon}\cdot\bar{F}\epsilon\bar{G}>: X \to D$$

pf. With hom-sets, the two given adjunctions yield a composite isomorphism, natural in $x \in ObjX,\ d \in ObjD \to D(\bar{F}Fx,d) \cong A(Fx,\bar{G}d) \cong X(x,G\bar{G}d)$. This makes the composite $\bar{F}F$ left adjoint to $G\bar{G}$. Setting $d = \bar{F}Fx$ and applying these two isomorphisms to the identity $1 : \bar{F}Fx \to \bar{F}Fx$, the unit of the composite adjunction is $x \xrightarrow{\eta_x} GFx \xrightarrow{G\bar{\eta}Fx} G\bar{G}\bar{F}Fx$ so is $G\bar{\eta}F\cdot\eta$ as asserted. By the dual argument, the counit is $\bar{\epsilon}\cdot\bar{F}\epsilon\bar{G}$.
$\to$ each hom-set $\mathbf{Adj}(X,A)$ may be regarded as a category with objects adjunctions and arrows the conjugate pairs $<\sigma,\tau>$.

Theorem 1.4.6. Adj *is a 2-dimensional category.*

1.5 Colimits and Limits

1.5.1 Colimits

For any category C, the diagonal functor is defined as $\begin{array}{rcl} \Delta : C & \to & C\times C \\ c & \mapsto & <c,c> \\ f & \mapsto & <f,f> \end{array}$ for all object c and arrow f.

Definition 1.5.1. *A universal arrow from an object* $<a,b>\in ObjC\times C$ *to the functor* Δ *is called a coproduct diagram. It consists of* $\begin{cases} \textit{object} & c\in ObjC \\ \textit{arrow} & <a,b>\to<c,c> \textit{ in } C \end{cases}$ *where* $\begin{cases} i:a\to c \\ j:b\to c \end{cases}$ *from* a,b *to* c.

$\to$it is universal in the sense that for any pair of arrows $\begin{cases} f:a\to d \\ g:b\to d \end{cases}$, there is a unique $h:c\to d$ with $\begin{cases} f=h\circ i \\ g=h\circ j \end{cases}$ i.e. $\exists!h$ s.t. make
$$\begin{tikzcd} a \arrow[r, "i"] \arrow[dr, "f"'] & a\coprod b \arrow[d, dotted, "h"] & b \arrow[l, "j"'] \arrow[dl, "g"] \\ & d & \end{tikzcd}$$
commute. Hence $<f,g>\mapsto h$ is a bijection $C(a,d)\times C(b,d)\cong C(a\coprod b,d)$ natural in d with inverse $h\mapsto<hi,hj>$.

More generally, we can define C^X for any discrete category X so the functor category C^X has as its objects the X-indexed families $a=\{a_x|x\in X\}$ of objects of C. The corresponding diagonal functor is defined as $\begin{array}{ccc} \Delta:C & \to & C^X \\ c & \mapsto & (c,c,c,\dots,c) \end{array}$. A universal arrow from a to Δ is an X-fold coproduct diagram with $\begin{cases} \text{object} & \coprod_x a_x\in C \\ \text{arrows} & i_x:a_x\to\coprod_x a_x \end{cases}$ with universality. Then $C(\coprod_x a_x,c)\cong\Pi_{x\in X}C(a_x,c)$ natural in c.

Definition 1.5.2. *Given in C, a pair* $f,g:a\to b$ *of arrows with the same domain and codomain, a coequaliser of* $<f,g>$ *is an arrow* $u:b\to e$ *s.t.*

1. $uf=ug$
2. *if* $h:b\to c$ *has* $hf=hg$, $\exists!\bar{h}:e\to c$ *s.t.* $h=hu'$

The diagram becomes
$$\begin{tikzcd} a \arrow[r, Rightarrow] & b \arrow[r, "u"] \arrow[dr, "h"'] & e \arrow[d, dotted, "\bar{h}"] & uf=ug \\ & & c & hf=hg \end{tikzcd}.$$

Form the functor category $C^{\downdownarrows}$ with an object a functor from $\cdot \rightrightarrows \cdot$ to C and an arrow a natural transformation between the corresponding functors. There is a diagonal functor $\Delta:C\to C^{\downdownarrows}$, defined on objects c and arrow $r:c\to c'$ of C as
$$\begin{tikzcd} c \arrow[r, Rightarrow] \arrow[d, "r"] & c \arrow[d, "r"] \\ c' \arrow[r, Rightarrow] & c' \end{tikzcd}.$$
Given the pair $<f,g>:a\to b$, an arrow $h:b\to c$ with $hf=hg$ is the same thing as an arrow $<hf=hg,h>:<f,g>\to<1_c,1_c>$ in the functor category

$C^{\downdownarrows}$, i.e. the arrow h which coequalises f and g are the arrows from $< f,g >$ to Δ. Thus a coequaliser $< e,u >$ of the pair $< f,g >$ is a universal arrow from $< f,g >$ to the functor Δ.

Let C,J be categories. The diagonal functor $\begin{array}{rcl} \Delta : C & \to & C^J \\ c & \mapsto & \Delta C \\ f & \mapsto & \Delta f \end{array}$, where $f : c \to c'$ is an arrow of C and $\Delta f : \Delta c \to \Delta c'$. Each functor $F : J \to C$ is an object of C^J.

Definition 1.5.3. *A universal arrow $< r,u >$ from F to Δ is called a colimit diagram for the functor F. It consists of an object $r \in ObjC$ and natural transformations $\tau : F \to \Delta c$.*

$\to$
$$\begin{tikzcd} F_1 \arrow[drrr, "\tau_1"] & F_2 \arrow[l] \arrow[r] \arrow[drr, "\tau_2"] & F_3 \arrow[r] \arrow[d, "\tau_3"] & F_4 \arrow[dl, "\tau_4"] \\ & & c & \end{tikzcd}$$
must commute. A natural transformation $\tau : F \to \Delta c$ is called a cone from the base F to the vertex c. Then a colimit of $F : J \to C$ consists of an object $Lim_{\to}F \in C$ and a cone $\mu : F \to \Delta(Lim_{\to}F)$ from the base F to the vertex $Lim_{\to}F$ which is universal.

e.g. Let $J = \omega = \{0 \to 1 \to 2 \to \ldots\}$, $F : \omega \to$ **Set** which maps arrow of ω to an inclusion. Then such functor F is a nested sequence of sets $F_0 \subset F_1 \subset \ldots$ and the union of all sets F_n with the cone given by the inclusion maps $F_n \to U$ is $Lim_{\to}F$, a colimit.

1.5.2 Limits

Definition 1.5.4. *Given categories C,J, and the diagonal functor $\Delta : C \to C^J$, a limit for a functor $F : J \to C$ is a universal arrow $< r,v >$ from Δ to F. It consists of an object $r = Lim_{\leftarrow}F \in ObjC$ and a natural transformation $v : \Delta r \to F$ which is universal among natural transformations $\tau : \Delta c \to F$.*

$\to$
$$\begin{tikzcd} & & c \arrow[dll, "\nu_1"'] \arrow[dl, "\nu_2"'] \arrow[d, "\nu_3"] \arrow[dr, "\nu_4"] & \\ F_1 \arrow[r] & F_2 \arrow[r] & F_3 \arrow[r] & F_4 \end{tikzcd}$$
must commute. $\nu : c \to F$ is called a cone to the base F from the vertex c. Then a limit of $F : J \to C$ consists of $Lim_{\leftarrow}F \in C$ and a cone $\nu : Lim_{\leftarrow}F \to F$ which is universal. It is the dual of the colimit.

The properties of the limit and colimit are summarised in the diagram

$$\begin{tikzcd} Lim_{\leftarrow}F \arrow[r, "\nu"] & F \arrow[r, "\mu"] \arrow[d, "\equiv"] & Lim_{\rightarrow}F \arrow[d, dotted] \\ c \arrow[u, dotted] \arrow[r, "\tau"] & F \arrow[r, "\sigma"] & c \end{tikzcd}$$

where the horizontal arrows are cones, the vertical arrows are arrows in C. When such limits exist, there are natural isomorphisms $\begin{cases} C(c,Lim_{\leftarrow}F) \cong Nat(\Delta c,F) = Cone(c,F) \\ Cone(F,c) = Nat(F,\Delta c) \cong C(Lim_{\rightarrow}F,c) \end{cases}$.

Definition 1.5.5. *A category C is called complete if all small diagrams in C have limits in C; if every functor $F : J \to C$ has a limit.*

Consider a functor $F : \omega^{op} \to$ **Set**. It is a list of sets F_n and of functions f_n, the first row of the diagram

$$\begin{tikzcd} F_0 & F_1 \arrow[l, "f_0"'] & F_2 \arrow[l, "f_1"'] & \dots \arrow[l] & F_n \arrow[l] & F_{n+1} \arrow[l, "f_n"'] & \dots \arrow[l] \\ \Pi_i F_i \arrow[u] \arrow[ur] \arrow[urr] & & & & Lim_{\leftarrow}F \arrow[llll, "incl"] \arrow[ulll] \arrow[ull] \arrow[u] \arrow[ur] & & \end{tikzcd}$$

. Given F, form the product set ΠF_i which consists of all strings $x = \{x_0, x_1, \dots\}$ of elements with $\forall x_n \in F_n$ and it has projections $p_n : \Pi F_i \to F_n$. A limit must be at least a vertex of a set of commuting triangles (cones). So take the subset L of those strings x who match under f, in the sense that $f_n x_{n+1} = x_n$, $\forall n$. Then for $\begin{array}{rcl} \mu_n : L & \to & F_n \\ x & \mapsto & x_n \end{array}$, $f_n \mu_{n+1} = \mu_n$ for all n, so $\mu : L \to F$ is a cone from the vertex $L \in$ **Set** to the base F. If $\tau : M \to F$ is any other cone from a set M, each $m \in M$ determines a string $\{\tau_n m\}$ which matches and hence a function $g : M \to L$ with $gm = \{\tau_n m\}$ so with $\mu g = \tau$. Since g is unique, μ is a universal cone to F and $L = Lim_{\leftarrow}F$.

Theorem 1.5.1. *If the category J is small, any functor $F : J \to$ **Set** has a limit which is the set $Cone(*,F)$ of all cones $\sigma : * \to F$ from the one point set $*$ to F, while the limiting cone ν with $\begin{array}{rcl} \nu_j : Cone(*,F) & \to & F_j \\ \sigma & \mapsto & \sigma_j \end{array}$ is for each j that function sending each cone σ to the element $\sigma_j \in F_j$.*

pf. Since J is small, $Cone(*,F)$ is a small set, hence an object of **Set**. If $u : j \to k$ is any arrow of J, then $F_u \sigma_j = \sigma_k$, given σ a cone. Thus ν defined in the theorem is a cone to the base F. Consider any other cone $\tau : X \to F$ to F. Then $\forall x \in X$, τx is a cone to F from one point, so there is a unique function $h : X \to Cone(*,F)$ sending each x to τx ($Cone(X,F) \cong$ **Set**$(X, Cone(*,F)) \Leftrightarrow Nat(\Delta X,F) \cong$ **Set**$(X, Cone(*,F)) =$ **Set**$(X, Lim_{\leftarrow}F)$).

Definition 1.5.6. *A functor $H : C \to D$ is said to preserve the limits of functors $F : J \to C$ when every limiting cone $\nu : b \to F$ in C for a functor F yields by composition* *with H a limiting cone $H\nu : Hb \to HF$ in D. A functor is called continuous when it preserves all small limits.*

Theorem 1.5.2. *For any category C with small hom-sets, each hom-functor $C(c,-)$: $C \to$ **Set** preserves all limits.*

pf. Let J be any category and $F : J \to C$ a functor with a limiting cone ν : $LimF \to F$ in C. By applying the hom-functor $C(c,-)$ to this, we get a cone $\nu_* =$ $C(c,\nu)$,
$$\begin{tikzcd} C(c,LimF) \arrow[r, "v_{*i}"] & C(c,F_i) \arrow[d, "\equiv"] \\ X \arrow[u, dotted, "k"] \arrow[r, "\tau_i"] & C(c,F_i) \end{tikzcd}$$
in **Set**. For any other cone τ to the same base from a vertex set X, $\forall x \in X$ we can get a cone $\tau_i x : c \to F_i$ in C. Because ν is universal, there is a unique arrow $h_x : c \to LimF$ with $v_i h_x = \tau_i x$. Then setting $kx = h_x$ for all x defines a function, and therefore an arrow k with $v_{*i}k = \tau_i$. Since k is unique, ν_* is a limiting cone.
$\to \lambda : c \to F$ in C is the same thing as a cone $\lambda : * \to C(c,F-)$ in **Set** by the definition of the functor $C(c,F-) : J \to$ **Set**. Then given $Cone(X,-) \cong \mathbf{Set}(X, Cone(*,-))$,

$$Cone(X,C(c,F-)) \cong \mathbf{Set}(X,Cone(*,C(c,F-))) = \mathbf{Set}(X,Cone(c,F)) \cong \mathbf{Set}(X,C(c,LimF))$$

Here $LimS$ for each $S : J \to$ **Set** is defined by the adjuction $Cone(X,S) \cong \mathbf{Set}(X,LimS)$. Therefore the above equations determine $LImS$ as $LimC(c,F-) \cong C(c,LimF)$.
$\Leftrightarrow$ the contravariant hom-functor may be written as $C(-,c) = C^{op}(c,-) : C^{op} \to$ **Set** and therefore by the previous theorem, the functor $C(-,c)$ carries all small colimits in C to the corresponding limits and limiting cones in **Set**. More generally, the colimit of any $F : J \to C$ is determined by

$$C(ColimF,c) \cong LimC(F-,c)$$

Theorem 1.5.3. *If $V : A \to X$ creates limits for $F : J \to A$ and the composite VF : $J \to X$ has a limit, then V preserves the limit of F.*

pf. Let $\begin{cases} \tau : a \to F \\ \sigma : x \to VF \end{cases}$ be limiting ones in $\begin{cases} A \\ X \end{cases}$. Since V creates limits, $\exists! \rho$: $b \to F$ in A with $V\rho : Vb \to VF$ equal to $\sigma : x \to VF$. Moreover ρ is a limiting cone. But limits are unique up to isomorphism, so there is an isomorphism $\theta : b \cong a$ with $\tau\theta = \rho$. Thus $V\theta : Vb = x \cong Va$ with $V\tau \circ V\theta = V\rho = \sigma$, so Va is a limit and V preserves limits as desired.

Definition 1.5.7. *In any category, an object p is called projective if every arrow h : $p \to c$ from p factors through every epi $g : b \to c$ as $h = g\bar{h}$ for some $\bar{h}$;*
$$\begin{tikzcd} & p \arrow[dl, dotted, "\bar{h}"'] \arrow[d, "h"] \\ b \arrow[r, "g"] & c \end{tikzcd}$$

$\Leftrightarrow$ g epi implies $\hom(p,g):\hom(p,b)\to\hom(p,c)$ epi in **Set**, i.e. p is projective exactly when $\hom(p,-)$ preserves epis. Dually, an object q is injective when $\hom(-,q)$ carries monics to epis.

Theorem 1.5.4. *If the functor $G : A \to X$ has a left adjoint, while the functor $T : J \to A$ has a limiting cone $\tau : a \to T$ in A, then GT has the limiting cone $G\tau : Ga \to GT$ in X.*

pf. By composition $G\tau$ is indeed a cone from the vertex Ga in X. If F is a left adjoint to G and if we apply the adjunction isomorphism to every arrow of a cone $\sigma : x \to GT$, we get arrows $(\sigma_i)^b : Fx \to Ti$ for all $i \in J$ which form a cone $\sigma^b : Fx \to T$ in A. But $\tau : a \to T$ is universal among cones to T in A, so there is a unique arrow $h : Fx \to a$ with $\tau h = \sigma^b$.
$$\begin{tikzcd} Ti \arrow[d, "Tu"'] & a \arrow[l, "\tau_i"'] \arrow[dl, "\tau_j"] \\ Tj & Fx \arrow[l, "\sigma_j^b"] \arrow[u, dotted, "h"'] \arrow[ul, "\sigma_i^b"'] \end{tikzcd}$$
in A. Taking adjuncts again, we get a unique arrow $h^\# : x \to Ga$ with $G\tau \cdot h^\# = (\tau h)^\# = (\sigma^b)^\# = \sigma$. The uniqueness of the arrow $h^\#$ gives $G\tau : Ga \to T$ universal.
$$\begin{tikzcd} GTi \arrow[d, "GTu"'] & Ga \arrow[l, "G\tau_i"'] \arrow[dl, "G\tau_j"] \\ GT_j & x \arrow[l, "\sigma_j"] \arrow[u, dotted, "h^\#"'] \arrow[ul, "\sigma_i"'] \end{tikzcd}$$
in X.

$\to$ Given an adjunction $<F,G,\eta,\epsilon>: X \to A$ and any index category J, one may form the functor category $<F^J,G^J,\eta^J,\epsilon^J>: X^J \to A^J$ where $\begin{cases} F^J(S) = FS \\ \eta^J S = \eta S : S \to GFS \end{cases}$ for each functor $S : J \to X$. The we get the diagram of adjoint pairs
$$\begin{tikzcd} X^J \arrow[r, shift left, "F^J"] \arrow[d, shift left, "Lim_{\leftarrow}"] & A^J \arrow[l, shift left, "G^J"] \arrow[d, shift left, "Lim_{\leftarrow}"] \\ X \arrow[u, shift left, "\Delta"] \arrow[r, shift right, "F"'] & A \arrow[u, shift left, "\Delta"] \arrow[l, shift right, "G"'] \end{tikzcd}$$
By the definition of the diagonal functors Δ, $F^J\Delta = \Delta F$, and therefore the diagram of left adjoints commutes. Since the right adjoint of a given functor is unique up to natural isomorphism, $Lim \circ G^J \cong G \circ Lim$. Thus for each functor $T : J \to A$ with limit a, $Ga = GLimT = LimG^J(T) = LimGT$.

$\Leftrightarrow$ the dual is that any functor P which has a right adjoint must preserve colimits.

Theorem 1.5.5. *Let D be a small-complete category with small hom-sets. Then D has an initial object iff it satisfies the following criterion:*
$\exists I$, a small set, and an I-indexed family k_i of objects of D s.t. $\forall d \in ObjD$, $\exists i \in I$ and $\exists$ an arrow $k_i \to d$ of D.

pf.

- if D has an initial object k, then k indexed by the one-point set realises the condition, since there is always a unique arrow $k \to d$.

-
  - Existence
    Assume the set met with the above criterion exists. Since D is small-complete, it contains a product object $w = \Pi k_i$ of the given I-indexed family. $\forall d \in ObjD$, there is at least one arrow $w \to d$, for exampel $w = \Pi k_i \to k_i \to d$, where the first arrow is a projection. Then by hypothesis, the set of endomorphisms $D(w,w,)$ of w is small and D is complete, so we can construct the equaliser $e : v \to w$ of the set of all endomorphisms of w. $\forall d \in ObjD$, $\exists v \to w \to d$ at least one arrow $v \to d$.
  - Uniqueness
    Suppose there were two $f,g : v \to d$. Take their equaliser e_1 so that
    $$\begin{tikzcd} u \arrow[r, "e_1"] & v \arrow[r, shift left, "g"] \arrow[r, shift right, "f"'] \arrow[d, "e"] & d \\ w \arrow[u, dotted, "s"] \arrow[r, "ee_1s"] & \Pi k_i \arrow[r] & k_i \arrow[u] \end{tikzcd}$$
    commutes. By the construction of w, $\exists s : w \to u$ so the composite ee_1s is like 1_w, an endomorphism of w. But e was defined as the equaliser of all endomorphisms of w so $ee_1se = 1_we = e1_v$. Given e an equaliser, it is monic, so by cancelling e on the left givens $e_1se = 1_v$. This implies e_1 has a right inverse. Like any equaliser e_1 is monic, hence is an isomorphism. Thus $f = g$.

Lemma 1.5.1. *If $G : A \to X$ preserves all small products / equalisers, then $\forall x \in X$, the projection* $\begin{array}{ccc} Q : (x \downarrow G) & \to & A \\ (x \mapsto_s Ga) & \mapsto & a \end{array}$ *of the comma category creates all small products / equalisers.*

pf.

- products
  Let J be a set and $f_j : x \to Ga_j$ a J-indexed family of objects of $(x \downarrow G)$ s.t. the product diagram $p_j : \Pi a_j \to a_j$ exists in A. Since G preserves products, $Gp_j : G\Pi a_j \to Ga_j$ is a product diagram in X, so there is a unique arrow $f : x \to G\Pi a_j$ in X with $(Gp_j)f = f_j$;
  $$\begin{tikzcd} & G\Pi a_j \arrow[d, "Gp_j"] \\ a \arrow[ur, dotted, "f"] \arrow[r, "f_j"] & Ga_j \end{tikzcd}$$
  . This states $p_j : f \to f_j$ is a cone of arrows $(x \downarrow G)$. This is the unique cone there which projects under Q to the given cone $p_j : \Pi a_j \to a_j$. Given this gone a product diagram in $(x \downarrow G)$, Q creates products.

- equalisers
  let $s,t : f \to g$ be two arrows in $(x \downarrow G)$. Then we have the equaliser e of Qs, Qt;
  $$\begin{tikzcd} a & d \arrow[l, dotted, "\bar{r}"] \arrow[dl, "r"] \\ b \arrow[u, leftarrow, "e"] \arrow[d, shift right, "s"'] \arrow[d, shift left, "t"] & \\ c & \end{tikzcd}$$
  . Since G preserves equalisers, Ge is then the equaliser of Gs and Gt. But $Gs \circ f = g = Gt \circ f$, so there is a unique arrow $h : x \to Ga$ making $Ge \circ h = f$;
  $$\begin{tikzcd} x \arrow[r, dotted, "h"] \arrow[d, "\equiv"] & Ga \arrow[d, "Ge"] & Gd \arrow[l, dotted, "G\bar{r}"'] \arrow[dl, "Gr"] \\ x \arrow[r, "f"] \arrow[d, "\equiv"] & Gb \arrow[d, shift right, "Gs"'] \arrow[d, shift left, "Gt"] & x \arrow[u, "k"] \\ x \arrow[r, "g"] & Gc & \end{tikzcd}$$
  . Thus $e : h \to f$ in $(x \downarrow G)$ is the unique arrow of $(x \downarrow G)$ with Q-projection $e : a \to b$.
  Consider $k : x \to Gd \in Obj(x \downarrow G)$ and an arrow $r : k \to f$ of $(x \downarrow G)$ with $sr = tr$. Then $sr = tr$ in A, so $\exists! r'$ in A with $r = er'$. Since $Ge(Gr' \circ k) = G(er') \circ k = Gr \circ k = f$, by the uniqueness of h, $Gr' \circ k = h$, which implies r' is an arrow of $(x \downarrow G)$ and e is an equaliser in $(x \downarrow G)$.

Theorem 1.5.6 (The Freyd Adjoint Functor Theorem). *Given a small complete category A with small hom-sets, a functor $G : A \to X$ has a left adjoint iff it preserves all small limits and satisfies the following criterion:*
$\forall x \in ObjX$, $\exists I$, a small set and an I-indexed family of arrows $f_i : x \to Ga_i$ s.t. every arrow $h : x \to Ga$ can be written as a composite $h = Gt \circ f_i$ for some index i and some $t : a_i \to a$.

pf.

- if G has a left adjoint F, it must preserves all the limits which exist in A. Moreover the universal arrow $\eta_x : x \to GFx$ which is the unit of the adjunction satisfies the criterion above for x with I the one-point set.

- Given the set I which met the criterion above, we can get the same name for $(x \downarrow I) = D$. Since A has small hom-sets, so does D. Then by the previous lemma, D is small-complete. Then D has initial object, which is an universal arrow. Then G has a left adjoint by the pointwise construction of adjoints.

e.g. $U :$ **Grp** $\to$ **Set** is the forgetful functor. Then U creates all limits, hence the **Grp** is small-complete and U is continuous. Thus by the previous theorem, it has a left-adjoint. Consider any function $f : X \to UG$ for $G \in Obj$**Grp** and take the subgroup $S \subset G$ generated by all elements fx, $\forall x \in X$. Every element of S is then a finite product, say $(fx_1)^{\pm 1} \ldots (fx_n)^{\pm 1}$ of these generators and their inverses, so the cardinal number of S is bounded, given X. Taking one copy of each isomorphism class of such groups S then gives a small set of groups, and the set of all functions $X \to US$ is then the set in the previous theorem. This left adjoint $F : \mathbf{Set} \to \mathbf{Grp}$ assigns to each set X the free group FX generated by X. For any two elements $x \neq y \in X$, take a function $f : X \to UH$ with $\begin{cases} fx = h \\ fy = k \end{cases}$, , $h \neq k$. Then since f must factor through the universal $X \to UFX$, the universal must be an injection.

1.5.3 Topology Category

Top is the category with $\begin{cases} \text{objects} & \text{topological spaces } X, Y, \ldots \\ \text{arrows} & \text{continuous maps } f : X \to Y \end{cases}$. The standard forgetful functor $\begin{array}{rcl} G : \mathbf{Top} & \to & \mathbf{Set} \\ X & \mapsto & GX \end{array}$ where GX is the set of points in X is faithful and has a left adjoint D which assigns to each set S the discrete topology on S. This implies G preserves all limits which may exist in **Top**. It also has a right adjoint D', which assigns to each set S the indiscrete topology on S, and therefore it preserves all colimits which may exist in **Top**.

<Construction>

Consider the subspace topology on a set $S \subset GX$. If X is a fixed topological space, G induces a functor

$$\begin{array}{ccc} G \downarrow X : (\mathbf{Top} \downarrow X) & \to & (\mathbf{Set} \downarrow GX) \\ \begin{array}{ccc} Y & \xrightarrow{f} & X \\ \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle \equiv} \\ Y' & \xrightarrow{f'} & X \end{array} & \mapsto & \begin{array}{ccc} GY & \xrightarrow{Gf} & GX \\ \downarrow{\scriptstyle Gh} & & \downarrow{\scriptstyle \equiv} \\ GY' & \xrightarrow{Gf'} & GX \end{array} \end{array}$$

where f, f' are objects and h an arrow of the comma category $(\mathbf{Top} \downarrow X)$. This has a right adjoint L. Indeed, an object $t : S \to GX$ in $(\mathbf{Set} \downarrow GX)$ is a set S and a function t on S to GX. Put on S the topology with open sets all $t^{-1}U$ for U open set in X, and call the resulting space LS. Then t is a continuous map $Lt : LS \to X$. This topology on LS has the familiar universal property; any continuous map $f : Y \to X$ which factors through t as $Gf = t \circ s$,

$$\begin{tikzcd} GY \arrow[r, "Gf"] \arrow[d, "s"'] & GX \\ S \arrow[ur, "t"'] & \end{tikzcd}$$

has $s : Y \to LS$ continuous in **Set** $\Leftrightarrow hom(Gf, t) \cong \hom(f, Lt)$. Since $(G \downarrow X) \circ L = Id$, L is a right adjoint right inverse to $(G \downarrow X)$.

Proposition 1.5.1. *If $G : C \to D$ is a faithful functor, if D has equalisers, and if $\forall x \in ObjC$, $(G \downarrow x) : (C \downarrow x) \to (D \downarrow Gx)$ has a right adjoint right inverse L, then C has equalisers.*

pf. For any parallel pair $f, f' : x \to y$, apply G, and take equaliser $t : s \to Gx$ of Gf, Gf' in D. Then we can apply L. The universality of the adjunction shows $Lt : Ls \to x$ is an equaliser in C.

Top is complete.
pf. The products of any family X_i, $i \in J$ of spaces is constructed by taking the product ΠGX_i of the underlying sets and putting on it the universal topology in which all projections $p_i : \Pi GX_i \to GX_i$, $i \in J$ are continuous. Then there is a universal topology with exactly those open sets on S required to make all t_i continuous.

For any space X, the functor $(X \downarrow G) : (X \downarrow \mathbf{Top}) \to (GX \downarrow \mathbf{Set})$ has a left adjoint M. Indeed an object of $(GX \downarrow \mathbf{Set})$ is a function $t : GX \to S$. Put on S the topology with open sets all subsets $V \subset S$ with $t^{-1}V$ open in S and call the resulting space MS. Then the function t is a continuous map $Mt : X \to MS$. Moreover, $f : X \to Y$ continuous and $Gf = k \circ t$ for some function k.

$$\begin{tikzcd} & X \arrow[dl, "Mt"'] \arrow[dr, "f"] & \\ MS \arrow[rr, dotted, "k"] & & Y \end{tikzcd} \qquad \begin{tikzcd} & GX \arrow[dl, "t"'] \arrow[dr, "Gf"] & \\ S \arrow[rr, dotted, "k"] & & GY \end{tikzcd}$$

. Thus $k : MS \to Y$ is continuous and $k \mapsto k$ is an adjunction $(X \downarrow \mathbf{Top})(Mt, f) \cong (GX \downarrow \mathbf{Set})(t, Gf)$ with unit the identity map, so M is left adjoint right inverse to $X \downarrow G$. By the previous proposition and the duality principle, **Top** has coequalisers.

<Construction>

Let $\{U_i : i \in J\}$ be an open cover of a space X. Each continuous $f : X \to Y$ determines a J-indexed family of restrictions $f|_{U_i} : U_i \to Y$. Conversely, a J-indexed family of continuous maps $f_i : U_i \to Y$ determines a map f continuous on all of X iff $f_i|_{(U_i \cap U_j)} = f_j|_{(U_i \cap U_j)}$, $\forall i, j \in J$. This can be written as $\mathbf{Top}(X, Y) \to \Pi_i(\mathbf{Top}(U_i, Y) \rightrightarrows \Pi_{i,j}\mathbf{Top}(U_i \cap U_j, Y)$ where the arrows are given by restrictions. Thus X is the colimit in **Top** with colimiting cone the inclusion maps $U_i \to X$, of the functor $U : J' \to \mathbf{Top}$ where J' is the category with objects the pairs of indices $< i, j >$, and the non-identity maps $< i, j > \to < i >, < i, j > \to < j >$, while U is the functor with $U < i, j > = U_i \cap U_j, U < i > = U_i$.

e.g. X/A is obtained from the space X by collapsing the subset A to a point. Then $* \overset{a}{\underset{a'}{\rightrightarrows}} X \longrightarrow X/A$ X/A is the coequaliser of the set of all the arrows sending the one point space $*$ to one of the point $a \in A$.

$\to \mathbf{Top}^{(2)}$ consists of $\begin{cases} \text{objects} & < X, A >,\ A \subseteq X \\ \text{arrows} & \text{continuous maps } X \to X' \text{ sending } A \to A' \end{cases}$. Then

$\mathbf{Top}_*(X/A, Y) = \mathbf{Top}^{(2)}(<X, A>, <Y, *>)$, i.e. $<X, Y> \mapsto X/A$ is left adjoint to the functor $Y \mapsto <Y, *>$, which sends each pointed space to the pair $<Y, *>$.

Proposition 1.5.2. **Haus**, *the full subcategory of all Hausdorff spaces in* **Top** *is complete and co-complete. The inclusion functor* **Haus** $\to$ **Top** *has a left adjoint* H*, as does the forgetful functor* **Haus** $\to$ **Set**.

pf.

- Any product of Hausdorff spaces or subspaces of a Hausdorff space is also Hausdorff, hence **Haus** is complete and the inclusion functor is continuous. Thus it is complete by the previous theorem. Any continuous map of X to a Hausdorff space Y factors through the image, a subspace of Y, hence Hausdorff. This image is a quotient set of X with some topology, so there is at most a small set surjection $X \to Y$ to a Hausdorff Y. This is the solution set condition. The left adjoint Hassigns to each space X a Hausdorff space HX and a continuous map $\eta : X \to HX$, universal from X to a Hausdorff space. Then η is a surjection, so HX may be described as the largest Hausdorff quotient of X. If X is Hausdorff, $HX = X$, $\eta = 1$ so H is a left adjoint left inverse to the inclusion.

- Since H is a left adjoint, it preserves colimits. Thus **Haus** is cocomplete. If particular, the coproduct in **Haus** is the coproduct in **Top** while a coequaliser in **Haus** is the largest Hausdorff quotient of the coequalsier in **Top**.

Definition 1.5.8. *An effective epimorphism in a category C is a map $X \to Y$ in C s.t. the pullback $X \times_Y X$ exists and the map $X \to Y$ is the coequaliser of the two projection maps $X \times_Y X \to X$.*

Lemma 1.5.2. *A map in* **Top** *is a quotient map iff it is an effective epimorphism.*

Definition 1.5.9. *Let X be a space. A subspace $F \subseteq X$ is said to be compactly closed, or k-closed if for any map $k : K \to X$ from a compact Hausdorff space K, the preimage $k^{-1}(F) \subseteq K$ is closed.*

Definition 1.5.10. *A topological space X is a k-space or is compactly generated if every compactly closed set is closed.*

$\Leftrightarrow$ X is compactly generated iff a map $X \to Y$ is continuous precisely when for every compact Hausdorff space K and map $k : K \to X$ the composite $K \to X \to Y$ is continuous.

Let k**Top** be the category of k-spaces, which is a full subcategory of **Top** with

the inclusion functor $j : k\mathbf{Top} \to \mathbf{Top}$. It has adjoint $k : \mathbf{Top} \to k\mathbf{Top}$ which k-ify a topological space; endow the underlying set of a space X with a new topology, one for which the closed sets are precisely the sets that are compactly closed w.r.t. the original topology. Then the resulting space, denoted kX is a k-space. This has adjunction relation $k\mathbf{Top}(X, kY) = \mathbf{Top}(jX, Y)$ with the unit $\eta : X \to kjX$ is a homeomorphism. Since it preserves limits, for any functor $X : J \to k\mathbf{Top}$, $\lim^{k\mathbf{Top}} X \xrightarrow{\equiv} \lim^{k\mathbf{Top}} kjX \xleftarrow[\equiv]{} k\lim^{\mathbf{Top}} jX$. Then this $k\mathbf{Top}$. In particular, the product in $k\mathbf{Top}$ is formed by k-ifying the product in $\mathbf{Top}$. Similarly colimit of any diagram of k-spaces give
$colim^{k\mathbf{Top}} X \xrightarrow{\equiv} kjcolim^{k\mathbf{Top}} X \xleftarrow[\equiv]{} kcolim^{\mathbf{Top}} jX$.

Proposition 1.5.3. *The category $k\mathbf{Top}$ is Cartesian closed.*

Proposition 1.5.4. *1. any locally compact Hausdorff space is compactly generated.*

2. *quotient spaces and closed subspaces of compactly generated spaces are compactly generated.*

3. *if X is a locally compact Hausdorff space and Y is compactly generated, then $X \times Y$ is again compactly generated.*

4. *the colimit of any diagram of compactly generated spaces is compactly generated.*

Definition 1.5.11. *A pointed space is a space X together with a specified point in it, to be called the basepoint.*

$\to k\mathbf{Top}_*$ is the category of pointed k-spaces.

The one-point space $*$ is both the initial and the terminal object in this category and therefore the category $k\mathbf{Top}_*$ is not Cartesian closed.

Define Y_*^X to be the subspace of Y^X consisting of the pointed maps. Given $\{*\}$ closed in Y, Y_*^X is closed in Y^X and therefore it is a k-space. For fixed $X \in k\mathbf{Top}_*$, the functor $Y \mapsto Y_*^X$ has a left adjoint (it is a closed symmetric monoidal category).
$\to \{f : \begin{matrix} f(w,*) = * & \forall w \in W \\ f(*,x) = * & \forall x \in X \end{matrix}\} = k\mathbf{Top}_*(W, Y_*^X) \subseteq k\mathbf{Top}(W, Y_*^X) = \{f : f(w,*) = *, \ \forall w \in W\} \subseteq k\mathbf{Top}(W, Y^X) = \{f : W \times X \to Y\}$. So the map $W \times X \to Y$ corresponding to $f : W \to Y_*^X$ sends the wedge $W \vee X \subseteq X \times W$ to the basepoint of Y, hence factors through the smash product $W \wedge X = \frac{W \times X}{W \vee X}$ obtained by pinching the axes in the product of the point. Thus $- \wedge X : k\mathbf{Top}_* \rightleftarrows k\mathbf{Top}_* : (-)_*^X$.

Operations over Topological Spaces

- Cylinder, Cone and Suspension

Denote the segment $[0,1]$ as I. The product $ZX = X \times I$ is called the cylinder over X; the subsets $\begin{cases} X \times 0 \\ X \times 1 \end{cases}$ of the cylinder are called its bases. Smashing the upper base of the cylinder into one point gives the cone CX of X; thus $CX = (X \times I)/(X \times 1)$. The base of the cylinder not affected by the factorisation is called the base of the cone, and the point of CX is called the vertex of the cone. If further factorise the cone over its base, we get the suspension ΣX over X. The suspension has two vertices. The image of the middle section $(X \times 1/2) \subset X \times I$ is called the base of the suspension. Smashing with S^1 is known as reduced suspension $\Sigma X = S^1 \wedge X = \frac{I \times X}{(\partial I \times X) \cup (I \times *)}$. Then by induction, the n-fold suspension is $\Sigma^n X = S^1 \wedge \Sigma^{n-1} X = S^1 \wedge (S^{n-1} \wedge X) = (S^1 \wedge S^{n-1}) \wedge X = S^n \wedge X$.

- Attachings

Let X, Y be topological spaces, let A be a subspace of Y, and let $\phi : A \to X$ be a continuous map. Take the sum $X \coprod Y$ and make an identification; attach $\forall a \in A \subset Y$ to $\phi(a) \in X$. The resulting space is denoted as $X \cup_\phi Y$ and the procedure for this is called "attaching Y to X by means of the map ϕ. Let $f : X \to Y$ be an arbitrary continuous map. Then space obtained by attaching the cylinder $X \times I$ to Y by means of the map $X \times 0 = X \xrightarrow{f} Y$ is called the cylinder of the map f, denoted $Cyl(f)$. The space obtained by attaching the cone CX to Y by means of the same map is called the cone of the map f, denoted $Con(f)$.

- Mapping Spaces

The set $C(X,Y)$ of all continuous maps of a space X into a space Y is furnished by compact open topology. The base of open sets of this topology consists of sets of the form $U(K,O)$ where K is a compact subset of X and O is an open subset of Y. The set $U(K,O)$ consists of continuous maps $f : X \to Y$ s.t. $f(K) \subset O$. Let X, Y, Z be topological spaces. The formula $\{f : C(X,Y)\} \mapsto \{(x,y) \mapsto [f(x)](y)\}$ defines a map.

Definition 1.5.12. *The loop space of a pointed space X is defined as*

$$\Omega X = X_*^{S^1}$$

$\to \Omega^n X = \Omega(\Omega^{n-1}X) = (X_*^{S^{n-1}})_*^{S^1} = X_*^{S^{n-1}\wedge S^1} = X_*^{S^n}$ so by induction, $\Omega^n X = X_*^{S^n}$.

Consider a pair (X,A), (Y,B). Then $(X/A)\wedge(Y/B) = \frac{X\times Y}{(A\times Y)\cup_{A\times B}(X\times B)}$. So if we think of $I^m/\partial I^m \sim S^n$ as a pointed space, $S^m \wedge S^n = (I^m/\partial I^m)\wedge(I^n/\partial I^n) = \frac{I^{m+n}}{(\partial I^m\times I^n)\cup(I^m\times\partial I^n)} = I^{m+n}/\partial I^{m+n} = S^{m+n}$. Smashing with S^1 gives a suspension $\Sigma X = S^1\wedge X = \frac{I\times X}{(\partial I\times X)\cup(I\times *)}$, i.e. the suspension is obtained from the cylinder by collapsing the top and the bottom to a point, as well as the line segment along a basepoint.

Ho**Top** is called the homotopy category, consisting of $\begin{cases} \text{objects} & k\text{-spaces} \\ \text{arrows} & [X,Y] \text{ a set of morphisms} \end{cases}$. If we have basepoints, a pointed homotopy between pointed maps is a function $h : I\times X \to Y$ s.t. $h(t,-)$ is pointed for all t, i.e. $\frac{I\times X}{I\times *} = I_+\wedge X$. Pointed homotopy is an equivalence relation, and we have the pointed homotopy category, denoted $Ho\mathbf{Top}_*$ with the set of morphisms $[X,Y]_*$.

Definition 1.5.13. *Let $(X,*)$ be a pointed space and n positive integer. The nth homotopy group of X is $\pi_n(X) = [S^n, X]_*$.*

Then for any k with $0\le k\le n$, $\pi_n(X,*) = [S^n,X]_* = [S^0,\Omega^n X]_* = [S^k,\Omega^{n-k}X]_* = \pi_k(\Omega^{n-k}X)$.

Chapter 2

Homotopy Theory

2.1 General Structure

2.1.1 Homotopy Groups

Definition 2.1.1. *Let X,Y be topological spaces. Continuous maps $f,g : X \to Y$ are called homotopic if there exists a family of maps $h_t : X \to Y$, $t \in I$ s.t.* $\begin{cases} h_0 = f, h_1 = g \\ H : X \times I \to Y,\ H(x,t) = h_t(x) \text{ continuous} \end{cases}$.

Definition 2.1.2. *The equivalence classes for the homotopy relation in $C(X,Y)$ are called homotopy classes. The set of homotopy classes in $C(X,Y)$ is denoted as $\pi(X,Y)$.*

$\to$ this can be regarded as the set of path components of $C(X,Y)$.

Let X,X',Y,Y' be topological spaces and $\begin{cases} \phi : X' \to X \\ \psi : Y \to Y' \end{cases}$ be continuous maps. Then for continuous maps $f,g : X \to Y$, $\begin{cases} f \sim g \Rightarrow f \circ \phi \sim g \circ \phi \\ f \sim g \Rightarrow \psi \circ f \sim \psi \circ g \end{cases}$ $\to \circ\phi$, $\psi\circ$ can be applied to the homotopy classes of maps $X \to Y$, which gives the map $\begin{cases} \phi^* : \pi(X,Y) \to \pi(X',Y) \\ \psi_* : \pi(X,Y) \to \pi(X,Y') \end{cases}$

$\to$ Let $\begin{cases} \phi_1 : X' \to X \\ \phi_2 : X'' \to X' \end{cases}$, $\begin{cases} \psi_1 : Y \to Y' \\ \psi_2 : Y' \to Y'' \end{cases}$ be the continuous maps. Then for continuous maps $f,g : X \to Y, f \sim g \Rightarrow f\circ(\phi_1\circ\phi_2) = (f\circ\phi_1)\circ\phi_2 \sim (g\circ\phi_1)\circ\phi_2 = g\circ(\phi_1\circ\phi_2)$ and similarly for ψ. This implies $(\phi_1\circ\phi_2)^* = \phi_2^*\circ\phi_1^*$ and similarly $(\psi_1\circ\psi_2)_* = \psi_{1*}\circ\psi_{2*}$.

Definition 2.1.3. *The spaces X,Y are called homotopy equivalent if there exists a continuous map* $\begin{cases} f : X \to Y \\ g : Y \to X \end{cases}$ *s.t. the composition* $\begin{cases} g\circ f : X \to X \\ f\circ g : Y \to Y \end{cases}$ *are homotopic* *to the identity map* $\begin{cases} Id_X : X \to X \\ Id_Y : Y \to Y \end{cases}$.

Definition 2.1.4. *$X \sim Y$ if there exists a way to define $\forall Z$, a space, a bijective map $\alpha^Z : \pi(Y,Z) \to \pi(X,Z)$ s.t. for any continuous map $\psi : Z \to W$,*
$$\begin{array}{ccc} \pi(X,Z) & \xleftarrow[\alpha^Z]{} & \pi(Y,Z) \\ \downarrow{\psi_*} & & \downarrow{\psi_*} \\ \pi(X,W) & \xleftarrow[\alpha^W]{} & \pi(Y,W) \end{array}$$
commute, i.e. $\alpha^W \circ \psi_* = \psi_* \circ \alpha^Z$.

Definition 2.1.5. *$X \sim Y$ if there exists a way to define for every space Z, a bijective map $\beta_Z : \pi(Z,X) \to \pi(Z,Y)$ s.t. for any continuous map $\phi : Z \to W$, the diagram*
$$\begin{array}{ccc} \pi(Z,X) & \xrightarrow{\beta_Z} & \pi(Z,Y) \\ {\phi^*}\uparrow & & {\phi^*}\uparrow \\ \pi(W,X) & \xrightarrow{\beta_W} & \pi(W,Y) \end{array}$$
commutes, i.e. $\beta_Z \circ \phi^* = \phi^* \circ \beta_W$.

Theorem 2.1.1. *The definition 2.1.2, 2.1.3, and 2.1.4 are equivalent.*

pf.

- assume $X \sim Y$ in the sense of the definition 2.1.3. Then $\exists \alpha^Y : \pi(X,Y) \to \pi(Y,Y)$ and we take for $f : X \to Y$ any representative of the homotopy class $(\alpha^Y)[Id_Y]$. Also there is a bijection $\alpha^X : \pi(Y,X) \to \pi(X,X)$ and we take for $g : Y \to X$ any representative of the homotopy class $(\alpha^X)^{-1}[Id_X]$. Then
$$\begin{array}{ccc} \pi(X,Y) & \xleftarrow[\alpha^Y]{} & \pi(Y,Y) \\ \downarrow{g_*} & & \downarrow{g_*} \\ \pi(X,X) & \xleftarrow[\alpha^X]{} & \pi(Y,X) \end{array}, \text{ and } \begin{array}{ccc} \pi(X,X) & \xleftarrow[\alpha^X]{} & \pi(Y,X) \\ \downarrow{f_*} & & \downarrow{f_*} \\ \pi(X,Y) & \xleftarrow[\alpha^Y]{} & \pi(Y,Y) \end{array}$$
$\Leftrightarrow \begin{cases} g_* \circ \alpha^Y = \alpha^X \circ g_* \\ f_* \circ \alpha^X = \alpha^Y \circ f_* \end{cases}$. Apply these to $\begin{cases} Id_Y \\ Id_X \end{cases}$, $\begin{cases} (g \circ f) = (Id_X) \\ (Id_Y) = (f \circ g) \end{cases}$ and the definition 2.1.2. follows.

- assume $X \sim Y$ in the sense of the definition 2.1.2. Then there exists a continuous map $\begin{cases} f : X \to Y \\ g : Y \to X \end{cases}$ s.t. $\begin{cases} g \circ f \sim Id_X \\ f \circ g \sim Id_Y \end{cases}$. For any space Z, let $\alpha^Z = f^* : \pi(Y,Z) \to \pi(X,Z)$. This is a bijection with the inverse map g^*
$\to \begin{cases} g^* \circ f^* = (f \circ g)^* = (Id_Y)^* = Id_{\pi(Y,Z)} \\ f^* \circ g^* = (g \circ f)^* = (Id_X)^* = Id_{\pi(X,Z)} \end{cases}$. Also, for any $\psi : Z \to W$,
$$\begin{array}{ccc} \pi(X,Z) & \xleftarrow[f^*]{} & \pi(Y,Z) \\ \downarrow{\psi_*} & & \downarrow{\psi_*} \\ \pi(X,W) & \xleftarrow[f^*]{} & \pi(Y,W) \end{array}$$
commutes, since for any $h : Y \to Z$,

$\begin{cases} \psi_* \circ f^*[h] = \psi_*[h \circ f] = [\psi \circ h \circ f] \\ f^* \circ \psi_*[h] = f^*[\psi \circ h] = [\psi \circ h \circ f] \end{cases}$. Thus $X \sim Y$ in the sense of the definition 2.1.3.

e.g. Let X be a circle and Y be an annulus. Then for $f : X \to Y$ the inclusion of X into Y as the outer boundary circle and put $g = f^{-1} \circ h : Y \to X$ where h is the radial projection of the annulus onto the outer boundary circle. Then $\begin{cases} g \circ f \sim Id_X \\ f \circ g \sim Id_Y \end{cases}$, i.e. they are homotopy equivalent while they are not homeomorphic.

Theorem 2.1.2. *The category is homotopy invariant; if $X \sim Y$, $catX = catY$.*

pf. Let $\begin{cases} f : X \to Y \\ g : Y \to X \end{cases}$ be mutually inverse homotopy equivalences and let $h_t : X \to X$ be a homotopy s.t. $\begin{cases} h_0 = Id_X \\ h_1 = g \circ f \end{cases}$. Let $\{U_1, \ldots, U_{catY}\}$ be a covering of Y by open sets contractible in Y, and let $k_{i,t} : U_t \to Y$ be a homotopy with $\begin{cases} k_0 \text{ being the inclusion map of } U_i \text{ into } Y \\ \qquad\qquad\qquad\qquad k_1 \text{ a constant map} \end{cases}$. Let $V_i = f^{-1}(U_i)$. Then the sets V_i form an open covering of X. Consider two homotopies $V_i \to X$, where the first consists of maps $x \mapsto h_t(x)$ and the second consists of maps $x \mapsto g(k_{i,t}(f(x)))$. The first homotopy joins the inclusion map $V_i \to X$ with the restriction map $(g \circ f)|_{V_i}$ and the second homotopy joins this restriction map with a constant map. Together V_i is contractible in X, and $catX \le catY$. Similarly, we can get $catY \le catX$ and therefore the theorem follows.

The definition of a homotopy and homotopy equivalence may be modified for basepoint spaces. The set of basepoint homotopy classes of maps between basepoint spaces X, Y is also denoted as $\pi(X, Y)$ or if necessary $\pi_b(X, Y)$. A further generalisation is a homotopy theory of pairs. A pair (X, Y) is a topological space X with a distinguished subspace A. A map of a pair $(X, A) \to (Y, B)$ is simply a continuous map $X \to Y$ taking A into B. Then homotopies, homotopy equivalences and so on are defined similarly.

Homotopy groups $\pi_n(X, x_0)$ of a space X with a basepoint x_0 are defined as a particular case of a general group-valued homotopy functor. The set $\pi_n(X, x_0)$ is defined as the set of basepoint homotopy classes of continuous maps of the sphere S^n into X, denoted $[S^n, X]_*$, called spheroids. $\Leftrightarrow$ a spheroid can be defined as a continuous map of the cube I^n into X taking the boundary ∂I^n of the cube into x_0 (given $S^n = I^n / \partial I^n$).

The sum of two spheroids, $f, g : S^n \to X$ is defined as the spheroid $f + g : S^n \to X$ s.t.

1. the equator of the sphere S^n containing the basepoint is collapsed into a point, so the sphere becomes the bouquet of two spheres.

2. each sphere is mapped into X according to f and g respectively

$\rightarrow$ it is not a group operation, but it becomes a group operation after the transition to homotopy classes. Thus $\pi_n(X,x_0)$ becomes a group.

Definition 2.1.6. *The space Y is called an H-space if there are maps*
$\begin{cases} \mu : Y\times Y \to Y & \textit{multiplication} \\ \nu : Y \to Y & \textit{inversion} \end{cases}$ *s.t.*

1. *the compositions* $\begin{cases} Y \xrightarrow{j_1} Y\times Y \xrightarrow{\mu} Y \\ Y \xrightarrow{j_2} Y\times Y \xrightarrow{\mu} Y \end{cases}$ *where* $\begin{cases} j_1(y) = (y, y_0) \\ j_2(y) = (y_0, y) \end{cases}$ *are homotopic to $Id_Y : Y \to Y$.*

2. *the compositions* $\begin{cases} Y\times(Y\times Y) \xrightarrow{Id\times\mu} Y\times Y \xrightarrow{\mu} Y \\ (Y\times Y)\times Y \xrightarrow{\mu\times Id} Y\times Y \xrightarrow{\mu} Y \end{cases}$ *are homotopic.*

3. *the maps* $\begin{cases} Y \longrightarrow Y\times Y \xrightarrow{\mu} Y \\ Y \longrightarrow Y\times Y \xrightarrow{\mu} Y \end{cases}$ *where the two left arrows mean the maps* $\begin{cases} y \mapsto (y, \nu(y)) \\ y \mapsto (\nu(y), y) \end{cases}$ *are homotopic to the constant map.*

Theorem 2.1.3. *If $n > 1$, the group $\pi_n(X,x_0)$ is commutative for any (X,x_0).*

pf.

- $n = 2$
  Let $Y = \Omega X$. Then it is a H-space and $\pi_1(Y,*)$ has extra structure$\rightarrow$ since $\pi_1(Y\times Y,*) = \pi_1(Y,*)\times\pi_1(Y,*)$, we get a group G together with a group homomorphism $\mu : G\times G \to G$ s.t.
  $$\begin{tikzcd} G \arrow[r, "in_1"] \arrow[dr, "Id"'] & G\times G \arrow[d, "\mu"] & G \arrow[l, "in_2"'] \arrow[dl, "Id"] \\ & G & \end{tikzcd}$$
  commutes. Then $\mu(a,1) = a, \mu(1,d) = d, \mu(ac,bd) = \mu(a,b)\cdot\mu(c,d)$. If we take $b = 1 = c, \mu(a,d) = ad$, i.e. the multiplication μ is non other than the group multiplication. Take $a = a = d$, then $\mu(c,b) = bc$; i.e. the group structure is commutative.

- $n > 2$
  $\pi_n(X) = \pi_2(\Omega^{n-2}X)$. So if $\pi_2(Y)$, $\forall Y \in k\mathbf{Top}_*$, it is abelian, it is abelian for all $n > 2$.

e.g. the loop space ΩZ of an arbitrary space Z where $\mu : \Omega Z \times \Omega Z \to \Omega Z$ is defined by the formula $[\mu(f,g)](t) = \begin{cases} f(2t) & t \le 1/2 \\ g(2t-1) & t \ge 1/2 \end{cases}$ and the map $\nu : \Omega Z \to \Omega Z$ defined by $[\nu(f)](t) = f(1-t)$.

Theorem 2.1.4. *The set $\pi_n(X,Y)$ possess a natural group structure iff Y is an H-space.*

pf.

- Let $\pi_n(X,Y)$ have a natural group structure w.r.t. X. Take $X = Y \times Y$ and consider the classes $[p_1][p_2] \in \pi_n(Y \times Y, Y)$ of the projections of $Y \times Y$ onto the factors. Set $[\mu] = [p_1] \cdot [p_2]$ and choose an arbitrary map $\mu : Y \times Y \to Y$ of the class $[\mu]$. For $\nu : Y \to Y$, we take an arbitrary map of the homotopy class $[Id_Y]^{-1}$. $\mu \circ j_1 \sim Id_Y$ and the map $j_1 : Y \to Y \times Y$ induces a map
$$\begin{array}{rcl} j_1^* : \pi_n(Y \times Y, Y) & \to & \pi_n(Y,Y) \\ p_1 & \mapsto & p_1 \circ j_1 \\ p_2 & \mapsto & p_2 \circ j_1 \end{array} \quad \text{where} \begin{cases} p_1 \circ j_1 = Id_Y \\ p_2 \circ j_1 = const \end{cases}.$$
Since the group structure in $\pi_b(X,Y)$ is natural, the map j_1^* takes products into products. Therefore $[\mu \circ j_1] = j_1^*[\mu] = j_1^*([p_1] \cdot [p_2]) = j_1^*[p_1] j_1^*[p_2] = [p_1 \circ j_1] \cdot [p_2 \circ j_1] = [Id_Y] \cdot [const] = [Id_Y] \to \mu \circ j_1 \sim Id_Y$.

- Let Y be an H-space and X be an arbitrary space in $k\mathbf{Top}_*$. The map $\mu : Y \times Y \to Y$ induces a map $\mu_* : \pi_n(X, Y \times Y) \to \pi_n(X,Y)$ which can be regarded as a map $\mu_* : \pi_n(X,Y) \times \pi_n(X,Y) \to \pi_n(X,Y)$ given the equality $\pi_n(X, Y \times Y) = \pi_n(X,Y) \times \pi_n(X,Y)$. Also the map $\nu : Y \to Y$ gives rise to a map $\nu_* : \pi_n(X,Y) \to \pi_n(X,Y)$. Together, the multiplication μ_* and the inversion ν_* determine in $\pi_n(X,Y)$ a natural w.r.t. X.

Definition 2.1.7. *Let $J_n = \partial I^{n-1} \times I \cup I^{n-1} \times 0 = \partial I^n \setminus I^{n-1}$. Let $(X,A,*)$ be a pointed pair. For $n \ge 1$, define a pointed set $\pi_n(X,A,*) = [(I^n, \partial I^n, J_n),(X,A,*)]$.*

The homotopy sequence of a pair is the sequence

$$\dots \xrightarrow{\partial} \pi_n(A,*) \xrightarrow{i_*} \pi_n(X,*) \xrightarrow{j_*} \pi_n(X,A,*) \xrightarrow{\partial} \pi_{n-1}(A,*) \longrightarrow \dots$$

$$\dots \xrightarrow{i_*} \pi_1(X,*) \xrightarrow{j_*} \pi_1(X,A,*) \xrightarrow{\partial} \pi_0(A,*) \xrightarrow{i_*} \pi_0(X,*)$$

where $\begin{cases} i_* \\ j_* \end{cases}$ is induced by $\begin{cases} \text{the inclusion map} & i : A \to X \\ \text{the identity map} & j : (X,*) \to (X,A) \end{cases}$ and the connecting homomorphism
$$\begin{array}{rcl} \partial : \pi_n(X,A,*) & \to & \pi_{n-1}(A,*) \\ f : (I^n, \partial I^n, \partial I^n \setminus I^{n-1}) \to (X,A,*) & \mapsto & f|_{I^{n-1}} : (I^{n-1}, \partial I^{n-1}) \to (A,*) \end{array}.$$
pf.

- $Imi_* \subseteq \ker j_*$.

  for any spheroid $f:(I^n,\partial I^n)\to(A,x_0)$, the class $j_*\circ i_*[f]$ is represented by the same map f, regarded as a map $(I^n,\partial I^n,J)\to(X,A,*)$ and the spheroid $\begin{array}{rcl} f_t:(I^n,\partial I^n,J_n) & \to & (X,A,*) \\ f(x_1,\dots,x_n) & \mapsto & f(x_1,\dots,x_{n-1},t+(1-t)x_n) \end{array}$ form a homotopy connecting f with a constant spheroid. So $j_*\circ i_*[f]=0$.

- $\ker j_* \subset Imi_*$.

  for any spheroid $f:(I^n,\partial I^n)\to(X,x_0)$ homotopic to the constant within the class of relative spheroids, let $F:I^n\times I\to X$ be a homotopy between f and the constant spheroid in the class of relative spheroids of the pair (X,A). Then F is the map $I^{n+1}\to X$ which is f on the face $I^n=\{x_{n+1}=0\}$ that maps the face $\{x_n=0\}$ into A and maps the remaining part of ∂I^{n+1} into $*$. Let $I^n_t\subseteq I^{n+1}$ be the intersection of the cube I^{n+1} with the plane $tx_n+(1-t)x_{n-1}=0$. Then $I^n_t\sim I^n$ and $F|_{I^n_t}:I^n_t=I^n\to X$ is a homotopy joining the spheroid f with g whose image is contained in A.

- $Imj_* \subset \ker\partial$

  if $f:I^n\to X$ is an absolute spheroid, $f|_{I^{n-1}}$ is a constant map.

- $\ker\partial \subset Imj_*$

  let $f:I^n\to X$ be a relative spheroid and let $g_t:I^{n-1}\to A$ be a homotopy connecting the absolute spheroid $f|_{I^{n-1}}$ of A with the constant spheroid. Consider the homotopy $f_t:\partial I^n\to X$ coinciding with g_t on I^{n-1} and taking $\partial I^n\setminus I^{n-1}$ into x_0 and extend it to a homotopy $h_t:I^n\to X$ of the spheroid f. Then h_t is a homotopy connecting f with the constant spheroid within the class of relative spheroid.

- $Im\partial \subset \ker i_*$

  if absolute spheroid $f:I^{n-1}\to A$ is a restriction of an absolute spheroid $g:I^n\to X$, $g_t=g|_{I^{n-1}\times t}:I^{n-1}\times t=I^{n-1}\to X$ is a homotopy connecting f with the constant spheroid.

- $\ker i_* \subset Im\partial$
  if $g_t:I^{n-1}$ is a homotopy connecting a spheroid $f=g_0:I^{n-1}\to A$ with a constant spheroid, then $\begin{array}{rcl} g:I^n & \to & X \\ (x_1,\dots,x_n) & \mapsto & g_{x_n}(x_1,\dots,x_{n-1}) \end{array}$ is a relative spheroid of the pair (X,A) whose restriction to A is f.

2.1.2 CW complexes

Definition 2.1.8. *A CW complex is a Hausdorff space X with a fixed partition $X = \cup_{q=0}^{\infty} \cup_{i \in I_q} e_i^q$ of X into pairwise disjoint cells e_i^q s.t. for every cell e_i^q, there exists a continuous map, called the characteristic map, $f_i^q : D^q \to X$, whose restriction to $IntD^q$ is a homeomorphism $IntD^q \sim e_i^q$ whose restriction to $S^{q-1} = D^q \setminus IntD^q$ maps S^{q-1} into the union of cells of dimensions $< q$.*

→ CW complexes are assumed to satisfy the axioms ;

1. (C:Closure Finite)
   the boundary of the cell $\bar{e}_i^q - e_i^q = f_i^q(S^{q-1})$ is contained in a finite union of cells

2. (W:Weak Topology)
   a set $F \subset X$ is closed iff for any cell e_i^q the intersection $F \cap \bar{e}_i^q$ is closed.

Definition 2.1.9. *A CW subcomplex of a CW complex X is a closed subset composed of whole cells.*

e.g. the nth skeleton X^n, denoted $sk_n X$, of X is the union of all cells e_i^q with $q \le n$. they are subcomplexes.

Definition 2.1.10. *A CW complex is called locally finite if every point has a nhood which is contained in some finite CW subcomplex.*

Theorem 2.1.5. *If X, Y are CW complexes, then the space Y^X is homotopy equivalent to a CW complex.*

Lemma 2.1.1.

$$sk_n X = (sk_{n-1} X) \cup_{g^n} \mathscr{D}$$

i.e. $sk_n X$ is obtained from $sk_{n-1} X$ by attaching n-dimensional balls by means of attaching maps corresponding to all n-dimensional cells of X.

Let $\{e_\alpha^n\}$ be the set of all n-dimensional cells of a CW complex X and let $f_\alpha^n : D^n \to X$ be corresponding characteristic maps. Since $f_\alpha^n(S^{n-1}) \subset sk_{n-1} X$, we can restrict f_α^n to an attaching map $g_\alpha^n : S^{n-1} \to sk_{n-1} X$. Take the disjoint union $\mathscr{D} = \coprod_\alpha D_\alpha^n$ of n-dimensional balls, one for each n-dimensional cell of X, and put $\mathscr{S} = \coprod_\alpha S_\alpha^{n-1} \subset \mathscr{D}$. Then $g^n : \mathscr{S} \to X$, $g^n|_{S_\alpha^{n-1}} = g_\alpha^n$. Then the above is a step of a universal inductive procedure which constructs an arbitrary CW complex from a discrete space or empty space by successively attaching balls of growing dimensions.

Definition 2.1.11. *A pair (X,A) is called a Borsuk pair if for every topological space Y, every continuous map $F : X \to Y$ and every homotopy $f_t : A \to Y$ s.t. $f_0 = F|_A$, there exists a homotopy $F_t : X \to Y$ s.t.* $\begin{cases} F_0 = F \\ F_t|_A = f_t \end{cases}$.

Theorem 2.1.6 (Borsuk)**.** *Every CW pair is a Borsuk pair.*

pf. Let (X,A) be a CW pair. There exist maps $\begin{cases} \Phi : A \times I \to Y \\ F : X \times 0 \to Y \end{cases}$ s.t. $F|_{A\times 0} = \Phi|_{A\times 0}$. To extend the homotopy f_t to a homotopy F_t, we need to extend the map F to a map $F' : X \times I \to Y$ s.t. $F'|_{A\times I} = \Phi$. Construct the extension by induction on the dimension of cells n.

- extension of the map Φ to $(A \cup X^0) \times I$ is

$$F'(x,t) = \begin{cases} F(x,0) & x \text{ 0-dimensional cell of } X,\ x \notin A \\ \Phi(x,t) & x \in A \end{cases}$$

- assume F' has been already defined on $(A \cup X^n) \times I$ and equal to $F' = \begin{cases} \Phi & A \times I \\ F & X \times 0 \end{cases}$. Take $n+1$ dimensional cell $e^{n+1} \subset X \setminus A$. Then F' is defined on the set $(\bar{e}^{n+1} \setminus e^{n+1}) \times I$ since the boundary is contained in X^n by the definition of a CW complex. Let $\psi = F' \circ f : (S^n \times I) \cup (D^{n+1} \times 0) \to Y$. Also let $f : D^{n+1} \to X$ be a characteristic map for the cell e^{n+1}. Let $\eta : D^{n+1} \times I \to (S^n \times I) \cup (D^{n+1} \times 0)$ be the projection of the cylinder $D^{n+1} \times I$ from a point slightly above the upper base of the cylinder. It is the identity on $(S^n \times I) \cup (D^{n+1} \times 0)$. Define the map $\psi' = \psi \circ \eta$. For all $n+1$-dimensional cells in $X \setminus A$, we do this simultaneously and we get an extension of the map F' to$(A \cup X^{n+1}) \times I$.

In this way we construct an extension of the map Φ to a map $F' : X \times I \to Y$. When $X \setminus A$ is infinite dimensional, the continuity of the map F' is obtained from the Axiom W.

Lemma 2.1.2 (Free point Lemma)**.** *Let U be an open subset of $\mathbb{R}^n$ and $\phi : U \to IntD^m$ be such a continuous map that the set $V = \phi^{-1}(d^m) \subset U$ where d^m is some closed ball in $IntD^m$ is compact. If $m > n$, there exists a continuous map $\psi : U \to IntD^m$ coinciding with ϕ in the complement of V and s.t. its image does not cover the whole ball d^m.*

Theorem 2.1.7. *Let f be a continuous map of a CW complex X into a CW complex Y s.t. the restriction $f|_A$ is cellular for some CW subcomplex A of X. Then there exists a cellular map $g : X \to Y$ s.t. $g|_A = f|_A$ and g is A-homotopic to f.*

pf. Assume the map f has already been made cellular not only on all cells from A, but also on all cells from X of dimension less than n. Take n-dimensional cell $e^n \subset X \setminus A$. Its image $f(e^n)$ has a nonempty intersection with only a finite set of cells of Y. Of these cells of Y, choose a cell of a maximal dimension, say ϵ^m with $\dim \epsilon^m = m$. If $m \leq n$, nothing to prove. So Assume $m > n$. By the free-point lemma, the restriction $f|_{A \cup X^{n-1} \cup e^n}$ is $(A \cup X^{n-1})$-homotopic to a map $f' : A \cup X^{n-1} \cup e^n \to Y$ s.t. $f'(e^n)$ has nonempty intersection with the same cells as $f(e^n)$, but $f'(e^n)$ does not cover the whole cell ϵ^m. Let $\begin{cases} h : D^n \to X \\ k : D^m \to Y \end{cases}$ be characteristic maps corresponding to the cells e^n, ϵ^m. Let $U = h^{-1}(f^{-1}(\epsilon^m \cap e^n))$ and define a map $\phi : U \to IntD^m$ as a composition $\begin{array}{ccccccc} U & \to & e^n \cup f^{-1}(\epsilon^m) & \to & e^m & \to & IntD^m \\ u & \mapsto_h & x & \mapsto_f & y & \mapsto_{k^{-1}} & v = \phi(u) \end{array}$.
Denote as d^m as a closed concentric subball of the ball D^m. The set $V = \phi^{-1}(d^m)$ is compact because it is a closed subset of a closed ball D^m. Let $\psi : U \to IntD^m$ be a map provided by the free-point lemma. Define the map

$$f' = \begin{cases} & & & & & & f & (h(U))^{\perp} \\ h(U) & \to & U & \to & IntD^m & \to & e^m \subset Y & h(U) \\ x & \mapsto_{h^{-1}} & u & \mapsto_{\psi} & v & \mapsto_k & y = f'(u) & \end{cases}$$

Then f' is continuous and $f'(e^n)$ does not cover ϵ^m. Then by Borsuk,s theorem, we can extend the homotopy fixed on $A \cup X^{n-1}$ between $f|_{A \cup X^{n-1} \cup e^n}$ and f' to the whole space X, which lets us assume the map f' is defined on the whole space X. Then we take a point $y_0 \in \epsilon^m \setminus f(e^n)$ and apply to $f'|_{e^n}$ a radial homotopy ($\begin{cases} x \in e^n \setminus f^{-1}(\epsilon^m), & f'(x) \text{ constant} \\ f'(x) \in \epsilon^m & f'(x) \text{ moving from } y_0 \text{ to the boundary of } \epsilon^m \text{ through } f'(x) \end{cases}$). Extend this to a homotopy of $f'|_A \cup X^{n-1} \cup e^n$ and then by the Borsuk's theorem, to a homotopy of the whole map $f' : X \to Y$. So we reduce the number of m-dimensional cells hit by $f'(e^n)$ by one, and repeating this until there is no cell of dimension m gives an $(A \cup X^{n-1})$-homotopy of f to a map cellular on $A \cup X^{n-1} \cup e^n$. If we do this simultaneously for all n-dimensional cells in $X \setminus A$, we get a map cellular on $A \cup X^n$ and $(A \cup X^{n-1})$-homotopic to f. Perform this construction successively for $n = 0, 1, 2, \ldots$. The number of steps may be infinite, but we can perform the nth homotopy at the parameter interval $1 - 2^{-n} \leq t \leq 1 - 2^{-n-1}$. The continuity of the whole homotopy is guaranteed by Axiom W.
$\to$ there are various ways to say the above theorem. **The skeletal approximation theorem** (for (X,A), (Y,B) relative CW complexes, any map $f : (X,A) \to (Y,B)$ is homotopic $relA$ to a skeletal map.) is one of them.

Definition 2.1.12. *A relative CW complex is a pair (X,A) together with a filtration*

$A = X_{-1} \subseteq X_0 \subseteq X_1 \subseteq \cdots \subseteq X$ *s.t.* $\forall n$,
$$\begin{tikzcd} \coprod_{\alpha\in\Sigma_n} S^{n-1}_\alpha \arrow[r] \arrow[d] & \coprod_{\alpha\in\Sigma_n} D^n_\alpha \arrow[d] \\ X_{n-1} \arrow[r] & X_n \end{tikzcd}$$
and $X = \lim X_n$ *topologically.*

Definition 2.1.13. *A space X is called n-connected if for $m \le n$, the set $\pi(S^m, X)$ consists of one element, i.e. any two continuous maps $S^m \to X$ with $m \le n$ are homotopic.*

$\Leftrightarrow \forall a, \pi_m(X,a)$ trivial $\forall m \le n$.

Theorem 2.1.8. *Let $n \ge 0$. An n-connected CW complex is homotopy equivalent to a CW complex with only one vertex and without cells of dimensions $1,2,\ldots,n$.*

pf. Choose a vertex $e_0 \in X$ and join all other vertices $e_1, e_2, \ldots$ with e_0 by paths $s_1, s_2, \ldots$. This is possible since X is n-connected and hence path-connected. Then every path s_i lies in the first skeleton of X. For every i, attach a two-dimensional disk to X by s_i, regarded as a map of the lower semicircle to X. Then we get a new CW complex $\hat{X}$ which contains X and also cells $\begin{cases} e_i^2 & \text{the interior} \\ e_i^1 & \text{upper semicircle} \end{cases}$ of the disk attached. The boundaries of the cells e_i^2 are contained in the first skeleton because the paths s_i have this property. Then $\hat{X}$ deformation retracts to X, since every attached disk can be retracted onto the corresponding path s_i. Denote Y the union of the closures of the cells e_i^1. Then Y contains all the vertices of $\hat{X}$ and is contractible. Hence $\hat{X}/Y$ has only one vertex and $\hat{X}/Y \sim \hat{X} \sim X$. Assume $X \sim X'$ where X' has only one vertex and has no cells of dimensions $1,2,\ldots,k-1$, where $k \le n$. Then the closure of every k-dimensional cell is a k-dimensional sphere. Since X is n-connected, the inclusion of the sphere into X' can be extended to a continuous map $D^{k+1} \to X'$ and by the previous theorem, we can assume the image of this map is contained in the $k+1$-st skeleton of X'. We attach a ball D^{k+2} to X' by this map and we do this for every k-dimensional cell of X'. The resulting X' acquires $k+1, k+2$-dimensional cells. The new CW complex $\hat{X}'$ is homotopy equivalent to X'. Denote Y' the union of all new $k+1$-dimensional cells, which contains the whole kth skeleton of $\hat{X}'$. The quotient $\hat{X}'/Y' \sim X' \sim X$ and it does not have cells of dimensions $\le k$, besides its only one vertex.

Definition 2.1.14. *A Moore space is a simple space with only one nonzero reduced homology group.*

Proposition 2.1.1. *Let π be an abelian group and n a positive integer. Then there is a CW complex M with cells in dimensions $0, n, n+1$ s.t.* $\bar{H}_q(M) = \begin{cases} \pi & q = n \\ 0 & otherwise \end{cases}$.

pf. if π is a free abelian group, we can pick generators and take a corresponding wedge of n-spheres.
otherwise, pick a free resolution $0 \longleftarrow \pi \longleftarrow F_0 \xleftarrow[d]{} F_1 \longleftarrow 0$. Pick generators for F_0, F_1, say $\begin{cases} \{\alpha_i : i \in I\} \\ \{\beta_j : j \in J\} \end{cases}$ respectively. Build the corresponding wedges of n-spheres. A pointed map from a wedge is given by pointed maps from each factor. The map d determined by $d\beta_j = \sum_i a_{ji}\alpha_i$ for some set of integers $\{a_{ji}\}$, finitely nonzero for fixed j. For each i, $i_i : S^n \to \bigvee_{i \in I} S^n$. The sum $\sum_i a_{ji} i_i$ determines a map $S^n \to \bigvee_i S^n$. Then $\bigvee_i S^n \leftarrow \bigvee_j S^n$ realises d. We can build M as an $n+1$ dimensional CW complex by taking the mapping cone of this map.

Definition 2.1.15. *An Eilenberg MacLane space, denoted $K(\pi, n)$ is a topological space with just one homotopy group in dimension n which is isomorphic to π.*

e.g. $K(\mathbb{Z}, 1) = S^1$, $K(\mathbb{Z}^n, 1) = (S^1)^n$.

Let π be an abelian group and n a positive integer. Pick a free resolution $0 \to F_1 \to F_0 \to \pi \to 0$. Pick generators for F_0, F_1 respectively and build the corresponding cofiber sequence $\bigvee_j S^n \to \bigvee_i S^n \to M$ where M is a Moore space with $H_n(M) = \pi$. The Postnikov tower is

$$\begin{tikzcd} & \vdots \arrow[d] \\ & P_2(X) \arrow[d] \\ & P_1(X) \arrow[d] \\ X \arrow[r] \arrow[ur] \arrow[uur] & P_0(X) \end{tikzcd}$$

, where $P_n(X)$ is X with cells of dimension at least $n+2$ attached so that $\pi_q(P_n(X)) = 0$, $\forall q > n$. This $P_n(X)$ is also denoted $\tau_{\leq n} X$ the truncation of X at dimension n.

Lemma 2.1.3. *Let n be a positive integer and let Y be any pointed space s.t. $\pi_q(Y, *) = 0$, $\forall q \neq n$ write $G \equiv \pi_n(Y, *)$. Then*

$$\pi_n : [\tau_{\leq n} M, Y]_* \to \hom(\pi, G)$$

is an isomorphism.

pf. Since the sequence defining M is co-exact, $\exists$ an exact sequence
$[\bigvee_j S^n, Y]_* \leftarrow [\bigvee_i S^n, Y]_* \leftarrow [M, Y]_* \leftarrow [\bigvee_j S^{n+1}, Y]_*$. Then this sequence reads $\hom(F_1, G) \leftarrow \hom(F_0, G) \leftarrow [M, Y]_* \leftarrow 0$. But a homomorphism $F_0 \to G$ that restricts to 0 on F_1 is exactly a homomorphism $\pi \to G$. Thus $\pi_n : [M, Y]_* \to \hom(\pi, G)$

is an isomorphism. Since $M \to \tau_{\leq n}M$ is universal among maps to spaces with homotopy concentrated in dimension at most n, the lemma follows.
$\to \tau_{\leq n}N \to Y$ inducing the identity in π_n is unique. This is a weak equivalence, and therefore if Y is a CW complex, the map is a homotopy equivalence.

Corollary 2.1.1. *For any positive integer n, there is a functor* $\begin{array}{ccc} \mathbf{Ab} & \to & Ho(\mathbf{CW}_*) \\ \pi & \mapsto & K(\pi,n) \end{array}$ *unique up to isomorphism.*

Let n be a positive integer and Y an $(n-1)$-connected space. Then $\bar{H}_q(Y) = 0$, $\forall q < n$. $\forall \pi \in \mathbf{Ab}$, the universal coefficient theorem asserts

$$0 \to Ext^1(H_{q-1}(Y),\pi) \to H^q(Y,\pi) \to \hom(H_q(Y),\pi) \to 0$$

s.e.s exists for any q. Then $H^q(Y,\pi) = 0$, $\forall q < n$. When $q = n$, the Ext term vanishes so the second map is an isomorphism. If we take $Y = K(\pi,n)$, we obtain a canonical class $\iota_n \in H^n(K(\pi,n);\pi)$, called the fundamental class.

Lemma 2.1.4. *For any CW complex X, the set of homotopy classes of maps $[X,K(\pi,n)]$ is canonically identified with the elements of the group $H^n(X;\pi)$. The bijection is induced by the natural correspondence $f : X \to K(\pi,n)$ maps to $f^*(\iota_{\pi,n})$.*

pf. Let X be a CW complex and let $\alpha \in H^n(X;\pi)$ be a class. Define the map $X^{(n-1)} \to \{x_0\} \in K(\pi,n)$. Fix a cocycle $\tilde{\alpha} \in C^n(X;\pi)$ representing α. Then for each n-cell e, orienting the cell gives an element $[e] \in C_n(X)$. Define the map f_n s.t. its restriction to e represents the element in $< \tilde{\alpha},[e] > \in \pi = \pi_n(K(\pi,n))$. There is no obstruction to extending this over $X^{(n+1)}$ since for every $n+1$-cell its attaching map gives a sphere representing a cycle in $C_n(X)$ and $\tilde{\alpha}$ evaluates on this sphere to give the value of α on the homology class of the sphere in X. Since the sphere bounds a disk in X, the homology class of the sphere is trivial. That the higher obstructions vanish means that we can extend this map to a map $f : X \to K(\pi,n)$ and therefore we get f_{n+1}. Then $f^*\iota_n = \alpha$. For maps from X to $K(\pi,n)$, denoted f,g to be homotopic, $f^*(\iota_n) = g^*(\iota_n)$. Conversely, suppose we have maps $f,g : X \to K(\pi,n)$ with $f^*(\iota) = g^*(\iota)$. Since the homotopy groups of $K(\pi,n)$ vanish in dimensions less than n, we can assume f,g both map $X^{(n-1)}$ to the basepoint. Then the obstruction to extending the constant homotopy on the $(n-2)$-skeleton to a homotopy defined over the n-skeleton is $f^*(\iota) - g^*(\iota)$. Thus if $f^*(\iota) = g^*(\iota)$, the restrictions of f,g to $X^{(n)}$ are homotopic. The obstructions to extending the homotopy from $X^{(n)}$ to all of X vanish since all the higher obstruction groups vanish.
$\to K(\pi,n)$ is a group object in the homotopy category, i.e. an H-space ($\iota_{\pi,n} \otimes Id + Id \otimes \iota_{\pi,n} \in H^n(K(\pi,n) \times K(\pi,n);\pi)$.)

2.1.3 Fibration and CoFibration

Definition 2.1.16. *An open cover $\mathscr{U}$ of a space X is said to be numerable if there exists a subordinate partition of unity, i.e. there is a family of functions $\phi_U : X \to [0,1] = I$, indexed by the elements of $\mathscr{U}$, s.t. $\phi_U^{-1}((0,1]) = U$ and $\forall x \in X$ belongs to only finitely many $U \in \mathscr{U}$. The space X is said to be paracompact if any open cover admits a numerable refinement.*

→ by the Axiom W, CW complexes are paracompact.

Definition 2.1.17. *A fibration is a map $p : E \to B$ that satisfies the HLP(homotopy lifting property; given any $\left\{ \begin{array}{c} f : W \to E \\ h : I \times W \to B \end{array} \right.$ with $h(0,\omega) = pf(\omega)$, $\exists \hat{h}$ that lifts h and extends f, i.e. the diagram*
$$\begin{tikzcd} W \arrow[r, "f"] \arrow[d, "i_0"] & E \arrow[d, "p"] \\ I \times W \arrow[r, "h"] \arrow[ur, dotted, "\hat{h}"] & B \end{tikzcd}$$
commutes.

e.g. for any space X, the unique map $X \to *$ is a fibration ($\hat{h}(t,\omega) = f(\omega)$).

$B^I \times_B E = \{(\omega, e) \in B^I \times E : \omega(0) = p(e)\}$ with $\begin{array}{rcl} \tilde{p} : E^I & \to & B^I \times_B E \\ \omega & \mapsto & (p\omega, \omega(0)) \end{array}$. Then the lift $\bar{h}$ is equivalent to a lift
$$\begin{tikzcd} & E^I \arrow[d, "\tilde{p}"] \\ W \arrow[r] \arrow[ur, dotted, "\tilde{h}"] & B^I \times_B E \end{tikzcd}$$
. When $W = B^I \times_B E$, we have the identity map. Then $\tilde{p}$ is a fibration iff a lift $\tilde{h} = \lambda$, a section of $\tilde{p}$ exists. The section λ is called a path lifting function. Then by the universality, we get the lift for all space W and homotopy h.

Theorem 2.1.9. *Let $p : E \to B$ be a continuous map. Assume there is a numberable cover of B, say $\mathbf{U}$ s.t. $\forall U \in \mathbf{U}$, the restriction $p|_{p^{-1}(U)} : p^{-1}(U) \to U$ is a fibration. Then p itself is a fibration.*

If $p : E \to B$ is a covering space, then for any path $\omega : a \to b$, the homeomorphism $F_a \to F_b$ depends only on the path homotopy class of ω.
$$\begin{tikzcd} F_a \arrow[rr] \arrow[d, "i_0"] & & E \arrow[d, "p"] \\ I \times F_a \arrow[r, "p_1"] \arrow[urr, dotted, "h"] & I \arrow[r, "\omega"] & B \end{tikzcd}$$
commutes since $\omega(0) = a$. By the HLP, there is a map $h : I \times F_a \to E$ that makes the entire diagram commute. If $x \in F_a$, $h(1,x) \in F_b$. Thus we can define a map $\begin{array}{rcl} f : F_a & \to & F_b \\ x & \mapsto & h(1,x) \end{array}$. Suppose we have two paths $\omega_0, \omega_1 : a \to b$

and a homotopy $g : I \times I \to b$ between them. Choose lifts h_0, h_1 and we get the diagram

$$\begin{tikzcd} ((\partial I \times I) \cup (I \times \{0\})) \times F_a \arrow[rr, "\phi"] \arrow[d, "i_0"] & & E \arrow[d, "p"] \\ I \times I \times F_a \arrow[r, "p_1"] \arrow[urr, dotted] & I \times I \arrow[r, "g"] & B \end{tikzcd}$$

where $\phi = \begin{cases} h_0, h_1 & \partial I \times I \times F_a \\ i \circ p_2 & \partial I \times \{0\} \times F_a \end{cases}$.

If the dotted lift exists, we can restrict it on $I \times \{1\} \times F_a$ to a homotopy between f_1, f_0. The subspace $(\partial I \times I) \cup (I \times \{0\})$ of $I \times I$ wraps around the three wedge of the square of homotopy, we can create a homeomorphism with the pair $(I \times I, \{0\} \times I)$ so the HLP gives the dotted lift. Thus the map $F_a \to F_b$ is well-defined up to homotopy by the path homotopy class of ω from a to b.

Then space B defines a category $\begin{cases} \text{objects} & \text{the points of } B \\ \text{arrows} & \text{homotopy class of paths from two points} \end{cases}$ with juxtaposition as composition $(\sigma \cdot \omega)(t) = \begin{cases} \omega(2t) & t \le 1/2 \\ \sigma(2t-1) & t \ge 1/2 \end{cases}$. Then this juxtaposition has identity, and satisfy the associativity and unit rule. Thus the space B, itself, is a groupoid, denoted $\Pi_1(B)$.

Proposition 2.1.2. *Formation of fibers of a fibration $p : E \to B$ determines a functor $\Pi_1(B) \to Ho\mathbf{Top}$.*

pf. Let $\begin{cases} \omega : a \sim b \\ \sigma : b \sim c \end{cases}$. Pick lifts

$$\begin{tikzcd} F_a \arrow[r] \arrow[d] & E \arrow[d] \\ I \times F_a \arrow[r, "\omega"] \arrow[ur, dotted, "h_\omega"] & B \end{tikzcd}, \quad \begin{tikzcd} F_b \arrow[r] \arrow[d] & E \arrow[d] \\ I \times F_b \arrow[r, "\sigma"] \arrow[ur, dotted, "h_\sigma"] & B \end{tikzcd}$$

so that $\begin{cases} f_\omega(e) = h_\omega(1, e) \\ f_\sigma(e) = h_\sigma(1, e) \end{cases}$. Then we can construct a lift in

$$\begin{tikzcd} F_a \arrow[r] \arrow[d] & E \arrow[d] \\ I \times F_a \arrow[r, "\sigma\omega"] \arrow[ur, dotted, "\bar{h}"] & B \end{tikzcd}$$

where $\bar{h} = \begin{cases} h_\omega \\ h_\sigma \circ f_\omega \end{cases}$. The resulting map $F_a \to F_b$ is $f_\sigma \circ f_\omega = f_{\sigma\omega}$.

Proposition 2.1.3. *Let $p : E \to Y$ be a fibration and $f_0, f_1 : X \to Y$ two maps. Write E_0, E_1 for pullbacks of E along f_0, f_1 respectively. If $f_0 \sim f_1$, $E_0 \cong E_1$.*

pf. Construct a fibration over Y^X whose fiber over f is f^*E, the pullback of $E \to Y$ along f;

$$\begin{tikzcd} f^*E \arrow[r] \arrow[d] & E \times_Y (Y^X \times X) \arrow[r] \arrow[d] & E \arrow[d, "p"] \\ * \times X \arrow[r, "i \times 1"] \arrow[d] & Y^X \times X \arrow[r, "ev"] \arrow[d, "p_1"] & Y \\ * \arrow[r, "i"] & Y^X & \end{tikzcd}$$

where the middle horizontal composite is

the map f. Thus the pullback is f^*E as in the diagram. A homotopy between f_0, f_1 is a path in Y^X from f_0 to f_1, so by the previous proposition, the fibers over them are homotopy equivalent.
$\rightarrow$ a fiber homotopy equivalence between E_0, E_1 holds if the above holds.

Definition 2.1.18. *A cofibration is a map $i : A \to X$ that satisfies HEP(homotopy extension property; for any solid arrow commutative diagram, a dotted arrow exists making the whole diagram commute.*

$$\begin{tikzcd} A \arrow[r, "i"] \arrow[d] & X \arrow[d] \arrow[ddr] & \\ A \times I \arrow[r] \arrow[drr] & X \times I \arrow[dr, dotted] & \\ & & Z \end{tikzcd}$$

$i : A \to X$ is a cofibration iff there exists an extension in

$$\begin{tikzcd} (X \times 0) \cup_A (A \times I) \arrow[r, "j"] \arrow[dr, "f"'] & X \times I \arrow[d, dotted] \\ & Z \end{tikzcd}$$

for every map f. If $Z = (X \times 0) \cup_A (A \times I)$, $f = Id$. In this case the map $j : (X \times 0) \cup_A (A \times I) \to X \times I$ admits a retraction $r : X \times I \to (X \times 0) \cup_A (A \times I)$ s.t. $rj = 1$. If $A \subseteq X$ is a closed subset, $A \times I \subseteq X \times I$ is a closed map, and $X \times 0 \subset X \times I$ is a closed map. Thus the map $(X \times 0) \cup_A (A \times I) \to imj \subseteq X \times I$ is a closed map and hence is a homeomorphism to its image. Then by universality, we get the cofibration for any space Z and any map f.

Definition 2.1.19. *A basepoint $* \in X$ is nondegenerate if $\{*\} \to X$ is a closed cofibration. Also $(X, *)$ is said to be well-pointed.*

e.g. any point in CW complex is nondegenerate.

If $*$ is a nondegenerate basepoint of A, the evaluation map $ev : X^A \to X$ is a fibration, and the fiber of ev is exactly the space of pointed maps X^A_*. Let $M(i) = (X \times 0) \cup_A (A \times I)$

Proposition 2.1.4. *Let $A \to X$ be a cofibration, and write X/A for the pushout of $* \longleftarrow A \longrightarrow X$. If A is contractible, $X \to X/A$ is a homotopy equivalence.*

pf. Pick a contracting homotopy $h : A \times I \to A$ so that $\begin{cases} h(a,0) = a \\ h(a,1) = * \in A \end{cases}$ for all $a \in A$. Then there exists an extension $f \circ h$ to a homotopy $g : X \times I \to X$ s.t. $g(x,0) = x$. Then $g(-,1)$ factors through the projection $p : X \to X/A, \exists r : X/A \to X$ s.t. $r \circ p \sim Id$. The homotopy $g : X \times I \to X$, so compose it with $p : X \to X/A$. Then it factors through the map $\bar{g}:(X/A)\times I \to X/A$. At $t=0$, it is the identity and at $t=1$ it is $p\circ r$.

Any pointed map $f:X\to Y$ admits a canonical factorisation as a closed pointed cofibration followed by a pointed homotopy equivalence
$$\begin{tikzcd} & X \arrow[dl, "f"'] \arrow[d, "i"] \\ Y & M(f) \arrow[l, "\cong"] \end{tikzcd}$$
where i imbeds X along $t=1$. Since i is a cofibration, we may form the pushout in
$$\begin{tikzcd} & X \arrow[dl, "f"'] \arrow[d, "i"] \arrow[r] & * \arrow[d] \\ Y & M(f) \arrow[l, "\cong"] \arrow[r] & C(f) \end{tikzcd}$$
where Cf is the mapping cone of f. Since i is a cofibration, $*\to C(f)$ is again a cofibration. $\Leftrightarrow$
$$\begin{tikzcd} & * \\ X \arrow[ur] \arrow[r, "i_1"] \arrow[d, "f"] & CX \arrow[u, "\cong"] \arrow[d] \\ Y \arrow[r, "i(f)"] & Cf \end{tikzcd}$$
homeomorphic to each other.

Definition 2.1.20. *A cofibration sequence is a diagram that is homotopy equivalent to*
$$X \xrightarrow{f} Y \xrightarrow{i(f)} C(f)$$
for some map f.

$\to$ the composite $X\to C(f)$ is nullhomotopic with $h;(x,t)\mapsto [x,t]$. This pair $(i(f),h)$ is universal ; given a map $g:Y\to Z$ along with a nullhomotopy of the composite $g\circ f$ is the same thing as giving a map $C(f)\to Z$ that extends g.

Lemma 2.1.5. *For any pointed map $f:X\to Y$ and any pointed space Z, the sequence of pointed sets*
$$[X,Z]_* \xleftarrow[f^*]{} [Y,Z]_* \xleftarrow[i(f)^*]{} [C(f),Z]_*$$
is exact in the sense that $im(i(f)^*)=\{g:Y\to Z : g\circ f\cong *\}$.

The map $f:X\to Y$ gives
$$\begin{tikzcd} X \arrow[r, "f"] & Y \arrow[r, "i(f)"] & C(f) \arrow[r, "i^2(f)"] \arrow[dr, "\pi(f)"] & C(i(f)) \arrow[r, "i^3(f)"] \arrow[d, "\cong"] \arrow[dr, "\pi(i(f))"] & \dots \\ & & & \Sigma X \arrow[r, "-\Sigma f"] & \Sigma Y \arrow[r] & \dots \end{tikzcd}$$
where $C(f)/Y=\Sigma X$, $C(i(f))\to C(f)/Y$ is a homotopy equivalence by the previous

lemma, and $-\Sigma : [t,x] \mapsto [1-t, f(x)]$. The resulting l.e.s. of maps is called the Barratt-Puppe sequence associated to the map f

$$X \to Y \to C(f) \to \Sigma X \to \Sigma Y \to \Sigma C(f) \to \Sigma^2 X \to \dots$$

Let $p : E \to B$ be a fibration of the pointed category and $\begin{cases} f : W \to E \\ g : W \to B \end{cases}$ a map for any pointed space W. If $g : W \to B$ is s.t. $pg \cong *$, then g is homotopic to a map that lands in the fiber $p^{-1}(*) = F$ of p over $*$. This gives the fiber sequence of pointed spaces $F \to E \to B$ is exact in the sense that for any well-pointed space W,

$$[W,F]_* \to [W,E]_* \to [W,B]_*$$

is exact. The fiber of p is the homotopy fiber of f, written $F(f) = \{(x,\omega) \in X \times Y_*^I : \omega(1) = f(x)\}$.

Lemma 2.1.6. *Let $p : E \to B$ be a fibration and $* \in B$. The natural map $p^{-1}(*) \to F(p)$ is a homotopy equivalence.*

pf. Regard $F(p)$ as the pullback of p along $PB \downarrow B$ in
$$\begin{tikzcd} F(p) \arrow[r] \arrow[d] & P(B) = B_*^I \arrow[d] \\ E \arrow[r] & B \end{tikzcd}.$$
The induced map on fibers is a homeomorphism. But PB is contractible, so the inclusion of the fiber of $F(p) \downarrow PB$ into $F(p)$ is a homotopy equivalence.
$\to$ continuing this way, the map $p(f) : F(f) \to X$ is a fibration with fiber ΩY and the Barratt-Puppe fibration sequence

$$Y \xleftarrow[f]{} X \xleftarrow[p]{} F(f) \xleftarrow[i]{} \Omega Y \xleftarrow[\Omega f]{} \Omega X \xleftarrow[\Omega p]{} \Omega F(f) \longleftarrow \dots$$

is exact.

Lemma 2.1.7. *Let $(X, A, *)$ be a pointed pair, and let F denote the homotopy fiber of the inclusion $A \to X$. For each $n \geq 1$, there exists a natural isomorphism*

$$\pi_n(X,A) \xrightarrow{\cong} \pi_{n-1}(F,*)$$

s.t.
$$\begin{tikzcd} \pi_n(X,*) \arrow[r] \arrow[d, "\cong"] & \pi(X,A,*) \arrow[d, "\cong"] \arrow[dr, "\partial"] & \\ & & \pi_{n-1}(A,*) \\ \pi_{n-1}(\Omega X,*) \arrow[r, "i"] & \pi_{n-1}(F,*) \arrow[ur, "p"] & \end{tikzcd}$$
commutes.

pf. Denote $\pi_1(X,A,*)$ the set of path components of the space of maps $(I,\partial I,J_1) \to (X,A,*)$. Then it is the space of paths in X from $*$ to some element of $a \in A$, i.e. $F(A \to X)$.
For any $n \geq 1, (I^n, \partial I^n, J_n) \to (X,A,*)$ is $\Omega^{n-1}F(A \to X)$ ($n = 2$, an element in the given space is a map $I^2 \to X$ which is the basepoint along the bottom and takes values in A along the top, i.e. a path in $F(A \to X)$ and also is the basepoint along the left and right edges, i.e. loop in $F(A \to X)$).
$\to$ the diagram commutes.

Lemma 2.1.8. *Let $p : E \to B$ be a fibration and $* \in E$. Write $*$ also the image of $*$ in B and let F be the fiber over $*$. Then $p_* : \pi_*(E,F,*) \to \pi_*(B,*)$ is an isomorphism.*

pf. $F(p) \to E$ is a fibration, so

$$hofib(F \to E) \cong hofib(F(p) \to E) \cong fib(F(p) \to E) = \Omega B$$

The resulting composite isomorphism $\pi_n(E,F) \to \pi_{n-1}(hofib(F \to E)) \to \pi_{n-1}(\Omega B)$ $\to \pi_n(B)$ is the isomorphism induced by p_*.

e.g. the fiber of $p : \mathscr{P}(K(\pi,n)) \to K(\pi,n)$ is $K(\pi,n-1)$. Thus there is a fibration $K(\pi,n-1) \to \mathscr{P}(K(\pi,n)) \to K(\pi,n)$, where $\mathscr{P}$ is a path-space.

2.2 Spectral Sequences

2.2.1 Obstruction Theory

For a relative CW complex (X,A), let $f : A \to Y$ be a map. Then we can build an extension $X \to Y$ skeleton by skeleton. Assume Y is path-connected and simple, so that $[S^n, Y] = \pi_n(Y,*)$.

- if $Y = \emptyset$, then so is A. There is an extension iff $X = \emptyset$.
- $Y \neq \emptyset$.
  we can extend the map $f : A \to Y$ to $X_0 \cup_A A \to Y$ by sending the new points anywhere in Y.
  assume we have constructed $f : X_n \to Y$, $n \geq 1$. Pick attaching maps for the $n+1$-cells, so that there exists a diagram
  $$\begin{tikzcd} \coprod_{i\in\Sigma_{n+1}} S^n \arrow[r, "\alpha"] \arrow[d] & X_n \arrow[r, "f"] \arrow[d] & Y \\ \coprod_{i\in\Sigma_{n+1}} D^n \arrow[r] & X_{n+1} \arrow[ur, dotted] & \end{tikzcd}.$$
  If the composite $S^n \to X_n \to Y$ is nullhomotopic for $i \in \Sigma_{n+1}$, the extension $X_{n+1} \to Y$ exists. Then this produces a map $\theta_f : \Sigma_{n+1} \to \pi_n(Y)$, a n-cochain $\theta_f \in C^{n+1}(X,A;\pi_n(Y))$.

- **θ_f is a cocycle in $C^{n+1}(X,A;\pi_n(Y))$.**
  pf. θ_f gives a map $H_{n+1}(X_{n+1},X_n) \to \pi_n(Y)$. A relative homotopy class is represented by a map $(I^q,\partial I^q,J_q) \to (X,A,*)$. The choice of orientation for $I^q/\partial I^q$ specifies a generator for $H_q(I^q,\partial I^q)$. Evaluation of H_n determines a map $h:\pi_q(X,A,*) \to H_q(X,A)$, the relative Hurewicz homomorphism. The characteristic maps in the cell structure for X give elements of $\pi_{n+1}(X_{n+1},X_n)$ that maps to the generators of $H_{n+1}(X_{n+1},X_n)$. Then

$$\begin{tikzcd}
\pi_{n+2}(X_{n+2},X_n) \arrow[r] \arrow[d, "\partial"] & H_{n+1}(X_{n+2},X_{n+1}) \arrow[d, "\partial"] \arrow[dd, bend left=60, "d"] \\
\pi_{n+1}(X_{n+1}) \arrow[r] \arrow[d] \arrow[dd, bend right=60, "0"'] & H_{n+1}(X_{n+1}) \arrow[d] \\
\pi_{n+1}(X_{n+1},X_n) \arrow[r] \arrow[d, "\partial"] & H_{n+1}(X_{n+1},X_n) \arrow[d, "\theta_f"] \\
\pi_n(X_n) \arrow[r, "f_*"] & \pi_n(Y)
\end{tikzcd}$$

  where the bottom square commutes by the definition of θ_f. Tracing around the left side goes through two successive maps in the homotopy l.e.s., and therefore sends these elements to 0.

$\to$ θ_f is called the obstruction cocycle associated to $f: X_n \to Y$.

- **$f|_{X_{n-1}}$ extends to X_{n+1} iff $[\theta_f] \in H^{n+1}(X,A;\pi_n(Y))$ is zero.**
  pf. Define a difference cochain δ associated to maps $f',f'': X_n \to Y$ with a homotopy from $f'|_{X_{n-1}}$ to $f''|_{X_{n-1}}$. Fix a cell structure on X. Give $X \times I$ the CW structure in which $(X \times I)_n = (X \times \partial I) \cup (X_{n-1} \times I)$. Each n-cell e in X produces in $X \times I$ $\left\{ \begin{array}{c} n+1\text{-cell } e \times I \\ n\text{-cells } e \times 0, e \times 1 \end{array} \right.$. Thus there is a map $- \times I : C_n(X) \to C_{n+1}(X \times I)$. This is not a chain map ($d(e \times I) = de \times I + (-1)^n(e \times 1 - e \times 0)$. This construction defines a map $\begin{array}{rcl} C^{n+1}(X;\pi_n(Y)) & \to & C^n(X;\pi_n(Y)) \\ c & \mapsto & (e \mapsto c(e \times I)) \end{array}$. Define a map
  $\begin{array}{rcc} g:(X \times I)_n & \to & Y \\ X_n \times 0, X_n \times 1 & \mapsto & f_0,f_1 \\ X_{n-1} \times I & \mapsto & \text{homotopy between } f_0,f_1 \end{array}$. Then we have the obstruction cocycle $\theta_g \in C^{n+1}(X \times I;\pi_n(Y))$ associated to the map g. Then the difference cochain $\delta = f' - f'' \in C^n(X,\pi_n(Y))$ is defined by $\delta(e) = \theta_g(e \times I)$. For any n-cell in X, $0 = (d\theta_g)(e \times I) = \theta_g(d(e \times I)) = \theta_g((de) \times I) \pm (\theta_g(e \times 0) - \theta_g(e \times 1)) = \delta(de) \pm (\theta_{f'}(e) - \theta_{f''}(e)) = d\delta e \pm (\theta_{f'}(e) - \theta_{f''}(e)) \Leftrightarrow d\delta = \pm(\theta_{f'} - \theta_{f''})$. So for a map $f: X_n \to Y$, the cohomology class of the

obstruction cocycle θ_f depends only on $f|_{X_{n-1}}$. In particular, if $f|_{X_{n-1}}$ does extend to a map from X_{n+1}, then this cohomology class vanishes. If $[\theta_{f'}] = 0 \in H^{n+1}(X;\pi_n(Y))$, pcik a nullhomotopy δ of $\theta_{f'}$ and pick f'' in such a way that δ is a difference cocycle between f' and f''. Then $\theta_{f''} = \theta_{f'} - d\delta = 0$ so f'' extends to X_{n+1}.

e.g. Obstruction in Fibration

Let $\pi : E \to B$ be a Serre fibration. Fix a basepoint $b_0 \in B$ and let $F_0 = \pi^{-1}(b_0)$ be the fiber over the basepoint. Let $h : \Sigma^k \to F_0$ be a singular cycle, i.e. Σ is an n dimensional simplicial complex and $H_n(\Sigma) = \mathbb{Z}$ with generator $[\Sigma]$. Fix a loop γ based at b_0. Then we can extend the map $\Sigma \to F_0$ to a map $\Sigma \times I \to E$ whose projection of B is the composition $\Sigma \times I \to I \to_\gamma B$. Also $H|_{\Sigma\times\{1\}} = \Sigma \to F_0$ representing a homology class, the action of γ on the class $h_*([\Sigma])$. Thus we can define the action of $\pi(B, b_0)$ on $H_*(F_0)$ s.t. $\pi_1(B, b_0) \times H_*(F_0) \to H_*(F_0)$. The action on the cohomology is the dual action.

Lemma 2.2.1. *Suppose $F \to P \to X$ is a fibration with X a simply connected CW complex and simply connected fiber. Suppose the fundamental group of the base acts trivially on the homotopy groups of the fiber. Then the obstruction to a section $X \to P$ lie in $H^*(X;\pi_{*-1}(F))$.*

$\to$if $F = K(\pi, n-1)$, there is only one obstruction to a section and it lies in $H^n(X;\pi_{n-1})$. This is classified by a map $X \to K(\pi, n)$. Here $F = K(\pi, n)$ is called a principal fibration if the action of the fundamental group of the base on the nontrivial homotopy group of the fiber is the trivial action.

2.2.2 Construction of Spectral Sequences

Definition 2.2.1. *A filtration of C is a family of subgroups $F_pC \subset C, p \in \mathbb{Z}$ s.t. if $p < q$, $F_pC \subset F_qC$. Assume $\begin{cases} \cup F_pC = C \\ \cap F_pC = 0 \end{cases}$. A filtration $\{F_pC\}$ of C is said to be finite if for some $m \le n$, $\begin{cases} F_pC = 0 & p < m \\ F_pC = C & p \ge n \end{cases}$. A filtration $\{F_pC\}$ is said to be positive if $F_pC = 0,\ \forall p < 0$.*

Definition 2.2.2. *A grading of C is a family of subgroups $C_r \subset C, r \in \mathbb{Z}$ s.t. $C = \oplus_r C_r$. Assume the grading is positive and finite, i.e. $C = \oplus_{r=0}^n C_r$ and $C_r = 0$, $\forall r < 0, r > n$. For a filtered group C, define the adjoint graded group $GrC = \oplus_r(F_rC/F_{r-1}C)$.*

$\to$ if an Abelian group C has a differential d, a filtration $\{F_pC\}$, and a grading $C = \oplus_r C_r$, then two structures are said to be compatible if $\forall p$, $F_pC = \oplus_r(F_pC \cap C_r)$ (they are compatible if every F_pC is generated by homogeneous elements).

Let C be a differential group with a differential d and a filtration $\{F_pC\}$ compatible with d. Assume the filtration is finite and positive. Since $d(F_pC) \subset F_pC$, d induces a differential $d_p^0 : F_pC/F_{p-1}C \to F_pC/F_{p-1}C$ and the direct sum of all d_p^0 becomes a homogeneous differential of degree 0, $d^0 : GrC \to GrC$. For $p, r \geq 0$,

$$E_p^r = \frac{F_pC \cap d^{-1}(F_{p-r}C)}{[F_{p-1}C \cap d^{-1}(F_{p-r}C)] + [F_pC \cap d(F_{p+r-1}C)]}$$

i.e. take elements of F_pC whose differential lies in a smaller group $F_{p-r}C$ and then factor out those in $F_{p-1}C$ and those with differential in F_{r+p-1}.
e.g.

- $r = 0$
$E_p^0 = \dfrac{F_pC \cap d^{-1}(F_pC)}{[F_{p-1}C \cap d^{-1}(F_pC)] + [F_pC \cap d(F_{p-1}C)]} = \dfrac{F_pC}{F_{p-1}C + d(F_{p-1}C)} = \dfrac{F_pC}{F_{p-1}C} \to$
$E^0 = GrC$

- $r = 1$
$E_p^1 = \dfrac{F_pC \cap d^{-1}(F_{p-1}C)}{[F_{p-1}C \cap d^{-1}(F_{p-1}C)] + [F_pC \cap d(F_pC)]} = \dfrac{F_pC \cap d^{-1}(F_{p-1}C)}{F_{p-1}C + d(F_pC)} = \dfrac{\ker d_p^0}{Imd_p^0} \to$
$E^1 = H(GrC, d^0)$

- $r = \infty$
$E_p^\infty = \dfrac{F_pC \cap \ker d}{[F_{p-1}C \cap \ker d] + [F_pC \cap Imd]} = \dfrac{F_pH(C)}{F_{p-1}H(C)} \to E^\infty = GrH(C).$

Define $d_p^r : E_p^r \to E_{p-r}^r$ as follows;
$\forall \alpha \in E_p^r$, $\exists a \in F_pC \cap d^{-1}(F_{p-r}C)$ a representative of α. Then $\begin{cases} da \in F_{p-r}C \\ dda = 0 \end{cases}$. Thus $da \in d^{-1}(0) \subset d^{-1}(F_{p-2r}C)$. Thus $da \in F_{p-r}C \cap d^{-1}(F_{p-2r}C)$. But E_{p-r}^r is the quotient of $F_{p-r}C \cap d^{-1}(F_{p-2r}C)$ over some subgroup; the class of da in E_{p-r}^R is taken for $d_p^r\alpha$.

Proposition 2.2.1. *1. d_p^r is well-defined.*

2. $d_{p-r}^r \circ d_p^r = 0$

3. $\ker d_p^r / Imd_{p+r}^r \cong E_p^{r+1}$.

pf.

- let a' be a different representative of α in $F_pC \cap d^{-1}(F_{p-r})$, i.e. $a' = a + b + c$, where
$\begin{cases} b \in F_{p-1}C \cap d^{-1}(F_{p-r}C) \\ c \in F_pC \cap d(F_{p+r-1}C) \end{cases}$. Then $da' = da + db + dc$, where

$\begin{cases} db \in F_{p-r}C \cap d(F_{p-1}C) = F_{p-r}C \cap d(F_{p-r+r-1}C) \\ dc = 0 \end{cases}$. Thus da' belongs to the same class in E^r_{p-r} as da given $F_{p-r}C \cap d(F_{p-r+r-1}C)$ is the part of denominator of E^r_{p-r}.

- $d^r_{p-r} \circ d^r_p \alpha = dda = 0$ in E^r_{p-2r} where a is a representative of α.
- let $\alpha \in E^r_p$ s.t. $d^r_p\alpha = 0$. Pick $a' \in F_pC \cap d^{-1}(F_{p-r}C)$, a representative of α. Since $d^r_p\alpha = 0$, da' belongs to the denominator of E^r_{p-r}, i.e. $da' = b + dc$, where $\begin{cases} c \in F_{p-1}C, dc \in F_{p-r}C \\ b \in F_{p-r-1}C, db \in F_{p-2r}C \end{cases}$. Put $a = a' - c$. Since $c \in F_{p-1}C \cap d^{-1}(F_{p-r}C)$, it is another representative of α. Since $da = da' - dc = b \in F_{p-r-1}C, \alpha$ has a representative $a \in F_pC \cap d^{-1}(F_{p-r}C)$ s.t. $da \in F_{p-r-1}C$. Then define a homomorphism $\ker d^r_p \to E^{r+1}_p$ as follows;
pick $\alpha \in \ker d^r_p$ and choose a representative $a \in F_pC$ with $da \in F_{p-r-1}C$. Then $a \in F_pC \cap d^{-1}(F_{p-r-1}C)$, i.e. a represents a certain element $\beta \in E^{r+1}_p$. Assign that element to α.
  - well-defined

    let $\tilde{a} \in F_pC$ be another representative of α with $d\tilde{a} \in F_{p-r-1}C$. Then $\tilde{a} - a \in [[F_{p-1}C \cap d^{-1}(F_{p-r}C)] + [F_pC \cap d(F_{p+r-1}C)]] \cap d^{-1}(F_{p-r-1}C)$. Therefore $\tilde{a} - a \in [F_{p-1}C \cap d^{-1}(F_{p-r-1}C)] + [F_pC \cap d(F_{p+r}C)]$. This implies $\tilde{a}, a$ present the same $\beta \in E^{r+1}_p$. So the homomorphism above is well-defined.
  - $\ker(\ker(d^r_p) \to E^{r+1}_p) = Imd^r_{p+r}$

    if $\alpha = d^r_{p+r}\beta$, $\exists a = db$, a representative of α, where $b \in F_{r+p}C$ represents β. But $db \in d(F_{p+r}C)$ which is a part of denominator of E^{r+1}_p. Thus a represents 0 in $E^{r+1}_p \to Imd^r_{p+r} \subseteq \ker[\ker d^r_p \to E^{r+1}_p]$. Conversely, take $\alpha \in \ker[\ker d^r_p \to E^{r+1}_p]$. Then α is representative by some $a \in [F_{p-1}C \cap d^{-1}(F_{p-r-1}C)] + [F_pC \cap d(F_{p+r}C)]$, i.e. $a = b + dc$, where $\begin{cases} b \in F_{p-1}C \cap d^{-1}(F_{p-r-1}C) \\ c \in F_{p+r}C \cap d^{-1}(F_pC) \end{cases}$. This c represents some element $\gamma \in E^r_{p+r}$ where $d^r_{p+r}\gamma$ is represented by $dc = a - b$. Since $b \in F_{p-1}C \cap d^{-1}(F_{p-r-1}C) \subset F_{p-1}C \cap d^{-1}(F_{p-r}C)$, b represents 0 in E^r_p and dc represents the same element of E^r_p as a. Thus $d^r_{p+r}\gamma = \alpha \to \ker[\ker d^r_p \to E^{r+1}_p] \subset Imd^r_{p+r}$.

Thus 3 follows.

Definition 2.2.3. *Given a differential group (C, d) with positive finite filtration $\{F_pC\}$ compatible with the differential, a spectral sequence associated with the* *above group is a sequence of differential graded groups*

$$E^r = \oplus_p E^r_p, d^r = \sum_p [d^r_p : E^r_p \to E^r_{p-r}]$$

$\to$ natural in the sense that if there exists another differential group (C', d') with a filtration compatible with a differential, then there exists a spectral sequence $\{E^{r'}, d^{r'}\}$ associated to this group and any homomorphism $C \to C'$ that is compatible with differentials and filtrations induces homomorphisms $E^r \to E^{r'}, 0 \le r \le \infty$ compatible with differentials and gradings and coinciding with the induced homomorphisms $GrH(C,d) \to GrH(C',d')$.

Assume additionally the group C has a grading $C = \oplus_m C_m$, compatible with filtration $F_pC = \oplus_m F_p C_m$ where $F_pC_m = F_pC \cap C_m$ and differential.

Let the differential d has degree i. Then for $p, r \ge 0, q \in \mathbb{Z}$, put

$$E^r_{pq} = \frac{F_pC_{p+q} \cap d^{-1}(F_{p-r}C_{p+q+i})}{[F_{p+i}C_{p+q} \cap d^{-1}(F_{p-r}C_{p+q+i})] + [F_pC_{p+q} \cap d(F_{p+r-i}C_{p+q-i})]}$$

$\to \begin{cases} p & \text{the filtration degree} \\ q & \text{the complementary degree} \end{cases}$ and $p+q$ is the full degree.

e.g. assume $i = -1$.

- $r = 0$
  $E^0_{pq} = \dfrac{F_pC_{p+q}}{F_{p-1}C_{p+q}}$
- $r = 1$
  $E^1_{pq} = \dfrac{\ker(\frac{F_pC_{p+q}}{F_{p-1}C_{p+q}} \to \frac{F_pC_{p+q+1}}{F_{p-1}C_{p+q-1}})}{Im(\frac{F_pC_{p+q+1}}{F_{p-1}C_{p+q+1}} \to \dfrac{F_pC_{p+q}}{F_{p-1}C_{p+q}})}$
- $r = \infty$
  $E^\infty_{pq} = \dfrac{F_pH_{p+q}(C)}{F_{p-1}H_{p+q}(C)}$

Define the differential $d^r_{pq} : E^r_{pq} \to E^r_{p-r,q+r+i}$ similarly;
$\forall \alpha \in E^r_{pq} \subset E^r$, $\exists a \in F_pC_{p+q}$ a representative of α with $da \in F_{p-r}C_{p+q-i}$. The class of da is taken for $d^r\alpha$. Then this differential reduces the filtration degree by r and reduces the full degree by i. Thus the above differential is defined.

$$E^{r+1}_{pq} = \frac{\ker[d^r_{pq} : E^r_{pq} \to E^r_{p-r,q+r+i}]}{Im[d^r_{p+r,q-r-i} : E^r_{p+r,q-r-i} \to E^r_{pq}]}$$

is a term of the spectral sequence.

2.2.3 Spectral Sequences in Topological Space

A positive finite filtration in a topological space X is a chain of subspaces $\emptyset = X_{-1} \subset X_0 \subset \cdots \subset X_n = X$. The full chain group has $\begin{cases} C = C_*(X;G) = \oplus_r C_r(X;G) & \text{a grading} \\ \partial : C_r(X;G) \to C_{r-1}(X;G) & \text{differential with degree } -1 \end{cases}$. Then the filtration $F_pC = C_*(X_p;G)$ is compatible with the grading. Then we can lead to a spectral sequence $\{E^r_{pq}, d^r_{pq} : E^r_{pq} \to E_{p-r,q+r-1}\}$ s.t.

$$E^r_{pq} = \frac{C_{p+q}(X_p;G) \cap \partial^{-1}(C_{p+q-1}(X_{p-r};G))}{[C_{p+q}(X_{p-1};G) \cap \partial^{-1}(C_{p+q-1}(X_{p-r};G))] + [C_{p+q}(X_p;G) \cap \partial(C_{p+q+1}(X_{p+r-1};G))]}$$

Let C be a cochain complex s.t. $C = C^*(X;G) = \oplus_r C^r(X;G)$ with $\begin{cases} \delta : C^r(X;G) \to C^{r+q}(X;G) & \text{the differential} \\ F^pC = C^*(X, X_{p-1};G) & \text{the filtration} \end{cases}$ where the differential degree is +1. Then the compatibility condition holds and we get a spectral sequence $E^{p,q}_r =$

$$\frac{C^{p+q}(X, X_{p-1};G) \cap \delta^{-1}(C^{p+q+1}(X, X_{p+r-1};G))}{[C^{p+q}(X, X_p;G) \cap \delta^{-1}(C^{p+q+1}(X, X_{p+r-1};G))] + [C^{p+q}(X, X_{p-1};G) \cap \delta(C^{p+q-1}(X, X_{p-r};G))]}$$

e.g. $\begin{cases} E^{pq}_0 = C^{p+q}(X_p, X_{p-1};G) & d^{pq}_0 = [\delta : C^{p+q}(X_p, X_{p-1};G) \to C^{p+q+1}(X_p, X_{p-1};G)] \\ E^{pq}_1 = H^{p+q}(X_p, X_{p-1};G) & d^{pq}_1 = [\delta^* : H^{p+q}(X_p, X_{p-1};G) \to H^{p+q+1}(X_{p+1}, X_p;G)] \end{cases}$

If there is a different space Y with a positive filtration $\{Y_p\}$ and a continuous map $f : X \to Y$ s.t. $f(X_p) \subset Y_p$, $\forall p$, then there are homomorphisms between homological and cohomological spectral sequences of the filtered spaces X and Y and these are compatible with the definition of spectral sequences above.

e.g. $\emptyset = X_{-1} \subset X_0 \subset X_1 \subset \cdots \subset X$ s.t. $X = \cup_p X_p$ and a weak topology(a set $F \subset X$ is closed in X iff $F \cap X_p$ is closed in X_p) has a spectral sequence of homology and cohomology.
$\to X$, a CW complex, and $X_p = sk_pX = X^p$ is its skeleton. If we filter X by its skeleton, we have $E^1_{pq} = H_{p+q}(X^p, X^{p-1}) = \begin{cases} \mathscr{C}_p(X) & q = 0 \\ 0 & q \neq 0 \end{cases}$ where $\mathscr{C}$ denote a cellular homology.

2.2.4 Spectral Sequences of Fibration

Definition 2.2.4. *Let $\xi = (E, B, F, p)$ be a locally trivial fibration with a CW base B with skeletons B^n. Consider a filtration $\{F_nE\}$ of the space E with $F_nE = p^{-1}(B^n)$. The spectral sequence corresponding to this filtration is called the spectral sequence of the fibration ξ.*

***Construction**

Let $x : I \to B$ be a path joining $b_0, b_1 \in B$. Then this path determines a homotopy equivalence $p^{-1}(b_0) \to p^{-1}(b_1)$ and therefore $H_n(p^{-1}(b_0);G) \leftarrow_{\cong} H_n(p^{-1}(b_1);G)$. If this homomorphism does not depend on the path for any n, G, the fibration ξ is called homologically simple.

- the fibration ξ is homologically simple.

Theorem 2.2.1. *1.* $E^1_{pq} = \mathscr{C}_p(B;H_q(F;G))$

2. $d^1_{pq} = [\partial_p \mathscr{C}_p(B;H_q(F;G)) \to \mathscr{C}_{p-1}(B;H_q(F;G))]$

3. $E^2_{pq} = H_p(B;H_q(F;G))$

pf. $E^1_{pq} = H_{p+q}(p^{-1}(B^p), p^{-1}(B^{p-1});G)$ by the previous subsection. Let $\{e_i : i \in I_p\}$ be the set of all p-dimensional cells of B and let $c_i \in e_i, d_i \subset e_i$ be the centre and a small ball around the centre of the cell e_i. Then the pair $(p^{-1}(B^p), p^{-1}(B^{p-1}))$ is homotopy equivalent to the pair $(p^{-1}(B^p), p^{-1}(B^p) \setminus \cup_i p^{-1}(c_i))$, and by the excision theorem,
$H_{p+q}(p^{-1}(B^p), p^{-1}(B^p) \setminus \cup_i p^{-1}(c_i);G) = H_{p+q}(\cup_i p^{-1}(d_i), \cup_i p^{-1}(d_i \setminus c_i);G)$
$= \oplus_i H_{p+q}(p^{-1}(d_i), p^{-1}(d_i \setminus c_i);G) = \oplus_i H_{p+q}(p^{-1}(d_i), p^{-1}(d_i \setminus Int d_i);G)$.
The fibration over the disk d_i is trivial with the fiber $F_i = p^{-1}(c_i) \cong F$ and the disk d_i may be regarded as a copy of the standard disk D^p.
$\to H_{p+q}(p^{-1}(d_i), p^{-1}(d_i \setminus Int d_i);G) = H_{p+q}(D^p \times F_i, S^{p-1} \times F_i;G) = H_q(F_i;G)$ by the Kunneth formula. This implies $E^1_{pq} = \oplus_i H_q(F_i;G) = \sum_i a_i e_i$, where $a_i \in H_q(F_i;G)$. Given ξ homologically simple, we can consider all a_i as belonging to one group $H_q(F;G)$, since if B is connected, there are paths joining every pair of points of B, each path establishes an isomorphism between homology groups of fibers over its endpoints and these do not depend on paths. So 1 follows. 2 and 3 follow from 1.

Then $E^2_{pq} = 0,\ \forall p < 0, q < 0$. If the base and the fiber are connected,
$\begin{cases} E^2_{p0} = H_p(B;H_0(F;G)) = H_p(B;G) \\ E^2_{0q} = H_0(B;H_q(F;G)) = H_q(F;G) \end{cases}$ and $E^2_{pq} = [H_p(B) \otimes H_q(F;G)] \oplus Tor(H_{p-1}(B), H_q(F;G))$.

e.g **Spectral Sequences for Serre Fibration**

Consider singular cubes $Cube_*(X)$ for a topological space X, instead of singular simplices. Define $Dcube_*(X)$ singular cube subcomplex generated by the degenerate singular cube. Then the quotient $Cube_*(X)/Dcube_*(X)$ is called the cubical chain complex, denoted $CC_*(X)$. Then this compute the singular homology of X.

Definition 2.2.5. *For each $0 \le p \le n$, define a map* $\begin{array}{rcl} proj_{p,n}: I^n & \to & I^p \\ (t_1,\dots,t_n) & \mapsto & (t_1,\dots,t_p) \end{array}$.
A singular $f: I^{p+q} \to E$ is in filtration level p if $\pi \circ f$ factors through $proj_{p,p+q}$,

i.e. $\begin{array}{ccc} I^{p+q} & \longrightarrow & E \\ \downarrow{\scriptstyle proj_{p,p+q}} & & \downarrow{\scriptstyle \pi} \\ I^p & \longrightarrow & B \end{array}$ *commutes.*

Then we get a filtration $0 = F_{-1}(Cube_n(E)) \subset F_0(Cube(E)) \subset \cdots \subset F_n(Cube_n(E)) = Cube_n(E)$. The boundary map is compatible with this filtration. Associated to any cube $\sigma: I^{p+q} \to E$ of filtration level p the unique map $\bar{\sigma}_p : I^p \to B$ s.t. $\pi \circ \sigma = \bar{\sigma}_p \circ proj_{p,p+q} : I^{p+q} \to I^p \to B$, i.e. $\begin{array}{ccc} I^{p+q} & \xrightarrow{\sigma} & E \\ {\scriptstyle proj_{p,p+q}}\downarrow & & \downarrow{\scriptstyle \pi} \\ I^p & \xrightarrow{\bar{\sigma}_p} & B \end{array}$ commutes.
$\to$ if σ is the singular cube of filtration level p, then it is in filtration level $p-1$ iff $\pi \circ \sigma = \bar{\sigma}_{p-1} \circ proj_{p-1,p+q}$. Such a map $\bar{\sigma}_{p-1}$ exists iff $\bar{\sigma}_p = \bar{\sigma}_{p-1} \circ proj_{p-1,p}$ for some map $\bar{\sigma}_{p-1}$, which is true iff $\bar{\sigma}_p$ is degenerate. Then we have a decomposition $F_p(\mathscr{C}_*(E))/F_{p-1}(\mathscr{C}_*(E)) = \oplus_{\tau_p}(F_p(\mathscr{C}_*(E))/F_{p-1}(\mathscr{C}_*(E)))_{\tau_p}$ where τ_p ranges over the non-degenerate p-cubes in B and the summand indexed by τ_p is the subgroup generated by the cube σ in E of the filtration level p with $\bar{\sigma}_p = \tau_p$. Then for $\sigma: I^n \to E$ of filtration level p with $\bar{\sigma}_p : I^p \to B$ nondegenerate, $\partial\sigma = \sum_{i=0}^n (-1)^i(\{t_1,\dots,t_{i-1},1,t_{i+1},\dots,t_n\} - \{t_1,\dots,t_{i-1},0,t_{i+1},\dots,t_n\})$. The terms in the summation indexed by $0 \le i \le p$ are in filtration level $p-1$ and therefore are 0 in $E^0_{p,q-1}$. The terms indexed by $p+1 \le i \le n$ are of filtration level p and their projection to B factors through $\bar{\sigma}_p$. Thus the differential d_0 preserves the decomposition. Then for each nondegenerate map $\tau_p : I^p \to B$, let $F_{\tau_p} = \pi^{-1}(\tau_p(0,\dots,0))$ be the fiber over the initial corner of τ_p.
For each such τ_p, define a map
$\begin{array}{rcl} R_{\tau_p} : (F_p(Cube_{p+q}(E))/F_{p-1}(Cube_{p+q}(E)))_{\tau_p} & \to & Cube_q(F_{\tau_p}) \\ \sigma & \mapsto & \sigma|_{\{0,\dots,0\}\times I^q} \end{array}$. Then for any σ^n degenerate, s.t. $\pi \circ \sigma^n$ factors through τ_p for a nondegenerate map $\tau_p : I^p \to B$, $R_{\tau_p}(\sigma)$ is the restriction of σ to a positive dimensional cube $\{0,\dots,0\} \times I^{n-p}$. Since σ is degenerate, so is its restriction to $\{0,\dots,0\} \times I^{n-p}$, i.e. $R_{\tau_p}(\sigma)$ is a degenerate cube in F_{τ_p}. For nondegenerate $\tau_p : I^p \to B$,
$d_0(\sigma^n) = (-1)^p \sum_{i=0}^{n-p}((t_1,\dots,t_{p+i},1,t_{p+i+2},\dots,t_n) - (t_1,\dots,t_{p+i},0,t_{p+i+2},\dots,t_n)) = (-1)^p \partial R_{\tau_p}(\sigma)$. Thus d_0 preserves the decomposition $\oplus_{\tau_p}(F_p(CC_*(E))/F_{p-1}(CC_*(E)))_{\tau_p}$ of E^0_{p*} and hence we have a decomposition $E^1_{pq} = \oplus_{\tau_p}(E^1_{pq})_{\tau_p}$ and for each nondegenerate cup $\tau_p : I^p \to B$, $(R_{\tau_p})_*(E^1_{pq})_{\tau_p} \to H_q(F_{\tau_p})$.

Define a map $J_{\tau_p}: CC_*(F_{\tau_p}) \to (F_p(CC_*(E))/F_{p-1}(CC_*(E)))_{\tau_p}$ as follows; for $I^r \times \{0,\dots,0\}$, we can extend any map $I^0 \to F_{\tau_p}$ to a map $I^p \to E$ projecting to τ_p induction on r by the Serre homotopy extension. Suppose $\forall s < q$, we have extension of all maps of $I^s \to F_{\tau_p}$ to maps $I^p \times I^s \to E$ covering τ_p compatible with the boundary maps. Let $I^q \to F_{\tau_p}$ be given. Then we have an extension $((0,\dots,0)\times I^q) \cup (I^p \times \partial I^q) \to E$ projecting to τ_p by the inductive hypothesis. Suppose we have an extension of this map over $I^r \times I^q$ so that on this subset, the map projects to $\tau_p|_{I^r}$. Then we have a map defined on $(I^r \times I^q) \cup (I^{r+1} \times \partial I^q)$ projecting to τ_p by the induction on r. Since there is a homeomorphism $((I^r \times I^q)\cup(I^{r+1}\times\partial I^q))\times I \to I^{r+q}\times I^q$ which is the inclusion on $((I^r \times I^q)\cup(I^{r+1}\times\partial I^q))\times\{0\}$.

$\to$ it is a chain map when we use the differential

$$\begin{cases} (-1)^p\partial & CC_*(F_{\tau_p}) \\ d_0 & (F_p(CC_*(E))/F_{p-1}(CC_*(E)))_{\tau_p} \end{cases}.$$

$R_{\tau_p} \circ J_{\tau_p}$ is the identity map of $CC_*(F_{\tau_p})$ and $J_{\tau_p} \circ R_{\tau_p} \sim Id$, i.e. R_{τ_p}, J_{τ_p} induce inverse isomorphism $(E^1_{pq})_{\tau_p} \to H_q(F_{\tau_p})$. Since the action of the fundamental group of B on the homology of the fiber is trivial, there is a canonical identification $\iota_{\tau_p} \in H_*(F_{\tau_p})$ with $H_*(F_0)$ where F_0 is the fiber over a chosen basepoint. For each nondegenerate $\sigma: I^p \to B$ we have an isomorphism

$$\begin{array}{ccc} E^1_{pq} & \to & CC_p(B)\otimes H_q(F_0) \\ \sigma & \mapsto & \tau_p \otimes \iota_{\tau_p} R_{\tau_p}(\sigma) \end{array}.$$

Corollary 2.2.1. *Let $p: E \to B$ be a Serre fibration with B path-connected with trivial action of $\pi_1(B,b_0)$ on the homology of the fiber $F_0 = p^{-1}(b_0)$. The E^2-term of the Serre spectral sequence for this Serre fibration is identified with*

$$E^2_{p,q} = H_p(B; H_q(F_0))$$

pf. Let ξ be a q-cycle in F_{τ_p} and consider $J_{\tau_p}(\xi)$. Also let $[\xi]_0 = \iota_{\tau_p}([\xi])$ $\in H_*(F_0)$. Then $[J_{\tau_p}(\xi)] \in E^1_{p,q}$ corresponds to $\tau_p \otimes [\xi]_0$ in $CC_*(B)\otimes H_*(F_0)$ under the isomorphism above. Denote $\partial\tau_p = \sum_a (-1)^{\epsilon(a)}(\tau_p)_a$ where a indexes the codimension 1-faces of I^r. Since ξ is a cycle $\partial J_{\tau_p}(\xi)$ is a cycle projecting to $\partial\tau_p$ and $d_1([J_{\tau_p}\xi]) = \sum_a(-1)^{\epsilon(a)}(\tau_p)_a \otimes [J_{\tau_p}(\xi)|_{F_{(\tau_p)_a}}]$. For any corner c on I^p, let γ be a path consisting of edges of I^p connecting $(0,\dots,0)$ to c. The restriction of $J_{\tau_p}(\xi)$ to the preimage under the projection mapping of the image under τ_p of this 1-chain produces a $(q+1)$-chain whose boundary is the difference of the cycle ξ in F_{τ_p} and the cycle in $F_c = p^{-1}(c)$ that is the intersection of $J_{\tau_p}(\xi)$ with F_c. Thus the resulting cycles in F_{τ_p} and F_c represent the same homology class under the canonical identifications of the homology of each fiber with $H_*(F_0)$. Thus

$$d_1(\tau_p \otimes [\xi]_0) = d_1([J_{\tau_p}(\xi)]) = \sum_a (-1)^{\epsilon(a)}(\tau_p)_a \otimes [\xi]_0 = \partial\tau_p \otimes [\xi]_0$$

where ∂ is the boundary map for $CC_*(B)$. Then this is exactly the boundary map of the chain complex $CC_*(B) \otimes H_*(F_0)$.

$\Leftrightarrow$ by duality, the similar arguments for the cohomology spectral sequence hold.

Proposition 2.2.2. *With the hypotheses in the previous corollary, the dual spectral sequence for $CC^*(E)$ has*

$$\begin{cases} E_1^{p,q}(CC^*(E)) = \hom(CC^p(B), H^q(F_0)) \\ E_2^{p,q} = H^p(B; H^q(F)) \end{cases}$$

- When B is not simply-connected

  Still the fibration defines functors $\begin{array}{rcl} H_q(p^{-1}(-)) : \Pi_1(B) & \to & \mathbf{Ab} \\ b & \mapsto & H_q(p^{-1}(b)) \end{array}$ which determines a local coefficient system. Assume X is path-connected and admits a universal cover $\tilde{X}$. Pick a basepoint $*$. Giving a local coefficient system is the same as giving a $\mathbb{Z}[\pi_1(X,*)]$-module. Write M for both. The fundamental group acts on $\tilde{X}$ and on its singular chain complex. Then

  $$H_*(X;M) = H_*(CC_*(\tilde{X}) \otimes_{\mathbb{Z}[\pi_1(X,*)]} M),\ H^*(X;M) = H^*(\hom_{\mathbb{Z}[\pi_1(X,*)]}(CC_*(\tilde{X}), M)$$

  Theorem 2.2.2. *Let $p : E \to B$ be a Serre fibration, R a commutative ring, and M an R-module. Then there is a first quadrant spectral sequence of R-modules with*

  $$E^2_{p,q} = H_p(B; H_q(p^{-1}(-); M))$$

  which converges to $H_*(E;M)$. It is natural from E^2 for maps of fibrations.

Chapter 3

Characteristic Classes

Let $\mathbb{R}^n$ denote the coordinate space consisting of all n-tuples $x = (x_1, \ldots, x_n)$ of real numbers. For $n = 0$, it is considered as $\mathbb{R}^0$ consisting of a single point.

Definition 3.0.1. *A function defined on an open set $U \subset \mathbb{R}^n$ with values in $\mathbb{R}^k$ is smooth if its partial derivatives of all orders exist and are continuous.*

Let A be any index set and let $\mathbb{R}^A$ denote the vector space consisting of all functions x from A to $\mathbb{R}$. Topologize this space $\mathbb{R}^A$ as a cartesian product of copies of $\mathbb{R}$. For any subset $M \subset \mathbb{R}^A$, we give M the relative topology, and therefore $f : Y \to M \subset \mathbb{R}^A$ is continuous iff each of the associated functions $f_\alpha : Y \to \mathbb{R}$ is continuous.

Definition 3.0.2. *For $U \subset \mathbb{R}^n$, a function $f : U \to M \subset \mathbb{R}^A$ is said to be smooth if each of the associated functions $f_\alpha : U \to \mathbb{R}$ is smooth. If f is smooth, then the partial derivative $\partial f / \partial u_i$ can be defined as the smooth function $U \to \mathbb{R}^A$ whose ith coordinate is $\partial f_i / \partial u_j$, $\forall j = 1, \ldots, n$.*

Definition 3.0.3. *A subset $M \subset \mathbb{R}^A$ is a smooth manifold of dimension $n \geq 0$ if for each $x \in M$, $\exists$ a smooth function $\begin{array}{ccc} h : U & \to & \mathbb{R}^A \\ U & \mapsto & V \end{array}$ where V is an open nhood of x in M, s.t. for each $u \in U$, $[\partial h_i(u) / \partial u_j]$ has rank n.*

$\to$ the image $h(U) = V$ is called a coordinate nhood in M and the triple (U, V, h) will be called a local parametrisation of M.

Lemma 3.0.1. *Let $\left\{ \begin{array}{c} (U, V, h) \\ (U', V', h') \end{array} \right.$ be two local parametrisations of M s.t. $V \cap V'$ is nonvacuous. Then the correspondence $u' \mapsto h^{-1}(h'(u'))$ defines a smooth mapping from the open set $(h')^{-1}(V \cap V') \subset \mathbb{R}^n$ to the open set $h^{-1}(V \cap V') \subset \mathbb{R}^n$.*

pf. Let $\bar{x} = h(\bar{u}) = h'(\bar{u'}) \in V \cap V'$. Choose indices $a_1, \ldots, a_n \in A$ so that the $n \times n$ matrix $[\partial h_{a_i}/\partial u_j]$ evaluated at $\bar{u}$ is nonsingular. Then by the inverse function theorem, $\exists u_j = f_j(h_{a_1}(u), \ldots, h_{a_n}(u))$ a smooth function for u in some nhood of $\bar{u}$. Writing these in vector notation gives the function $u' \mapsto h^{-1}h'(u') = f(h'_{a_1}(u'), \ldots, h'_{a_n}(u'))$ smooth throughout some nhood of u'.

Let $\bar{x} \in M$ fixed, $(-\epsilon, \epsilon)$ denote the set of real numbers t within that interval. A smooth path through $\bar{x}$ in M is a smooth function $p : (-\epsilon, \epsilon) \to M \subset \mathbb{R}^A$ with $p(0) = \bar{x}$.

Definition 3.0.4. *A vector $v \in \mathbb{R}^A$ is tangent to M at x if v can be expressed as the velocity vector of some smooth path through x in M. The set of all such tangent vectors is called the tangent space of M at x, denoted DM_x.*

Lemma 3.0.2. *A vector $v \in \mathbb{R}^A$ is tangent to M at $\bar{x}$ iff v can be expressed as a linear combination of the vectors*

$$\frac{\partial h}{\partial u_1}(\bar{u}), \ldots, \frac{\partial h}{\partial u_n}(\bar{u})$$

Thus $DM_{\bar{x}}$ is an n-dimensional vector space over the real numbers.

$\to$ the tangent manifold of M is defined to be the subspace $DM \subset M \times \mathbb{R}^A$ consisting of all pairs (x, v) where $\left\{ \begin{array}{c} x \in M \\ v \in DM_x \end{array} \right.$.

Consider two smooth manifolds $\left\{ \begin{array}{c} M \subset \mathbb{R}^A \\ N \subset \mathbb{R}^B \end{array} \right.$ and a function $f : M \to N$. Let $\bar{x}$ be a point of M and (U, V, h) a local parametrisation of M with $\bar{x} = h(\bar{u})$.

Definition 3.0.5. *The function h is said to be smooth at $\bar{x}$ if the composition*

$$f \circ h : U \to N \subset \mathbb{R}^B$$

is smooth throughout some nhood of $\bar{u}$.

Definition 3.0.6. *The function $f : M \to N$ is smooth if $\forall x \in M$, it is smooth. A function $f : M \to N$ is called a diffeomorphism if f is one to one and if both f and the inverse function $f^{-1} : N \to M$ are smooth.*

Lemma 3.0.3. *The identity map of M is always smooth. Furthermore, the composition of two smooth maps $M \xrightarrow{g} M' \xrightarrow{f} M''$ is smooth.*

pf. The composition with identity map $Id \circ h = h : U \to N \subset \mathbb{R}^B$ is smooth for all $x \in M$ in some nhood u. So the lemma follows.
$(g \circ f) \circ h = g \circ (f \circ h)$ by associativity and $f \circ h$ is smooth. We may consider $f \circ h$ as another local parametrisation and by the smoothness of g, $g \circ (f \circ h)$ is smooth.

Any map $f : M \to N$ which is smooth at x determines a linear map $\begin{array}{rcl} Df_X : DM_X & \to & DN_{f(X)} \\ v = (\frac{dp}{dt})|_{t=0} & \mapsto & (\frac{d(f\circ p)}{dt})|_{t=0} \end{array}$.
Then this does not depend on the choice of p and Df_X is a linear mapping ($Df_X \sum c_i \partial h/\partial u_i = \sum c_i \partial(f \circ h)/\partial u_i$)

Definition 3.0.7. *The linear transformation Df_X is called the derivative or the Jacobian of f at x.*

Suppose $f : M \to N$ is smooth everywhere. Then we can define the function

$$\begin{array}{rcl} Df : DM & \to & DN \\ (x,v) & \mapsto & (f(x), Df_X(v)) \end{array}$$

Lemma 3.0.4. *D is a functor from the category of smooth manifolds and smooth maps into itself.*

pf. The functoriality follows from the associativity of composition of maps.

It satisfies the Leibniz's Rule ($D(fg)_X = f(x)Dg_X + g(x)Df_X$)

Definition 3.0.8. *For any tangent vector $v \in DM_X$, the real number $Df_X(v)$ is called the directional derivative of the real valued function f at x in the direction v.*

→ given fixed (x,v), letting f vary over the vector space $C^\infty(M,\mathbb{R})$ gives a linear differential operator $\begin{array}{rcl} X : C^\infty(M,\mathbb{R}) & \to & \mathbb{R} \\ f & \mapsto & Df_x(v) \end{array}$, which satisfies the Leibniz's rule.

Given a smooth manifold $M \subset \mathbb{R}^A$, let $F = C^\infty(M,\mathbb{R})$ denote the set of all smooth functions from M to the real numbers $\mathbb{R}$. Define the embedding $\begin{array}{rcl} i : M & \to & \mathbb{R}^F \\ x & \mapsto & (f(x))_{f\in F} \end{array}$.
Let M_1 denote the image $i(M) \subset \mathbb{R}^F$.

Lemma 3.0.5. *The image M_1 is a smooth manifold in $\mathbb{R}^F$ and the canonical map $i : M \to M_1$ is a diffeomorphism.*

→ any smooth manifold has a canonical embedding in an associated coordinate space.

Let M be a set and let F be a collection of real valued functions on M which separates points ($\forall x \neq y \in M, \exists f \in F$ s.t. $f(x) \neq f(y)$). Then M can be identified with its image under the canonical imbedding $i : M \to \mathbb{R}^F$.

Definition 3.0.9. *The collection F is a smoothness structure on M if the subset $i(M) \subset \mathbb{R}^F$ is a smooth manifold, and if F is precisely the set of all smooth real valued functions on this smooth manifold.*

Let $\begin{cases} M_1 \subset \mathbb{R}^A \\ M_2 \subset \mathbb{R}^B \end{cases}$ **be smooth manifolds. Then $M_1 \times M_2 \subset \mathbb{R}^A \times \mathbb{R}^B$ is a smooth manifold, and the tangent manifold $D(M_1 \times M_2)$ is canonically diffeomorphic to the product $DM_1 \times DM_2$.**

$\rightarrow$ given the product topology, we get the same local coordinate system and projection map continuous. Thus $M_1 \times M_2$ is a smooth manifold. Continuity of the projection maps imply the continuity of the product map, which gives a diffeomorphism between $D(M_1 \times M_2) \rightarrow DM_1 \times DM_2$.

Let $\mathbb{R}P^n$ denote the set of all lines through the origin in the coordinate space $\mathbb{R}^{n+1}$. Define a function $\begin{array}{ccc} q : \mathbb{R}^{n+1} \setminus \{0\} & \rightarrow & \mathbb{R}P^n \\ x & \mapsto & \mathbb{R}x \end{array}$**. Let F denote the set of all functions $f : \mathbb{R}P^n \rightarrow \mathbb{R}$ s.t. $f \circ q$ is smooth. Then F is a smoothness structure on $\mathbb{R}P^n$.**

pf. By the construction F is the set of all smooth functions from M to the real number $\mathbb{R}$. Then the image of the embedding $i : M \rightarrow \mathbb{R}^F$, $i(M)$ is a smooth manifold of $\mathbb{R}^F$ and $i : M \rightarrow i(M)$ is a diffeomorphism by the previous lemma. For any $x \neq y \in M$, $i(x) \neq i(y) \in i(M)$,i.e. two different points are separated. Thus F is a smooth structure of $\mathbb{R}P^n$.

3.1 Vector Bundles

3.1.1 Basic Definitions and Constructions

Let B be a topological space. Define the category of spaces over B, denoted **Top**/B, consisting of $\begin{cases} \text{objects} & \xi : E \downarrow B \\ \text{arrows} & f \text{ homomorphisms} \end{cases}$, where
$$\begin{tikzcd} E' \arrow[rr, "f"] \arrow[dr, "p'"'] & & E \arrow[dl, "p"] \\ & B & \end{tikzcd}$$
commutes. This category has products, given by the fiber product over $B \rightarrow E' \times_B E = \{(e', e) : p'e' = pe\} \subseteq E' \times E$. Define an operation called addition $E \times_B E \rightarrow E$ with zero section $0 : B \rightarrow E$.

Definition 3.1.1. *A real vector bundle ξ over B consists of the followings*

1. *a topological space $E = E(\xi)$ called the total space*
2. *a continuous map $p : E \rightarrow B$ called the projection map*
3. *$\forall b \in B$, the structure of a vector space over the real numbers in the set $p^{-1}(b)$*

Condition of Local Triviality : $\forall b \in B$, $\exists U \subset B$, $n \in \mathbb{N} \cup \{0\}$ and a homeomorphism $h : U \times \mathbb{R}^n \to p^{-1}(U)$ so that $\forall b \in U$, $x \mapsto h(b,x)$ defines an isomorphism between the vector space $\mathbb{R}^n$ and the vector space $p^{-1}(b)$.
$\to (U,h)$ in the above is called a local coordinate system for ξ about b.

Definition 3.1.2. *When B,E smooth manifolds and p a smooth map, if $\forall b \in B$, $\exists$ local coordinate system (U,h) with $b \in U$, h a diffeomorphism, then it is a smooth vector bundle.*

Definition 3.1.3. *$\xi \sim \eta$ if there exists a homeomorphism $f : E(\xi) \to E(\eta)$ between the total spaces which maps each vector space $F_b(\xi)$ isomorphically onto the corresponding vector space $F_b(\eta)$.*

e.g.

- the trivial bundle with
  $\begin{cases} \text{the total space} & B \times \mathbb{R}^n \\ \text{the projection map} & p(b,x) = b \\ \text{vector space structure} & t_1(b,x_1) + t_2(b,x_2) = (b, t_1x_1 + t_2x_2) \end{cases}$, denoted ϵ_B^n.
- the tangent bundle τ_M of a smooth manifold M
  $\begin{cases} \text{the total space} & DM \\ \text{the projection map} & \pi : (x,v) \mapsto x \\ \text{vector space structure} & t_1(x,v_1) + t_2(x,v_2) = (x, t_1v_1 + t_2v_2) \end{cases}$.
  When τ_M is a trivial bundle, the manifold M is called parallelisable.
- the normal bundle ν of a smooth manifold $M \subset \mathbb{R}^n$
  $\begin{cases} \text{the total space} & E \subset M \times \mathbb{R}^n \\ \text{the projection map} & p : E \to M \\ \text{vector structure} & \text{defined as above} \end{cases}$ where $(x,\nu) \in E$ s.t. ν is orthogonal to the tangent space DM_X.
- the real projective space $\mathbb{P}^n$
  $\begin{cases} \text{the total space} & E(\gamma_n^1) \subset \mathbb{P}^n \times \mathbb{R}^{n+1} \\ \text{the projection map} & p(\{\pm x\}, v) = \{\pm x\} \end{cases}$ where $(\{\pm x\}, v) \in E(\gamma_n^1)$, v is a multiple of x.

Lemma 3.1.1. *Let ξ,η be vector bundles over B and let $f : E(\xi) \to E(\eta)$ be a continuous function which maps each vector space $F_b(\xi)$ isomorphically onto the corresponding vector space $F_b(\eta)$. Then f is necessarily a homeomorphism, and hence $\eta \sim \xi$.*

pf. Given any point $b_0 \in B$, choose local coordinate system $\begin{cases} (U,g) & \xi \\ (V,h) & \eta \end{cases}$ s.t. $b_0 \in U \cap V$. Setting $h^{-1}(f(g(b,x)) = (b,y)$, $y = (y_1, \ldots, y_n)$ can be expressed in the form

$y_i = \sum_j f_{ij}(b)x_j$ where $[f_{ij}(b)]$ denote a nonsingular matrix of real numbers. Also the entries of $f_{ij}(b)$ depend continuously on b. Let $[F_{ji}(b)]$ denote the inverse matrix. Then $g^{-1} \circ f^{-1} \circ h(b, y) = (b, x)$, where
$x_j = \sum_i F_{ji}(b)y_i$. Since the numbers $F_{ji}(b)$ depend continuously on $[f_{ij}(b)]$, they continuously depend on b. Thus $g^{-1} \circ f^{-1} \circ h$ is continuous and the lemma follows.

Definition 3.1.4. *A cross-section of a vector bundle ξ with base space B is a continuous function* $\begin{array}{rcl} s: B & \to & E(\xi) \\ b & \mapsto & F_b(\xi) \end{array}$. *Such a cross-section is nowhere zero if $s(b)$ is a nonzero vector of $F_b(\xi)$ for each b. The cross-sections $s_1, \ldots, s_n$ are nowhere dependent if $\forall b \in B$, the vectors $s_1(b), \ldots, s_n(b)$ are L.I.*

Theorem 3.1.1. *An $\mathbb{R}^n$-bundle ξ is trivial iff ξ admits n cross-sections $s_1, \ldots, s_n$ which are nowhere dependent.*

Definition 3.1.5. *A vector bundle over B is a space E over B that is locally trivial, i.e. $\forall b \in B$, $\exists$ a nhood over which E is isomorphic to a trivial bundle. Also its fiber vector spaces are of finite dimension.*

Definition 3.1.6. *A metric on a vector bundle is a continuous choice of inner products on fibers.*

Lemma 3.1.2. *Any numerable vector bundle ξ admits a metric.*

pf. Pick a trivialising open cover $\mathscr{U}$ for ξ, and for each $U \in \mathscr{U}$, an isomorphism $\xi|_U \cong U \times V_U$. Pick an inner product g_U on each of the vector space V_U, and pick a partition of unity subordinate to $\mathscr{U}$, i.e. functions $\phi_U : U \to [0,1]$ s.t. the preimage of $(0,1]$ is U and $\sum_{x \in U} \phi_U(x) = 1$. Then $g = \sum_U \phi_U g_U$ is a metric on ξ.

e.g.

- **Euclidean Vector Bundle**

  Definition 3.1.7. *A Euclidean vector space is a real vector space V together with a positive definite quadratic function $\mu : V \to \mathbb{R}$. The real number $v \cdot w$ is called the inner product of the vectors v, w.*

  Definition 3.1.8. *A Euclidean vector bundle is a real vector bundle ξ together with a continuous function $\mu : E(\xi) \to \mathbb{R}$ s.t. the restriction of μ to each fiber of ξ is positive definite and quadratic. The function μ is called a Euclidean metric on the vector bundle.*

  e.g. for the tangent bundle τ_M of a smooth manifold M, a Euclidean metric $\mu : DM \to \mathbb{R}$ is called a Riemannian metric and M together with μ is called a Riemannian manifold. (the trivial bundle ϵ_B^n can be given the Euclidean metric $\mu(b,x) = x_1^2 + \cdots + x_n^2$. Since the tangent bundle of $\mathbb{R}^n$ is trivial, the smooth manifold $\mathbb{R}^n$ possesses a standard Riemannian metric. Additionally for any smooth manifold $M \subset \mathbb{R}^n$, $DM \xrightarrow{in} D\mathbb{R}^n \xrightarrow{\mu} \mathbb{R}$ makes M into a Riemannian manifold.

Lemma 3.1.3. *Let ξ be a trivial vector bundle of dimension n over B and let μ be any Euclidean metric on ξ. Then there exist n cross-sections $s_1,\ldots,s_n$ of ξ which are normal and orthogonal in the sense that*

$$s_i(b)\cdot s_j(b) = \delta_{ij}, \ \forall b \in B$$

3.1.2 Derivation of New Vector Bundles out of Old

- Restricting a bundle to a subset of the base space

Let ξ be a vector bundle with projection $p : E \to B$ and let $B' \subset B$. Setting $\begin{cases} E' = p^{-1}(B') \\ p' : E' \to B' = p|_{E'} \end{cases}$. Then we obtain a new vector bundle, denoted $\xi|_{B'}$, called the restriction of ξ to B'.

- Induced bundles

Let ξ be a vector bundle with projection $p : E \to B$ and let B_1 be an arbitrary topological space. Given any map $f : B_1 \to B$, one can construct the induced bundle $f^*\xi$ over B_1 s.t.

$$\begin{cases} \text{the total space } E_1 \subset B_1 \times E & \\ \text{the projection map} & \begin{array}{rcl} p_1 : E_1 & \to & B_1 \\ (b,e) & \mapsto & b \end{array} \\ \text{vector space structure} & t_1(b,e_1) + t_2(b,e_2) = (b, t_1e_1 + t_2e_2) \end{cases} \quad \text{s.t.} \quad \begin{array}{ccc} E_1 & \xrightarrow{\hat{f}} & E \\ \downarrow{p_1} & & \downarrow{p} \\ B_1 & \xrightarrow{f} & B \end{array}$$

commutes with $\hat{f}(b,e) = e$.
$\to \hat{f}$ carries each vector space $F_b(f^*\xi)$ isomorphically onto the vector space $F_{f(b)}(\xi)$.
If (U,h) is a local coordinate system for ξ, set $U_1 = f^{-1}(U)$ and define $\begin{array}{rcl} h_1 : U_1 \times \mathbb{R}^n & \to & p_1^{-1}(U_1) \\ (b,x) & \mapsto & (b, h(f(b),x)) \end{array}$. Then (U_1,h_1) is a local coordinate system for $f^*\xi$.

- Cartesian products

Given two vector bundles ξ_1,ξ_2, with projection maps $p_i : E_i \to B_i$, the Cartesian product $\xi_1 \times \xi_2$ is defined to be the bundle with projection map $p_1 \times p_2$:

$E_1 \times E_2 \to B_1 \times B_2$ where each fiber $(p_1 \times p_2)^{-1}(b_1, b_2) = F_{b_1}(\xi_1) \times F_{b_2}(\xi_2)$ is given the similar vector space structure.

- Whitney sums

Given two vector bundles ξ_1, ξ_2 over the same base space B, let $\Delta : B \to B \times B$ be the diagonal embedding. The bundle $\Delta^*(\xi_1 \times \xi_2)$ over B is called the Whitney sum of ξ_1, ξ_2, denoted $\xi_1 \oplus \xi_2$. Then each fiber $F_b(\xi_1 \oplus \xi_2)$ is canonically isomorphic to the direct sum $F_b(\xi_1) \oplus F_b(\xi_2)$.

Lemma 3.1.4. *Let ξ_1, ξ_2 be subbundles of η s.t. each vector space $F_b(\eta)$ is equal to the direct sum of the subspaces $F_b(\xi_1), F_b(\xi_2)$. Then $\eta \sim \xi_1 \oplus \xi_2$.*

pf. Define $\begin{array}{rcl} f : E(\xi_1 \oplus \xi_2) & \to & E(\eta) \\ (b, e_1, e_2) & \mapsto & e_1 + e_2 \end{array}$. Then it is an isomorphism by the lemma 3.1.1.

$$\begin{array}{ccc} E \oplus E & \longrightarrow & E \times E \\ \downarrow & & \downarrow \\ B & \xrightarrow{\Delta} & B \times B \end{array}$$ the Whitney sum is the pullback in the diagram.

- Orthogonal complements

Given a subbundle $\xi \subset \eta$, let $F_b(\xi^\perp)$ denote the subspace of $F_b(\eta)$ consisting of all vectors v s.t. $v \cdot w = 0$, $\forall w \in F_b(\xi)$. Let $E(\xi^\perp) \subset E(\eta)$ denote the union of $F_b(\xi^\perp)$.

Definition 3.1.9. *$\xi^\perp$ is called the orthogonal complement of ξ.*

Theorem 3.1.2. *$E(\xi^\perp)$ is the total space of a subbundle $\xi^\perp \subset \eta$. Furthermore, $\eta \sim \xi \oplus \xi^\perp$.*

pf. Each vector space $F_b(\eta)$ is the direct sum of the subspaces $F_b(\xi)$ and $F_b(\xi^\perp)$. Given $\forall b_0 \in B$, let U be a nhood of b_0 which is sufficiently small that both $\xi|_U, \eta|_U$ are trivial. Let $s_1, \dots, s_m$ be normal orthogonal cross-sections of $\xi|_U$ and let $s'_1, \dots, s'_n$ be normal orthogonal cross-sections of $\eta|_U$. Then the $m \times n$ matrix $[s_i(b_0) \cdot s'_j(b_0)]$ has rank m. Reordering the s'_j if necessary, we may assume the first m columns are linearly independent. Let $V \subset U$ be the open set consisting of all points b for which the first m columns of the matrix $[s_i(b_0) \cdot s'_j(b_0)]$ are linearly independent. Then the n cross-sections $s_1, \dots, s_m, s'_{m+1}, \dots, s'_n$ of $\eta|_U$ are not linearly dependent at any point of V. By the Gram-Schmidt process to this sequence of cross-sections, we get

the normal orthogonal cross-sections $s_1,\dots,s_n$ of $\eta|_V$. Then a new local system is given by the formula $\begin{array}{rcl} h : V\times\mathbb{R}^{n-m} & \to & E(\xi^{\perp}) \\ (b,x) & \mapsto & x_1s_{m+1}(b)+\cdots+x_{n-m}s_n(b) \end{array}$.
The identity $h^{-1}(e)=(pe,(e\cdot s_{m+1}(pe),\dots,e\cdot s_n(pe)))$ shows h is a homeomorphism.

Corollary 3.1.1. *For any smooth submanifold M of a smooth Riemannian manifold N, the normal bundle ν is defined and $\tau_M\oplus\nu\cong\tau_N|_M$. More generally, a smooth map $f:M\to N$ between smooth manifolds is called an immersion if the Jacobian $Df_X:DM_X\to DN_{f(X)}$ maps the tangent space DM_x injectively for each $x\in M$.*

pf. $M\subset N\subset\mathbb{R}^A$, and N is given a Riemannian metric. Then the tangent bundle τ_M is a subbundle of the restriction $\tau_N|_M$. Then the orthogonal complement $\tau_M^{\perp}\subset\tau_N|_M$ is called the normal bundle ν of M in N. Thus the corollary follows.

Corollary 3.1.2. *For any immersion $f:M\to N$ with N Riemannian, there is a Whitney sum decomposition*

$$f^*\tau_N\cong\tau_M\oplus\nu_f$$

where ν_f is called the normal bundle of the immersion f.

3.1.3 Principal Bundles and Associated Bundles

Denote $Vect(B)$ the set of isomorphism classes of vector bundles over B, and $Vect_n(B)$ the set of n-plane bundles.

Definition 3.1.10. *Let G be a topological group. A principal G-bundle is a right action of G on a space P s.t.*

1. *G acts freely.*
2. *the orbit projection $P\to P/G$ is a fiber bundle.*

e.g. Suppose G is discrete. Then the fibers of the orbit projection $P\to P/G$ are all discrete. Thus the condition that $P\to P/G$ is a fiber bundle is that it is a covering projection.

The fundamental group at $*$ acts on the fiber over $*$ of any covering projection to produce a left $\pi_1(X)$-set. A functor in the other direction is given as follows; let F be any set with left $\pi_1(X)$-action, and from the balanced product $\tilde{X}\times_{\pi_1(X)}F=$

$\tilde{X} \times F/\sim$ where $(y, gz) \sim (yg, z)$, $\forall y \in \tilde{X}, z \in F, g \in \pi_1(X)$. The composite $p \circ p_1 : \tilde{X} \times F \to X$ factors to give a map $\tilde{X} \times_{\pi_1(X)} F \to X$, that is a covering projection.
$\to$ let $P \downarrow B$ be a principal G-bundle. If F is a left G-space, we can define a new fiber bundle associated to $P \downarrow B$, $\begin{array}{c} P \times_G F \\ \downarrow q \\ B \end{array}$ (Let $x \in B$, and pick $y \in P$ over x. Then $\exists$ $\begin{array}{ccc} F & \to & q^{-1}(x) \\ z & \mapsto & [y,z] \end{array}$. The inverse is given as $\begin{array}{ccc} q^{-1}(x) & \to & F \\ (y',z') = (y, gz') & \mapsto & gz' \end{array}$, where $y' = yg$ for some g. These two maps are inverse homeomorphism. If F is a finite dimensional vector space on which G acts linearly, then we get a vector bundle).

e.g. Let $\xi : E \downarrow B$ be an n-plane bundle. Construct a principal $GL_n(\mathbb{R})$-bundle $P(\xi)$ by defining $P(\xi)_b = \{$ ordered bases for $E(\xi)_b = Iso(\mathbb{R}^n, E(\xi)_b)\}$. Regard $P(\xi)$ as a quotient of the disjoint union of trivial bundles over the open sets in a trivialising cover for ξ; while for trivial bundles $P(B \times \mathbb{R}^n) = B \times Iso(\mathbb{R}^n, \mathbb{R}^n)$ topologically, where $Iso(\mathbb{R}^n, \mathbb{R}^n) = GL_n(\mathbb{R})$ is given the usual topology as a subspace of $\mathbb{R}^{n^2}$. Then $GL_n(\mathbb{R})$ acts on the right on $P(\xi)$ by precomposition and it is free and simply transitive on fibers. Thus we have a principal action of $GL_n(\mathbb{R})$ on $P(\xi)$ which is called the principalisation of ξ. Given the principalisation $P(\xi)$, we can recover the total space $E(\xi) \cong P(\xi) \times_{GL_n(\mathbb{R})} \mathbb{R}^n$. These two constructions are inverses.

Fix a topological group G. Define $Bun_G(B)$ as the set of isomorphism classes of G-bundles over B. An isomorphism is a G-equivariant homeomorphism over the base. Then we get a contravariant functor $Bun_G : \mathbf{Top} \to \mathbf{Set}$ and $Bun_{GL_n(\mathbb{R})}(B) \cong Vect(B)$ by the previous example.

Definition 3.1.11. *The functor D is I-invariant if the projection $p_1 : X \times I \to X$ induces an isomorphism $D(X) \to_{\cong} D(X \times I)$.*

Definition 3.1.12. *A G-CW complex is a G-space of the form $D^n \times H \setminus G$ where H is a closed subgroup of G and $H \setminus G$ is the orbit space of the action of H on G by left translation, viewed as a right G-space. The boundary of the G-cell $D^n \times H \setminus G$ is just $\partial D^n \times H \setminus G$.*

$\to$a relative G-CW complex is a right G-space X with a filtration $A = X_{-1} \subseteq X_0 \subseteq \cdots \subseteq X$ by G-subspaces s.t. $\forall n \geq 0$, $\exists$ a pushout square of G-spaces

$$\begin{array}{ccc} \coprod \partial D_i^n \times H_i \setminus G & \longrightarrow & \coprod D_i^n \times H_i \setminus G \\ \downarrow & & \downarrow \\ X_{n-1} & \longrightarrow & X_n \end{array}$$

and X has the direct limit topology.

A CW complex is a G-CW complex for the trivial group G. If G is discrete, the skeleton filtration provides X with the structure of a CW complex by neglecting the G-action. If X is a G-CW complex, X/G inherits a CW structure whose n-skeleton is given by $(X/G)_n = X_n/G$. If $P \downarrow X$ is a principal G-bundle, a CW structure on X lifts to a G-CW structure on P.

Theorem 3.1.3. *If G is a compact Lie group and M a smooth manifold on which G acts by diffeomorphisms, then M admits G-CW structure.*

Theorem 3.1.4. *Bun_G is I-invariant, and hence is a homotopy functor.*

pf.

- I-invariance

  Assume wlog X is a CW complex and denote $Y = X \times I$. Filter Y by subcomplexes s.t.
  $$\begin{cases} Y_0 = X \times 0 \\ Y_n = \begin{tikzcd} \coprod(\partial D^{n-1} \times I \cup D^{n-1} \times 0) \arrow[r] \arrow[d] & \coprod(D^{n-1} \times I) \arrow[d] \\ Y_{n-1} \arrow[r] & Y_n \end{tikzcd} \end{cases}$$
  via a pushout. The restriction of P to Y_n is a principal bundle with total space $P_n = p^{-1}(Y_n)$. Then $P_0 \downarrow Y_0$ is just $i_0^* P \downarrow X$.

  - $P \cong p_1^* i_0^* P$ over Y
    pf. Build P_n from P_{n-1} by lifting the pushout construction of Y_n from Y_{n-1}, i.e.
    $$\begin{tikzcd} \coprod(\partial D^{n-1} \times I \cup D^{n-1} \times 0) \times G \arrow[r] \arrow[d] & \coprod(D^{n-1} \times I) \times G \arrow[d] \\ P_{n-1} \arrow[r] & P_n \end{tikzcd}$$
    so to extend $P_{n-1} \to P_0$ to $P_n \to P_0$,
    $$\begin{tikzcd} \coprod(\partial D^{n-1} \times I \cup D^{n-1} \times 0) \arrow[r] \arrow[d] & \coprod(D^{n-1} \times I) \arrow[d] \arrow[ddr, dotted, bend left] & \\ \coprod(\partial D^{n-1} \times I \cup D^{n-1} \times 0) \times G \arrow[r] \arrow[d] & \coprod(D^{n-1} \times I) \times G \arrow[d] \arrow[dr, dotted, "f"] & \\ P_{n-1} \arrow[r] \arrow[rr, bend right=10] & P_n \arrow[r, dotted] & P_0 \end{tikzcd}$$
    we need an equivariant map f in covering the map $Y_n \to Y_0$ so that the diagram commutes. Since the action is free, define f on $D^{n-1} \times I$

for each cell, in such a way that

$$\begin{tikzcd}
\partial D^{n-1}\times I\cup D^{n-1}\times 0 \arrow[r] \arrow[d] & D^{n-1}\times I \arrow[d, dotted] \arrow[dd, bend left] \\
P_{n-1} \arrow[r] \arrow[d] & P_0 \arrow[d] \\
Y_{n-1} \arrow[r] & Y_0
\end{tikzcd}$$

commutes. Then extend it by equivariance. Since $(D^{n-1}\times I, \partial D^{n-1}\times I\cup D^{n-1}\times 0)\cong(D^{n-1}\times I, D^{n-1}\times 0)$,

$$\begin{tikzcd}
D^{n-1}\times 0 \arrow[r] \arrow[d] & P_0 \arrow[d] \\
D^{n-1}\times I \arrow[r] \arrow[ur, dotted] & Y_0
\end{tikzcd}$$

the dotted map exists, given $P_0\to Y_0$ is a fibration.

- homotopy functor

  Let $\xi\downarrow B$ be a vector bundle and suppose $H: B'\times I\to B$ a homotopy between two maps f_0, f_1. Then

$$\begin{tikzcd}
& B' \arrow[dl, "\equiv"'] \arrow[d, "i_0"] \arrow[dr, "f_0"] & \\
B' \arrow[dr, "\equiv"'] & B'\times I \arrow[l, "p_1"] \arrow[r, "h"] & B \\
& B' \arrow[u, "i_1"] \arrow[ur, "f_1"'] &
\end{tikzcd}$$

  the map p_1 induces a surjection in $Vect$. Thus $i_0^*=i_1^*$ so $f_0^*=i_0^*\circ h^*=i_1^*\circ h^*=f_1^*$.

$\to Vect$ is I-invariant and it is a homotopy functor.

Proposition 3.1.1. *Let E be a G-space that is weakly contractible as a space. Let (P,A) be a free relative G-CW complex. Then any equivariant map $f: A\to E$ extends to an equivariant map $P\to E$ and this is unique up to an equivariant homotopy $rel A$.*

Theorem 3.1.5. *Let G be a topological group and $\xi: E\downarrow B$ a principal G-bundle s.t. E is weakly contractible. For any CW complex X, the map* $\begin{array}{ccc}[X,B] & \to & Bun_G(X)\\ f & \mapsto & f^*\xi\end{array}$ *is bijective.*

pf. Let $f_0, f_1: P\to E$ be two equivariant maps. Then $\exists A = P\times\partial I\to E$ which we extend to an equivariant map from $P\times I$ by the previous proposition. Thus they are homotopic by an equivariant homotopy. Thus it is injective. Surjectivity is immediate.

Any choice of contractible free G-CW comoplex is written EG, and its orbit space BG. Then $EG \downarrow BG$ is the universal principal G-space and BG classifies principal G-bundles.

- The Grassmannian Construction

  Definition 3.1.13. *The grassmann manifold $Gr_n(\mathbb{R}^{n+k})$ is the set of all n-dimensional planes through the origin of the coordinate space $\mathbb{R}^{n+k}$.*

  $\to$ An n-frame in $\mathbb{R}^{n+k}$ is an n-tuple of linearly independent vectors of $\mathbb{R}^{n+k}$. The collection of all n-frames in $\mathbb{R}^{n+k}$ forms an open subset of the n-fold Cartesian product $\mathbb{R}^{n+k} \times \cdots \times \mathbb{R}^{n+k}$, called the Stiefel manifold $V_n(\mathbb{R}^{n+k})$. Then, a canonical function $q : V_n(\mathbb{R}^{n+k}) \to Gr_n(\mathbb{R}^{n+k})$ maps each n-frame to the n-plane it spans. Then a subset $U \subset Gr_n(\mathbb{R}^{n+k})$ is open iff its inverse image $q^{-1}(U) \subset V_n(\mathbb{R}^{n+k})$ is open.
  OR
  Denote by $V_n^0(\mathbb{R}^{n+k}) \subset V_n(\mathbb{R}^{n+k})$ the subset consisting of all orthonormal n-frames. Then $Gr_n(\mathbb{R}^{n+k})$ can be considered as an identification space of $V_n^0(\mathbb{R}^{n+k})$.

  Lemma 3.1.5. *The Grassmann manifold $Gr_n(\mathbb{R}^{n+k})$ is a comopact topological manifold of dimension nk. The correspondence $X \to X^\perp$ which assigns to each n-plane its orthogonal k-plane defines a homeomorphism between $Gr_n(\mathbb{R}^{n+k})$ and $Gr_k(\mathbb{R}^{n+k})$.*

  pf.

  - for a fixed $w \in \mathbb{R}^{n+k}$, let $\rho_W(X)$ be a square of the Euclidean distance from w to X. If $x_1, \ldots, x_n$ is an orthonormal basis for X, $\rho_W(X) = w \cdot w - (w \cdot x_1)^2 - \cdots - (w \cdot x_n)^2$ shows $V_n^0(\mathbb{R}^{n+k}) \to_{q_0} Gr_n(\mathbb{R}^{n+k}) \to_{\rho_W} \mathbb{R}$ is continuous, and hence ρ_W is continuous. If X, Y are distinct n-planes s.t. w belongs to X but not to Y, then $\rho_W(X) \neq \rho_W(Y)$. So the Grassmann is Hausdorff.
  - the set $V_n^0(\mathbb{R}^{n+k})$ of orthonormal n-frames is a closed bounded subset of $\mathbb{R}^{n+k} \times \cdots \times \mathbb{R}^{n+k}$, and therefore is compact. Thus $Gr_n(\mathbb{R}^{n+k}) = q_0(V_n^0(\mathbb{R}^{n+k}))$ is compact.
  - $\forall X_0 \in Gr_n(\mathbb{R}^{n+k})$, $\exists U$ nhood which is homeomorphic to $\mathbb{R}^{nk}$.

    regard $\mathbb{R}^{n+k}$ as the direct sum $X_0 \oplus X_0^\perp$. Let U be the open subset of $Gr_n(\mathbb{R}^{n+k})$ consisting of all n-planes Y s.t. the orthogonal projection $p : X_0 \oplus X_0^\perp \to X_0$ maps Y onto X_0. Then each $Y \in U$ can be considered as the graph of a linear transformation $T(Y) : X_0 \to X_0^\perp$. This defines a one-to-one correspondence $T : U \to \hom(X_0, X_0^\perp) \cong \mathbb{R}^{nk}$. Let $x_1, \ldots, x_n$ be a fixed orthonormal basis for X_0. Each n plane $Y \in U$ then has a unique basis $y_1, \ldots, y_n$ s.t. $p(y_1) = x_1, \ldots, p(y_n) = x_n$. Then the n-frame $(y_1, \ldots, y_n)$ depends continuously on Y. We have the identity $y_i = x_i + T(Y)x_i$. Since y_i depends continuously on Y, the image $T(Y)x_i \in X_0^\perp$ depends continuously on Y, and therefore the linear transformation $T(Y)$ depends on Y continuously. On the other hand, n-frame $(y_1, \ldots, y_n)$ depends continuously on $T(Y)$, and hence Y depends continuously on $T(Y)$. Thus T^{-1} is also continuous.

- $Y^\perp$ depends continuously on Y.

  let $(\bar{x}_1, \ldots, \bar{x}_k)$ be a fixed basis for $X_0^\perp$. Define a function $f : q^{-1}U \to V_k(\mathbb{R}^{n+k})$ as follows; $\forall (y_1, \ldots, y_n) \in q^{-1}U$, by the application of the Gram-Schmidt process to the vectors $(y_1, \ldots, y_n, \bar{x}_1, \ldots, \bar{x}_k)$, we get an orthonormal $n+k$-frame $(y'_1, \ldots, y'_{n+k})$ with $y'_{n+1}, \ldots, y'_{n+k} \in Y^\perp$. Setting $f(y_1, \ldots, y_n) = (y'_{n+1}, \ldots, y'_{n+k})$, we have a commutative diagram

  $$\begin{array}{ccc} q^{-1}U & \xrightarrow{f} & V_k(\mathbb{R}^{n+k}) \\ \big\downarrow q & & \big\downarrow q \\ U & \xrightarrow{\perp} & Gr_k(\mathbb{R}^{n+k}) \end{array}$$

  . Here, f is continuous, $q \circ f$ is continuous, and therefore $Y \mapsto Y^\perp$ is continuous.

Then a canonical vector bundle $\gamma^n(\mathbb{R}^{n+k})$ over $Gr_n(\mathbb{R}^{n+k})$ is constructed as follows; let $E = E(\gamma^n(\mathbb{R}^{n+k}))$ be the set of all pairs (n-planes in $\mathbb{R}^{n+k}$, the vector in the n-plane). This is topologized as a subset of $Gr_n(\mathbb{R}^{n+k}) \times \mathbb{R}^{n+k}$. The projection map $\begin{array}{rcl} p : E & \to & Gr_n(\mathbb{R}^{n+k}) \\ (X, x) & \mapsto & X \end{array}$ and the vector space structure in the fiber over X is defined by $t_1(X, x_1) + t_2(X, x_2) = (X, t_1x_1 + t_2x_2)$.

Lemma 3.1.6. *The vector bundle $\gamma^n(\mathbb{R}^{n+k})$ constructed above satisfies the local triviality condition.*

pf. Let U be the nhood of X_0 constructed in the previous lemma. Define the coordinate homeomorphism $\begin{array}{rcl} h : U \times X_0 & \to & p^{-1}U \\ (Y, x) & \mapsto & (Y, y) \end{array}$ where y denotes the unique vector in Y which is carried into x by the orthogonal projection $P : \mathbb{R}^{n+k} \to X_0$. The identities $\left\{ \begin{array}{r} h(Y, x) = (Y, x + T(Y)x) \\ h^{-1}(Y, y) = (Y, Py) \end{array} \right.$ shows h, h^{-1} are continuous. So the lemma follows.

Definition 3.1.14. *Given a smooth n-manifold $M \subset \mathbb{R}^{n+k}$, the generalised*

Gauss map
$\begin{array}{rcl} \bar{g}: M & \to & Gr_n(\mathbb{R}^{n+k}) \\ x & \mapsto & DM_x \end{array}$ *where* $DM_x \in Gr_n(\mathbb{R}^{n+k})$.

$\to$ it is covered by a bundle map $\begin{array}{rcl} g: E(\tau_M) & \to & E(\gamma^n(\mathbb{R}^{n+k})) \\ (x,v) & \mapsto & (DM_x, v) \end{array}$.

Lemma 3.1.7. *For any n-plane bundle ξ over a compact base space B, there exists a bundle map $\xi \to \gamma^n(\mathbb{R}^{n+k})$ provided that k is sufficiently large.*

pf. In order to construct a bundle map $f: \xi \to \gamma^n(\mathbb{R}^m)$, it is sufficient to construct a map $\hat{f}: E(\xi) \to \mathbb{R}^m$ which is linear and injective on each fiber of ξ. Then the required function f can be defined by
$f(e) = (\hat{f}(\text{ fiber through } e), \hat{f}(e))$.
$\to$choose open sets $U_1, \dots, U_r$ covering B so that each $\xi|_{U_i}$ is trivial. Since B is normal, there exist open sets $V_1, \dots, V_r$ covering B with
$\bar{V}_i \subset U_i$. Similarly construct $W_1, \dots, W_r$ with $\bar{W}_i \subset V_i$. Let $\lambda_i : B \to \mathbb{R}$ denote a continuous function which takes the value $\begin{cases} 1 & \bar{W}_i \\ 0 & \text{outside of } V_i \end{cases}$. Since $\xi|_{U_i}$ is trivial, $\exists h_i : p^{-1}U_i \to \mathbb{R}^n$ which maps each fiber of $\xi|_{U_i}$ linearly onto $\mathbb{R}^n$. Define $\begin{array}{rcl} h_i' : E(\xi) & \to & \mathbb{R}^n \\ e & \mapsto & \begin{cases} 0 & \pi(e) \notin V_i \\ \lambda_i(p(e)) h_i(e) & \pi(e) \in U_i \end{cases} \end{array}$. Then h_i' is continuous and linear on each fiber. If we define
$\begin{array}{rcl} \hat{f}: E(\xi) & \to & \mathbb{R}^n \oplus \cdots \oplus \mathbb{R}^n \cong \mathbb{R}^{rn} \\ e & \mapsto & (h_1'(e), \dots, h_r'(e)) \end{array}$, it is continuous and it maps each fiber injectively. Thus the lemma follows.
$\to \gamma^n(\mathbb{R}^{n+k})$ is called a universal bundle.

Let $\mathbb{R}^\infty$ denote the vector space consisting of those infinite sequences $x = (x_1, x_2, \dots)$ of real numbers for which all but a finite number of the x_i are zero. For fixed k, the subspace consisting of all $x = (x_1, \dots, x_k, 0, \dots)$ will be identified with the coordinate space $\mathbb{R}^k$. Thus $\mathbb{R}^1 \subset \mathbb{R}^2 \subset \dots$ with union $\mathbb{R}^\infty$.

Definition 3.1.15. *The infinite Grassman manifold $Gr_n = Gr_n(\mathbb{R}^\infty)$ is the set of all n-dimensional linear subspaces of $\mathbb{R}^\infty$, topologised as the direct limit of the sequence*

$$Gr_n(\mathbb{R}^n) \subset Gr_n(\mathbb{R}^{n+1}) \subset \dots$$

A canonical bundle γ^n over Gr_n is constructed, just as in the finite dimensional case as follows; let $E(\gamma^n) \subset Gr_n \times \mathbb{R}^\infty$ be the set of all pairs (n-planes in $\mathbb{R}^\infty$, vector in that n-plane), topologised as a subset of the Cartesian product. Define $\begin{array}{rcc} p: E(\gamma^n) & \to & Gr_n \\ (X,x) & \mapsto & X \end{array}$ and define the vector space structures in the fibers as before.

Lemma 3.1.8. *Let* $\left\{\begin{array}{l} A_1 \subset A_2 \subset \ldots \\ B_1 \subset B_2 \subset \ldots \end{array}\right.$ *be sequences of locally compact spaces with direct limits A, B respectively. The the Cartesian product topology on $A \times B$ coincides with the direct limit topology which is associated with the sequence $A_1 \times B_1 \subset A_2 \times B_2 \subset \ldots$.*

pf. Let W be open in the direct limit topology and let (a,b) be any point of W. Suppose $(a,b) \in A_i \times B_i$. Choose a compact nhood K_i of a in A_i and a compact nhood L_i of b in B_i so that $K_i \times L_i \subset W$. Then it is possible to choose compact nhood $\left\{\begin{array}{l} K_{i+1} \text{ of } K_i \text{ in } A_{i+1} \\ L_{i+1} \text{ of } L_i \text{ in } B_{i+1} \end{array}\right.$ so that $K_{i+1} \times L_{i+1} \subset W$. Continue by induction, construct nhoods $\left\{\begin{array}{l} K_i \subset K_{i+1} \subset K_{i+2} \subset \ldots \text{ with union } U \\ L_i \subset L_{i+1} \subset L_{i+2} \subset \ldots \text{ with union } V \end{array}\right.$. Then U, V are open sets and $(a,b) \in U \times V \subset W$ and therefore W is open in the product topology.

Lemma 3.1.9. *This vector bundle γ^n satisfies the local triviality condition.*

pf. Let $X_0 \subset \mathbb{R}^\infty$ be a fixed n-plane, and let $U \subset Gr_n$ be the set of all n-planes Y which project onto X_0 under the orthogonal projection $P: \mathbb{R}^\infty \to X_0$. This set U is open since $\forall k < \infty$, $U_k = U \cap Gr_n(\mathbb{R}^{n+k})$ is known to be an open set. Defining $h: U \times X_0 \to p^{-1}U$ as previously, $h|_{U_k \times X_0}$ is continuous for each k. Then by the previous lemma, h is continuous. The identity $h^{-1}(Y,y) = (Y, Py)$ implies h^{-1} is continuous and therefore h is a homeomorphism.

Definition 3.1.16. *A topological space B is paracompact if B is a Hausdorff space and if for every open covering $\{U_\alpha\}$ of B, there exists an open covering $\{V_\beta\}$ which is* $\left\{\begin{array}{c} \textit{a refinement of } \{U_\alpha\} \\ \textit{locally finite} \end{array}\right.$

Lemma 3.1.10. *For any fiber bundle ξ over a paracompact space B, there exists a locally finite covering of B by countably many ope sets $U_1, U_2, \ldots$ so that $\xi|_{U_i}$ is trivial for each i.*

pf. Choose a locally finite open covering $\{V_\alpha\}$ so that each $\xi|_{V_\alpha}$ is trivial; and choose an open covering $\{W_\alpha\}$ with $\bar{W}_\alpha \subset V_\alpha$ for each α. Let $\lambda_\alpha: B \to \mathbb{R}$ be a continuous function which takes the value $\left\{\begin{array}{ll} 1 & \bar{W}_\alpha \\ 0 & \text{outside of } V_\alpha \end{array}\right.$. For each

non-vacuous finite subset S of the index set $\{\alpha\}$, let $U(S) \subset B$ denote the set of all $b \in B$ s.t. $\min_{\alpha \in S} \lambda_\alpha(b) > \max_{\alpha \notin S} \lambda_\alpha(b)$. Let U_k be the union of those sets $U(S)$ for which S has precisely k elements. Then each U_k is an open set and $B = U_1 \cup U_2 \cup \ldots$. For a given $b \in B$, if precisely k of the numbers $\lambda_\alpha(b)$ are positive, then $b \in U_k$. If
$\alpha \in S, U(S) \subset V_\alpha$. Since the covering $\{V_\alpha\}$ is locally finite, $\{U_k\}$ is locally finite. Furthermore, since each $\xi|_{V_\alpha}$ is trivial, each $\xi|_{U(S)}$ is trivial. But the set U_k is equal to the disjoint union of its open subsets $U(S)$. Thus $\xi|_{U_k}$ is also trivial.

Theorem 3.1.6. *Any $\mathbb{R}^n$-bundle ξ over a paracompact base space admits a bundle map $\xi \to \gamma^n$.*

pf. The bundle map $f : \xi \to \gamma^n$ can be constructed as previously in the lemma 1.1.6. and the theorem follows by the previous lemma.
$\to \gamma^n$ over Gr_n is a universal $\mathbb{R}^n$-bundle!

Definition 3.1.17. *Two bundle maps $f, g : \xi \to \gamma^n$ are called bundle-homotopic if there exists one-parameter family of bundle maps*

$$h_t : \xi \to \gamma^n, \ 0 \le t \le 1$$

with $\begin{cases} h_0 = f \\ h_1 = g \end{cases}$ *s.t. h is continuous as a function of both variables, i.e. the associated function $h : E(\xi) \times [0,1] \to E(\gamma^n)$ must be continuous.*

Theorem 3.1.7. *Any two bundle maps from an $\mathbb{R}^n$-bundle to γ^n are bundle-homotopic.*

pf. Any bundle map $f : \xi \to \gamma^n$ determines a map $\hat{f} : E(\xi) \to \mathbb{R}^\infty$ whose restriction to each fiber of ξ is linear and injective. Conversely, $\hat{f}$ determines f by the identity $f(e) = (\hat{f}(\text{fiber through } e), \hat{f}(e))$. Let $f, g : \xi \to \gamma^n$ be any two bundle maps.

- the vector $\hat{f}(e) \in \mathbb{R}^\infty$ is never equal to a negative multiple of $\hat{g}(e)$ for $e \neq 0, e \in E(\xi)$

  $\hat{h}_t(e) = (1-t)\hat{f}(e) + t\hat{g}(e), \ 0 \le t \le 1$ defines a homotopy between $\hat{f}, \hat{g}$.
  By the lemma 1.1.7. $\hat{h}_t(e) \neq 0$ if e is a nonzero vector of $E(\xi)$. Hence we can define $\begin{array}{rcl} h : E(\xi) \times [0,1] & \to & E(\eta) \\ (e,t) & \mapsto & (\hat{h}_t(\text{fiber through } e), \hat{h}_t(e)) \end{array}$. This has the corresponding function $\hat{h} : B(\xi) \times [0,1] \to Gr_n$. If this is continuous, h is continuous. Let U be an open subset of $B(\xi)$ with $\xi|_U$ trivial and let $s_1, \ldots, s_n$ be nowhere dependent cross-sections of $\xi|_U$. Then $\hat{h}|_{U \times [0,1]}$ can

be considered as the composition of $\begin{array}{rcr} U \times [0,1] & \to & V_n(\mathbb{R}^\infty \\ (b,t) & \mapsto & (\hat{h}_1 s_1(b), \ldots, \hat{h}_t s_n(b)) \end{array}$
and $q : V_n(\mathbb{R}^\infty \to Gr_n$. Since those two are continuous, $\bar{h}$ is continuous, hence the bundle homotopy h is continuous.

- general case

  Let $f, g : \xi \to \gamma^n$ be arbitrary bundle maps. Bundle maps $\begin{cases} d_1 : \gamma^n \to \gamma^n \\ d_2 : \gamma^n \to \gamma^n \end{cases}$
  are induced by the linear transformations
  $\begin{cases} \mathbb{R}^\infty \to \mathbb{R}^\infty & \text{carries the } i\text{th basis vector of } \mathbb{R}^\infty \text{ to the } 2i-1th \\ \gamma^n \to \gamma^n & \text{carries the } i\text{th basis vector to the } 2ith \end{cases}$. The $f \sim d_1 \circ f \sim d_2 \circ g \sim g \to f \sim g$.

Corollary 3.1.3. *Any $\mathbb{R}^n$-bundle ξ over a paracompact space B determines a unique homotopy class of maps*

$$\bar{f}_\xi : B \to Gr_n$$

pf. Let $f_\xi : \xi \to \gamma^n$ be any bundle map, and let $\bar{f}_\xi$ be the induced map of base spaces. Then the corollary follows by the previous theorem.

Definition 3.1.18. *Let Λ be a coefficient group or ring and let $c \in H^i(Gr_n; \Lambda)$ be any cohomology class. Then ξ and c together determines a cohomology class $\bar{f}_\xi^* c \in H^i(B; \Lambda)$, denoted by $c(\xi)$ and called the characteristic cohomology class of ξ determined by c.*

$\to x \mapsto c(\xi)$ is natural w.r.t. the bundle maps and the converse also holds (given any $\xi \mapsto c(\xi) \in H^i(B(\xi); \Lambda)$ natural w.r.t. the bundle maps, we have $c(\xi) = \bar{f}_\xi^* c(\gamma^n)$).

*** A cell structure of Grassmann manifolds**

Recall D^p denote the unit disk in $\mathbb{R}^p$ consisting of all vectors v with $|v| \le 1$. The interior of D^p is defined to be the subset consisting of all v with $|v| < 1$.

Definition 3.1.19. *Any space homeomorphic to D^p is called a closed p-cell, and any space homeomorphic to the interior of D^p is called an open p-cell.*

Then the cell structure for the Grassmann manifold $Gr_n(\mathbb{R}^m)$ is obtained as follows;
consider $\mathbb{R}^k$ consists of all vectors of the form $v = (v_1, \ldots, v_k, 0, \ldots, 0)$. Then

$\mathbb{R}^0 \subset \mathbb{R}^1 \subset \cdots \subset \mathbb{R}^m$. Any n-plane $X \subset \mathbb{R}^m$ gives rise to a sequence of integers $0 \leq \dim(X \cap \mathbb{R}^1) \leq \cdots \leq \dim(X \cap \mathbb{R}^m) = n$, where two consecutive integers differ by at most 1 (given $0 \to X \cap \mathbb{R}^{k-1} \to X \cap \mathbb{R}^k \to_{pr_k} \mathbb{R}$ exact). Thus the above sequence of integers contains precisely n jumps. By a Schubert symbol $\sigma = (\sigma_1, \ldots, \sigma_n)$ is meant a sequence of n integers satisfying $1 \leq \sigma_1 < \cdots < \sigma_n \leq m$. For each σ, let $e(\sigma) \subset Gr_n(\mathbb{R}^m)$ denote the set of all planes X s.t. $\begin{cases} \dim(X \cap \mathbb{R}^{\sigma_i}) = i \\ \dim(X \cap \mathbb{R}^{\sigma_i - 1}) = i - 1 \end{cases}$ for all i. Clearly $\forall X \in Gr_n(\mathbb{R}^m)$ belongs to precisely one of the sets $e(\sigma)$.

CLAIM : $e(\sigma)$ **is an open cell of dimension** $\dim\sigma = (\sigma_1 - 1) + \cdots + (\sigma_n - n)$**.**
pf. Let $H^k \subset \mathbb{R}^k$ denote the open half-space consisting of all $x = (\xi_1, \ldots, \xi_k, 0, \ldots, 0)$ with $\xi_k > 0$. Then an n-plane X belongs to $e(\sigma)$ iff a basis $x_1, \ldots, x_n$ s.t. $x_1 \in H^{\sigma_1}, \ldots, x_n \in H^{\sigma_n}$ (if X possesses such a basis, the exact sequence above shows $\dim(X \cap \mathbb{R}^{\sigma_i}) > \dim(X \cap \mathbb{R}^{\sigma_i - 1})$, hence $X \in e\sigma$ and the converse holds similarly).

Lemma 3.1.11. *Each n-plane $X \in e(\sigma)$ possess a unique orthonormal basis $(x_1, \ldots, x_n)$ which belongs to $H^{\sigma_1} \times \cdots \times H^{\sigma_n}$.*

pf. The vector x_1 is required to lie in the 1-dimensional vector space $X \cap \mathbb{R}^{\sigma_1}$ and to be a unit vector. This leaves only two possibilities for x_1; $\begin{cases} +1 \\ -1 \end{cases}$. Since the σ_1th coordinate need to be positive, only one of these is possible. Now x_2 is required to be a unit vector in the 2-dimensional space $X \cap \mathbb{R}^{\sigma_2}$ and to be orthogonal to x_1. This leaves only two possibilities again, and the condition that the σ_2th coordinate be positive specifies one of these. Continuing by induction, we get $x_3, \ldots, x_n$ which are uniquely determined.

Definition 3.1.20. *Let $e'(\sigma) = V_n^0(\mathbb{R}^m) \cap (H^{\sigma_1} \times \cdots \times H^{\sigma_n})$ denote the set of all orthonormal n-frames $(x_1, \ldots, x_n)$ s.t. $\forall x_i \in H^{\sigma_i}$. Let $\bar{e}'(\sigma)$ denote the set of orthonormal frames $(x_1, \ldots, x_n)$ s.t. $\forall x_i \in \bar{H}^{\sigma_i}$.*

Lemma 3.1.12. *The set $\bar{e}'(\sigma)$ is topologically a closed cell of dimension $d(\sigma) = (\sigma_1 - 1) + \cdots + (\sigma_n - n)$ with interior $e'(\sigma)$. Furthermore q maps the interior $e'(\sigma)$ homeomorphically onto $e(\sigma)$.*

pf.

- the set $\bar{e}'(\sigma)$ is topologically a closed cell of dimension $d(\sigma) = (\sigma_1 - 1) + \cdots + (\sigma_n - n)$ with interior $e'(\sigma)$
  When $n = 1$, $\bar{e}'(\sigma_1)$ consists of all vectors $x_1 = (x_{11}, \ldots, x_{1\sigma_1}, 0, \ldots, 0)$ with

$\begin{cases} \sum x_{1i}^2 = 1 \\ x_{1\sigma_1} \geq 0 \end{cases}$. Since $\bar{e}'(\sigma_1)$ is a closed hemisphere of dimension $\sigma_1 - 1$, it is homeomorphic to the disk $D^{\sigma_1 - 1}$.
Given unit vectors $u, v \in \mathbb{R}^m$ with $u \neq -v$, let $T(u,v)$ denote the unique rotation of $\mathbb{R}^m$ which carries u to v and leaves everything orthogonal to u and v fixed. Thus $\begin{cases} T(u,u) & \text{the identity} \\ T(v,u) & T(u,v)^{-1} \end{cases} \Leftrightarrow T(u,v)x = x - \frac{(u+v)\cdot x}{1+u\cdot v}(u + v) + 2(u\cdot x)v$. Then it is continuous in u,v,x and if $u,v \in \mathbb{R}^k$, $T(u,v)x \equiv x \mod \mathbb{R}^k$. Let $b_i \in H^{\sigma_i}$ denote the vector with σ_ith coordinate equal to 1 and all other coordinates zero. Thus $(b_1,\ldots,b_n) \in e'(\sigma)$. For any n-frame $(x_1,\ldots,x_n) \in \bar{e}'(\sigma)$, $\exists T = T(b_n,x_n)\circ\cdots\circ T(b_1,x_1)$ of $\mathbb{R}^m$, a rotation which carries the n vectors $b_1,\ldots,b_n$ to the vectors $x_1,\ldots,x_n$ respectively. Here the rotations $T(b_1,x_1),\ldots,T(b_{i-1},x_{i-1})$ leaves b_i fixed and the rotation $T(b_i,x_i)$ carries b_i to x_i and the rotations $T(b_{i+1},x_{i+1}),\ldots,$ $T(b_n,x_n)$ leaves x_n fixed. Then given an integer $\sigma_{n+1} > \sigma_n$, let D denote the set of all unit vectors $u \in \bar{H}^{\sigma_{n+1}}$ with $b_1\cdot u = \cdots = b_n\cdot u = 0$. Then D is a closed hemisphere of dimension $\sigma_{n+1} - n - 1$ and hence is topologically a closed cell. Now construct a map $\begin{array}{ccc} \bar{e}'(\sigma_1,\ldots,\sigma_n)\times D & \to & \bar{e}'(\sigma_1,\ldots,\sigma_{n+1}) \\ ((x_1,\ldots,x_n),u) & \mapsto & (x_1,\ldots,x_n,Tu) \end{array}$, where the rotation T depends on $x_1,\ldots,x_n$ as above. Then $\begin{cases} x_i\cdot Tu = Tb_i\cdot Tu = b_i\cdot u = 0 & \forall i \leq n \\ \qquad Tu\cdot Tu = u\cdot u = 1 & Tu \in \bar{H}^{\sigma_{n+1}} \end{cases}$ given $Tu \equiv u \mod \mathbb{R}^{\sigma_n}$. Then f maps $\bar{e}'(\sigma_1,\ldots,\sigma_n)\times D$ continuously to $\bar{e}'(\sigma_1,\ldots,\sigma_{n+1})$. Similarly, the formula $u = T^{-1}x_{n+1} = T(x_1,b_1)\circ\cdots\circ T(x_n,b_n)x_{n+1} \in D$ shows f^{-1} is well-defined and continuous. Thus f is a homeomorphism. By inductive hypothesis, each $\bar{e}'(\sigma)$ is a closed cell of dimension $d(\sigma)$. Then the homeomorphism $f : \bar{e}'(\sigma_1,\ldots,\sigma_n)\times D \to \bar{e}'(\sigma_1,\ldots,\sigma_{n+1})$ carries the product $e'(\sigma,\ldots,\sigma_n)\times IntD \to e'(\sigma_1,\ldots,\sigma_{n+1})$.

- q maps the interior $e'(\sigma)$ homeomorphically onto $e(\sigma)$
  q carries $e'(\sigma)$ in one-to-one fashion onto $e(\sigma)$, but if $(x_1,\ldots,x_n)$ belongs to the boundary $\bar{e}'(\sigma)\setminus e'(\sigma)$, the n-plane $X = q(x_1,\ldots,x_n)$ does not belong to $e(\sigma)$ for one of the vectors x_i must lie in the boundary $\mathbb{R}^{\sigma_i - 1}$ of the half-space $\bar{H}^{\sigma_i}$. This implies $\dim(X\cap\mathbb{R}^{\sigma_i-1}) \geq i \to X \notin e(\sigma)$. Let $A \subset e'(\sigma)$ be a relatively closed subset. Then $\bar{A}\cap e'(\sigma) = A$ where $\bar{A} \subset \bar{e}'(\sigma)$ is compact, hence $q(\bar{A})$ is closed. Similarly, $q(\bar{A})\cap e(\sigma) = q(A)$ and therefore $q(A) \subset e(\sigma)$ is a relatively closed set. Thus q maps the cell $e'(\sigma)$ homeomorphically onto $e(\sigma)$.

Thus the claim follows.

Theorem 3.1.8. *The $\binom{m}{n}$ sets $e(\sigma)$ from the cells of a CW-complex with un-*

derlying space $Gr_n(\mathbb{R}^m)$. Similarly taking the direct sum as $m \to \infty$, one obtains an infinite CW-complex with underlying space $Gr_n = Gr_n(\mathbb{R}^\infty)$.

pf. Since $\bar{e}'(\sigma)$ is compact, the image $q\bar{e}'(\sigma)$ is equal to $\bar{e}(\sigma)$. Hence every n-plane X in the boundary $\bar{e}(\sigma) \setminus e(\sigma)$ has a basis $(x_1, \dots, x_n)$ belonging to $\bar{e}'(\sigma) \setminus e'(\sigma)$. Since the vectors $x_1, \dots, x_n$ are orthonormal with $x_i \in \mathbb{R}^{\sigma_i}$, $\dim(X \cap \mathbb{R}^{\sigma_i}) \geq i$, then the Schubert symbol $(\tau_1, \dots, \tau_n)$ associated with X must satisfy $\tau_1 \leq \sigma_1, \dots, \tau_n \leq \sigma_n$. One of the vectors x_i must actually belong to $\mathbb{R}^{\sigma_i - 1}$, and therefore the corresponding integer τ_i must be strictly less than σ_i. This implies $\dim \tau < \dim \sigma$. Then by the previous lemma, $Gr_n(\mathbb{R}^m)$ is a finite CW-complex. Similarly $Gr_n(\mathbb{R}^\infty)$ is a CW-complex since $\forall X \in Gr_n(\mathbb{R}^\infty)$ belongs to a finite subcomplex $Gr_n(\mathbb{R}^m)$ and it has the direct limit topology by definition.

Corollary 3.1.4. *The infinite projective space $\mathbb{R}P^\infty = Gr_1(\mathbb{R}^\infty)$ is a CW-complex having one r-cell $e(r+1)$ for integer $r \geq 0$. The closure $\bar{e}(r+1) \subset \mathbb{R}P^\infty$ is equal to the finite projective space $\mathbb{R}P^r$.*

3.1.4 Simplicial Sets and Classifying Spaces

Definition 3.1.21. *A simplicial set is a contravariant functor $\Delta \to$ **Set**, consisting of*
$\begin{cases} \textit{objects} & \textit{the sets } \{K_n\}_{n\geq 0} \\ \textit{arrows} & \textit{a set function } K_m \to K_n \end{cases}$*, where $K_m \to K_n$ corresponds to each weakly order-preserving map $\mathbf{n} \to \mathbf{m}$.*

$\to$ all morphisms are the compositions of $\begin{cases} \text{the boundary maps :} & \partial_i : K_n \to K_{n-1} \\ \text{the degeneracy :} & s_i : K_{n-1} \to K_n \end{cases}$
corresponding to the strictly order preserving function
$\begin{cases} \{0,1,\dots,n-1\} \mapsto \{0,\dots,n\} \text{ whose image misses } i \\ \{0,\dots,n\} \to \{0,\dots,n-1\} \text{ with the preimage of } i = \{i, i+1\} \end{cases}$. These maps satisfy

$$\begin{cases} \partial_i \partial_j = \partial_{j-1} \partial_i, i < j \\ s_i s_j = s_{j+1} s_i, i \leq j \\ \partial_i s_j = s_{j-1} \partial_i, i < j \\ \partial_j s_j = \partial_{j+1} s_j = Id \\ \partial_i s_j = s_j \partial_{i-1}, i > j+1 \end{cases}.$$

Definition 3.1.22. *Let $|\Delta^n|$ be the geometric n-simplex. Points of $|\Delta^n|$ are given by $(t_0, \dots, t_n)$ with the property that $\begin{cases} t_i \geq 0, \ \forall i \leq n \\ \sum_{i=0}^n t_i = 1 \end{cases}$. The set $Sing_n(X)$ is the set of continuous maps of the geometric n-simplex $|\Delta^n|$ to X. The boundary map $\begin{array}{rcl} \partial_i : Sing_n(X) & \to & Sing_{n-1}(X) \\ f & \mapsto & f|_{ith\ face\ of\ |\Delta^n|} \end{array}$ by the simplicial map that is order preserving*

on the vertices. The degeneracy
$$\begin{array}{rcl} s_i : Sing_n(X) & \to & Sing_{n+1}(X) \\ f & \mapsto & f\circ\phi \end{array} \text{ where } \begin{array}{rcl} \phi : |\Delta^{n+1}| & \to & |\Delta^n| \\ (t_0,\dots,t_{n+1}) & \mapsto & (t_0,\dots,t_{i-1},t_i+t_{i+1},t_{i+2},\dots,t_{n+1}) \end{array}.$$

Definition 3.1.23. *The geometric realisation of a simplicial set K is obtained as follows;*
let $|\tilde{K}| = \coprod_n K_n \times |\Delta^n|$ where each K_n is given the discrete topology. Define an equivalence relation s.t.

- *$\forall x \in K_n$ with $\partial_i x = y$, the ith face of $\{x\}\times|\Delta^n|$ is glued to $\{y\}\times|\Delta^{n-1}|$ by the simplicial isomorphism that is order-preserving on vertices*

- *$\forall z \in K_n$ with $z = s_i y$, $\{z\}\times|\Delta^n|$ is collapsed onto $\{y\}\times|\Delta^{n-1}|$ by the linear projection parallel to the edge with vertices $(i, i+1)$.*

Then the geometric realisation $|K| = |\tilde{K}|/\sim$, the quotient space by the above equivalence relation.

Let $\Lambda_{n,k} \subset \Delta^n$ be the sub simplicial set consisting of all weakly order-preserving maps $\mathbf{m} \to \mathbf{n}$, whose image does not include the kth face of Δ^n. Then a simplicial set K satisfies the Kan condition if $\forall n \geq 0$, $\forall 0 \leq k \leq n$, any map of $\Lambda_{n,k} \to K$ extends to a map $\Delta^n \to K$. A map of simplicial sets $f : K \to L$ is a Kan fibration if a relative version of the above condition holds; given $\begin{cases} \alpha : \Lambda_{n,k} \to L \\ \beta : \Delta^n \to L \end{cases}$ with

$f\circ\alpha = \beta|_{\Lambda_{n,k}}$, $\exists\tilde{\alpha} : \Delta^n \to K$ extending α with $f\circ\tilde{\alpha} = \beta$.
$$\begin{tikzcd} \Delta^n \arrow[r, "\beta"] \arrow[dr, dotted, "\tilde{\alpha}"] & L \\ \Lambda_{n,k} \arrow[u, "i"] \arrow[r, "\alpha"] & K \arrow[u, "f"'] \end{tikzcd}.$$

For any $e \in K_0$, denote $\mathbf{e}$ the sub-simplicial set of K consisting of all iterated degeneracies of e.

Definition 3.1.24. *Suppose K is a Kan complex and $e \in K_0$. Consider the set $P_n(\mathbf{e})$ of $y \in K_n$ s.t. $\partial_i y \in \mathbf{e}$, $\forall i \leq n$. Define an equivalence relation on this set by setting*
$$y \cong z \text{ if } \exists w \in K_{n+1} \text{ s.t. } \begin{cases} \partial_i w \in \mathbf{e},\ \forall i \leq n-1 \\ \partial_n w = y \\ \partial_{n+1} w = z \end{cases}.$$

$\to$ it is an equivalence relation on the set $P_n(\mathbf{e})$.

pf.

- Transitivity
  For each k, denote e_k the unique k-simplex in $\mathbf{e}$. Suppose $y, y', y'' \in P_n$ and

$\begin{cases} y \sim y' \\ y' \sim y'' \end{cases}$. Then $\exists v, v' \in K_{n+1}$ s.t. $\begin{cases} \partial_i v = \partial_i v' = e_n,\ i \leq n-1 \\ \partial_n v = y,\ \partial_{n+1} = y' \\ \partial_n v' = y',\ \partial_{n+1} v' = y'' \end{cases}$ by the definition. By the Kan condition, $\exists e \in K_{n+2}$ with $\begin{cases} \partial_i w = e_n,\ \forall i \leq n-1 \\ \partial_n w = v \\ \partial_{n+2} = v' \end{cases}$. Set $v'' = \partial_{n+1} w$. Then $\begin{cases} \partial_i v'' = e_n,\ \forall i \leq n-1 \\ \partial_n v'' = \partial_n v = y \\ \partial_{n+1} v'' = \partial_{n+1} v' = y'' \end{cases}$. Thus $y \sim y''$.

- Reflexivity
  $\forall y \in P_n,\ s_n y$ has the boundaries $e_n, \ldots, e_n, y, y$, showing $y \sim y$,
- Symmetry
  $\forall y, y' \in P_n$ with $y \sim y'$, $\exists v \in K_{n+1}$ s.t. $\begin{cases} \partial_i v = e_i,\ \forall i \leq n-1 \\ \partial_n v'' = \partial_{n+1} v = y' \\ \partial_{n+1} v = \partial_{n+1} s_n(y) = y \end{cases}$
  $\rightarrow y' \sim y$.

Definition 3.1.25. *Define the nth homotopy set of K, denoted $\pi_n(K, \mathbf{e})$ to be the set of equivalence classes of P_n under this equivalence relation.*

$\rightarrow [y],[z] \in \pi_n(K, \mathbf{e}),\ [y] * [z] = [u]$where $u = \partial_n w,\ w \in K_{n+1}$ is any element given by the Kan condition applied to $(e_n, \ldots, e_n, y, -, z)$ with e_n being the unique element of degree n in $\mathbf{e}$,

$\begin{cases} Sing : \mathbf{Top} \rightarrow \mathbf{n} \\ |\cdot| : \mathbf{n} \rightarrow \mathbf{Top} \end{cases}$ are adjoint functors, in the sense that

$$\hom(K, Sing(X)) = \hom(|K|, X)$$

holds.

Proposition 3.1.2. *For any topological space X, there is a natural continuous map $|Sing(X)| \rightarrow X$. For each point $e \in X$, the simplicial set $Sing(e)$ is naturally a sub-simplicial set of $Sing(X)$. Its only nondegenerate simplex is a 0-simplex so that the geometric realisation $|Sing(e)|$ is a point. The map of pairs $(|Sing(X)|, |Sing(e)|) \rightarrow (X, e)$ induces an isomorphism $\pi_i(|Sing(X)|, |Sing(e)|) \rightarrow \pi_i(X, e)$, $\forall i$. In particular, if X is a CW complex, $|Sing(X)| \rightarrow X$ is a homotopy equivalence.*

pf. For each simplex $\{x_n\} \times |\Delta^n| \in Sing_n(X) \times |\Delta^n|$, define a map on this simplex using the map $x_n : |\Delta^n| \rightarrow X$. These maps are compatible with the equivalence relation and hence define a continuous map $|Sing(X)| \rightarrow X$. Fix a point $e \in X$.

Then we have the induced relative map $(|Sing(X)|,|Sing(e)|) \to (X,e)$. Any element in $\pi_n(X,e)$ is represented by a map $(|\Delta^n|,\partial|\Delta^n|) \to (X,e)$. Thus this is an element $x \in Sing_n(X)$ s.t. $\forall \partial_j x$ is the unique element of $Sing_{n-1}(e)$. This element produces a map $(|\Delta^n|,\partial|\Delta^n|) \to (|Sing(X)|,|Sing(e)|)$ representing an element in $\pi_n(|Sing(X)|,|Sing(e)|)$ mapping to x. Thus π_n map is onto. Since $Sing(X)$ is a Kan complex, any element in $\pi_n(|Sing(X)|,|Sing(e)|)$ is represented as an element in $Sing_n(X)$ whose boundary lies in $Sing(e)$. If such an element represents the trivial element in $\pi_n(X,e)$, there exists an extension of the map $|\Delta^n| \to X$ to a map $|\Delta^{n+1}| \to X$ sending all the faces except the first one to e. Thus π_n map is one to one. Thus it is an isomorphism.
The last follows from the Whitehead theorem.

Definition 3.1.26. *Let K,L be simplicial sets. A homotopy of maps from K to L is a map $H : K \times \Delta^1 \to L$.*

$\to \exists i_0, i_1 : K \subseteq K \times \Delta^1$ two embeddings s.t. $\begin{cases} k \mapsto (k, s_0^n(0)) \\ k \mapsto (k, s_0^n(1)) \end{cases}$ respectively for all $k \in K_n$. We say H is a homotopy from $H \circ i_0 : K \to L$ to $H \circ i_1 : K \to L$.

Denote $SK(\pi,n)$ be a simplicial subset whose geometric realisation is the Eilenberg-MacLane space $K(\pi,n)$. Its k-simplexes are the ordered simplicial cocycles of degree n on $|\Delta^k|$ with values in π. The face and degeneracy maps are induced by pullback cocycles using the boundary and degeneracy maps.

Lemma 3.1.13. *$SK(\pi,n)$ is a Kan complex and its homotopy groups are trivial in all degrees except n. The nth homotopy group is identified with π.*

A singular cochain $\alpha \in Sing^*(X)$ has compact supports if there is a compact subset $K \subset X$ s.t. α vanishes on any singular simplex $\sigma : |\Delta^n| \to X$ whose image is disjoint from K. The singular cochains with compact support, denoted $Sing_c^*(X)$ form a subcomplex of the singular cochain complex. The cohomology of this is the cohomology of X with compact supports and is denoted $H_c^*(X)$. Suppose $U \subset X$ is an open subset. Let α be a cocycle with compact support on U and denote by $K \subset U$ a compact set containing the support of α. Let $V = X \setminus K$. Then $\{U,V\}$ is an open cover of X. Let $Sing_*^{\{U,V\}}(X) \subset Sing_*(X)$ denote the subcomplex generated by all singular simplexes whose image is contained either in U or V. The inclusion of this subcomplex into $Sing_*(X)$ induces an isomorphism on homology, and consequently the dual map between the algebraic duals induces an isomorphism on cohomology. Define $\tilde{\alpha}$ as a dual element of $Sing_*^{\{U,V\}}(X)$ whose value on a singular simplex σ is $<\tilde{\alpha}_K,\sigma> = \begin{cases} <\alpha,\sigma> & \sigma \subset U \\ 0 & \sigma \subset V \end{cases}$. Then $\tilde{\alpha}_K$ is a cocycle and hence its cohomology class $[\tilde{\alpha}_K] \in H^*(Sing_*^{\{U,V\}}(X))$ determines a class, denoted $ext([\tilde{\alpha}_K]) \in H_c^*(X)$.

3.2 The Stiefel-Whitney Classes

3.2.1 The Whitney Product

Definition 3.2.1. *Given cochains α^p, β^q, the Whitney cup product is given by the following formula for any $(p+q)$ singular simplex σ*

$$<\alpha^p \cup \beta^q, \sigma_{p+q}> = <\alpha^p, fr_p(\sigma)> \cdot <\beta^q, bk_q(\sigma)>$$

where $\begin{cases} fr_p(\sigma) & \text{the front } p\text{-face of } \sigma \\ bk_q(\sigma) & \text{the back } q\text{-face of } \sigma \end{cases}$.

The tensor product $Sing_*(X) \otimes Sing_*(X)$ is a chain complex with boundary map $\partial(a_p \otimes b_q) = \partial a_p \otimes b_q + (-1)^p a_p \otimes \partial b_q$. The dual $Sing^* \otimes Sing^*(X)$ is a cochain complex with $d(\alpha^p \otimes \beta^q) = d\alpha^p \otimes \beta^q + (-1)^p \alpha^p \otimes d\beta^q$. The duality pairing between these is given by

$$<\alpha \otimes \beta, a \otimes b> = <\alpha, a><\beta, b>$$

Definition 3.2.2. *Define the Whitney coproduct as follows;*
for each n-simplex $\sigma \in Sing_*(X)$, $\begin{array}{rcl} Wh : Sing_*(X) & \to & Sing_*(X) \otimes Sing_*(X) \\ \sigma & \mapsto & \sum_k fr_k(\sigma) \otimes bk_{n-k}(\sigma) \end{array}$.

$\to \forall \sigma$ a singular simplex, $<\alpha \otimes \beta, Wh(\sigma)> = <\alpha \cup \beta, \sigma>$ by the duality pairing.

Wh is a map of chain complexes.
pf. Let σ be a singular n-simplex. Then $\partial Wh(\sigma) = \sum_{k=0}^n \partial(fr_k(\sigma) \otimes bk_{n-k}(\sigma))$
$= \sum_{k=0}^n (\sum_{l=0}^k (-1)^l \partial_k fr_k(\sigma) \otimes bk_{n-l}(\sigma) + (-1)^k \sum_{l=0}^{n-k} (-1)^l fr_k(\sigma) \otimes \partial_l bk_{n-k}(\sigma))$
$= \sum_{k=0}^n (\sum_{l=0}^{k-1} (-1)^l \partial_l fr_k(\sigma) \otimes bk_{n-k}(\sigma) + (-1)^k \sum_{l=0}^{n-k} (-1)^l fr_k(\sigma) \otimes \partial_l bk_{n-k}(\sigma))$
$Wh\partial\sigma = \sum_l (-1)^l Wh\partial_l \sigma = \sum_k \sum_l fr_k(\partial_k \sigma) \otimes bk_{n-k-1}(\partial_l \sigma)$
$\to$ those are equivalent, and hence Wh is a map of chain complexes.

Lemma 3.2.1. *The Whitney cup product is bilinear, associative and satisfies*

$$d(\alpha^p \cup \beta^q) = (d\alpha^p) \cup \beta^q + (-1)^p \alpha^p \cup (d\beta^q)$$

The 0-cocycle that evaluates 1 on each singular 0-simplex is a two-sided unit for this product.

pf.

- Associativity

  $< (\alpha^p \cup \beta^q) \cup \gamma^r, \sigma^{p+q+r} >=< \alpha^p \cup \beta^q, fr_{p+q}(\sigma) > \cdot < \gamma^r, bk_r(\sigma) >$
  $= (< \alpha^p, fr_p(\sigma) >< \beta^q, bk_q(fr_{p+q}(\sigma)) >) < \gamma^r, bk_r(\sigma) >$
  $=< \alpha^p, fr_p(\sigma) > (< \beta^q, fr_q(bk_{q+r}(\sigma)) >< \gamma^r, bk_r(bk_{q+r}(\sigma)) >)$
  $=< \alpha^p \cup (\beta^q \cup \gamma^r), \sigma >$.

- $< \alpha^p \cup \beta^q, \sigma_{p+q} >=< \alpha^p, fr_p(\sigma) > \cdot < \beta^q, bk_q(\sigma) >$

  for cochains α^p, β^q and for a singular $p+q$-simplex σ,
  $< d\alpha \otimes \beta + (-1)^p \alpha \otimes d\beta, Wh(\sigma) >=< \alpha \cup \beta, \partial\sigma >=< d(\alpha \cup \beta), \sigma >$.

Corollary 3.2.1. *The Whitney cup product induces a well-defined product on cohomology*
$$H^p(X) \otimes H^q(X) \to H^{p+q}(X)$$
which defines an associative ring structure. This ring has a two-sided unit which is the cohomology class of the 0-cocycle evaluating 1 on each singular 0-simplex.

Proposition 3.2.1. *For cohomology classes* $[\alpha^p], [\beta^q]$,
$$[\alpha^p] \cup [\beta^q] = (-1)^{pq} [\beta^q] \cup [\alpha^p]$$

pf. Define $\begin{array}{rcl} T : Sing_*(X) \otimes Sing_*(X) & \to & Sing_*(X) \otimes Sing_*(X) \\ \sigma_p \otimes \tau_q & \mapsto & (-1)^{pq} \tau_q \otimes \sigma_p \end{array}$. Then $T\partial = \partial T$, i.e. it is a map of chain complexes. Then $\left\{ \begin{array}{r} < \alpha \cup \beta, \sigma >=< \alpha \otimes \beta, Wh(\sigma) > \\ (-1)^{pq} < \beta \cup \alpha, \sigma >=< \alpha \otimes \beta, T(Wh(\sigma)) > \end{array} \right.$.
For any σ, $Wh(\sigma) = (\sigma_* \otimes \sigma_*)(Wh|\Delta|^n)$.

- $T \circ Wh \sim Wh$
  pf. Construct a map $H : Sing_*(X) \to Sing_*(X) \otimes Sing_*(X)$ that raises degree by 1 and satisfies $\partial H + H\partial = T \circ Wh - Wh$.
  $\to$ begin with $H_0 = 0 \in Sing_*|\Delta^0| \otimes Sing_*|\Delta^0|$ given $\left\{ \begin{array}{r} Wh|\Delta^0| = Id \otimes Id \\ T \circ Wh|\Delta^0| = (-1)^0 Id \otimes Id \end{array} \right.$.
  $\exists \left\{ \begin{array}{r} Wh(|\Delta^n|) = \sum_k fr_k(\sigma) \otimes bk_{n-k}(\sigma) \\ T \circ Wh(|\Delta^n|) \end{array} \right. \in Sing_*|\Delta^n| \otimes Sing_*|\Delta^n|$. Inductively $\exists H_n \in (Sing_*|\Delta^n| \otimes Sing_*|\Delta^n|)_{n+1}$ s.t. $\partial H_n + H_{n-1}\partial|\Delta^n| = T \circ Wh|\Delta^n| - Wh|\Delta^n|$ where $H_{n-1}\partial|\Delta^n| = \sum_{r=0}^n (-1)^r (f_r \otimes f_r)_* H_{n-1}$, $fr_r : |\Delta^{n-1}| \to |\Delta^n|$, the inclusion as the rth face. By inductive hypothesis, $\partial H_{n-1}\partial|\Delta^n| + H_{n-2}\partial\partial|\Delta^n| = T \circ Wh\partial|\Delta^n| - Wh\partial|\Delta^n| = \partial H_{n-1}\partial|\Delta^n|$. So the condition above means ∂H_n is a cycle ($\partial H_n = -H_{n-1}\partial|\Delta|^n + T \circ Wh|\Delta^n| - Wh|\Delta^n| = 0$, i.e. ∂H_n is a cycle). Since $|\Delta^n|$ is contractible, $Sing_*|\Delta^n| \otimes Sing_*|\Delta^n|$ is acyclic. Thus like the acyclic carrier approach there is an element H_n as required. So $\forall n \geq 0, H_n$

exists.

for each $n \geq 0$, $\forall \sigma$, a singular n-simplex, define $H(\sigma) = \sigma_* H_n$. This defines a linear map $H : Sing_*(X) \to Sing_*(X) \otimes Sing_*(X)$ s.t. $\partial H \sigma + H \partial \sigma = T \circ Wh\sigma - Wh\sigma$.

* **the Acyclic Carrier Theorem**

Definition 3.2.3. *Let K,L be simplicial complexes and $f_* : C_*(K) \to C_*(L)$ be a chain map between their associated chain complexes. A carrier for f_* is a function $\Phi : K \to \{$sub-simplicial complexes of $L\}$ s.t.*

1. *$f_n([a_0,\ldots,a_n])$ may be represented by a sum of simplexes in $\Phi(< a_0,\ldots,a_n >)$*
2. *if $\tau \leq \sigma$, $\Phi(\tau) \subseteq \Phi(\sigma)$*

A carrier Φ is called acyclic if $\Phi(\sigma)$ has the homology of the one-point simplicial complex for every simplex $\sigma \in K$.

Definition 3.2.4. *For a simplicial complex K, define* $\begin{array}{rcl} \epsilon_K : C_0(K) & \to & \mathbb{Z} \\ a_0 & \mapsto & 1 \end{array}$. *A chain map $f_* : C_*(K) \to C_*(L)$ is said to be augmentation-preserving if $\epsilon_K = \epsilon_L \circ f_0$.*

Theorem 3.2.1 (the Acyclic Carrier Theorem). *Let $f_*, g_* : C_*(K) \to C_*(L)$ be augmentation-preserving chain maps and Φ be a carrier for both of them which is acyclic. Then $f_* \cong g_*$.*

pf. When $n = 0$, choose a total order $\preceq$ on V_K so that a basis for $C_n(K)$ is given by
$\{[a_0,\ldots,a_n] | < a_0,\ldots,a_n > \in K,\ a_0 \prec \cdots \prec a_n\}$. For each vertex $a_0 \in K$, $x = (f_0 - g_0)([a_0])$ is a sum of 0-simplexes of $\Phi(< a_0 >)$ so represents a class $[x] \in H_0(\Phi(< a_0 >)) \cong \mathbb{Z}$. But $(\epsilon_{\Phi(<a_0>)})_* = \epsilon_L((f_0 - g_0)([a_0])) = \epsilon_L(f_0([a_0])) - \epsilon_L(g_0([a_0])) = 1 - 1 = 0$ and $(\epsilon_{\Phi(<a_0>)})_*$ is an isomorphism, so $[x] = 0$. Thus $\exists y \in C_1(\Phi(< a_0 >))$ so that $x = \partial_1 y$ and we define $H_0([a_0]) = y$. By construction, $(\partial_1 \circ H_0 + H_{-1} \circ \partial_0)([a_0]) = \partial_1 y = x = (f_0 - g_0)([a_0])$.
When $n \geq 0$, assume we have defined homomorphisms $H_i : C_i(K) \to C_{i+1}(L)$ s.t. $\begin{cases} \partial_{i+1} \circ H_i + H_{i-1} \circ \partial_i = f_i - g_i \\ H_i(\sigma) \text{ represented by a sum of simplexes in } \Phi(\sigma) \end{cases}$. Consider $x = (f_n - g_n - H_{n-1} \circ \partial_n)([a_0,\ldots,a_n])$. Both $\begin{cases} f_n([a_0,\ldots,a_n]) \\ g_n([a_0,\ldots,a_n]) \end{cases}$ are sums of simplexes in $\Phi(< a_0,\ldots,a_n >)$ as both maps are carried by Φ. The chain $H_{n-1} \circ \partial_n([a_0,\ldots,a_n])$ is a sum of simplexes each in $\Phi(\tau)$ for some $\tau \leq < a_0,\ldots,a_n > \Leftrightarrow \Phi(\tau) \subseteq \Phi(< a_0,\ldots,a_n >)$ and therefore $H_{n-1} \circ \partial_n([a_0,\ldots,a_n])$ is also a sum of simplexes in $\Phi(< a_0,\ldots,a_n >)$. Thus $x \in C_n(\Phi(< a_0,\ldots,a_n >))$. Then

$\partial_n x = (\partial_n \circ f_n - \partial_n \circ g_n - \partial_n \circ H_{n-1} \circ \partial_n)([a_0,\dots,a_n])$
$= (f_{n-1} \circ \partial_n - g_{n-1} \circ \partial_n - \partial_n \circ H_{n-1} \circ \partial_n)([a_0,\dots,a_n])$
$= (f_{n-1} \circ \partial_n - g_{n-1} \circ \partial_n - (f_{n-1} - g_{n-1} - H_{n-2} \circ \partial_{n-1}) \circ \partial_n)([a_0,\dots,a_n])$
$= H_{n-2} \circ \partial_{n-1} \circ \partial_n([a_0,\dots,a_n]) = 0$. Thus as Φ is acyclic, $H_n(\Phi(< a_0,\dots,a_n >)) = 0$ and so $x = \partial_{n+1} y$, $\exists y \in C_{n+1}(\Phi(< a_0,\dots,a_n >))$. Define $H_n([a_0,\dots,a_n]) = y$. Then $\partial_{n+1} \circ H_n + H_{n-1} \circ \partial_n = f_n - g_n$.

Let α^p, β^q be cocycles. Then
$(-1)^{pq} < \beta \cup \alpha, \sigma > - < \alpha \cup \beta, \sigma > = < \alpha \otimes \beta, T \circ Wh(\sigma) - Wh(\sigma) >$
$= < \alpha \otimes \beta, \partial H\sigma + H\partial\sigma > = < \alpha \otimes \beta, H\partial\sigma >$
$\Leftrightarrow (-1)^{pq} < \beta \cup \alpha, \sigma > - < \alpha \cup \beta, \sigma > = < dH^*(\alpha \otimes \beta), \sigma >$
$\Leftrightarrow (-1)^{pq} \beta \cup \alpha - \alpha \cup \beta = dH^*(\alpha \otimes \beta)$, i.e. $\alpha \cup \beta, (-1)^{pq} \beta \cup \alpha$ represent the same cohomology class.

Proposition 3.2.2. *If α, β are cocycles on X, then $\alpha \otimes \beta \in Sing^*(X) \otimes Sing^*(X)$ defines a singular cohomology class on $X \times X$. Define $\Delta_X : X \to X \times X$ as the diagonal map. Then we have*
$$\Delta_X^*([\alpha \otimes \beta]) = [\alpha] \cup [\beta]$$
in $H^*(X)$.

pf. The geometric realisation $|Sing(X)|$ has a natural map to X which is a weak homotopy equivalence. Thus $|Sing(X)| \times |Sing(X)|$ has a map to $X \times X$ which is a weak homotopy equivalence. The product of the geometric realisations has the product cell structure from the cell structure on the factors. $Sing_*(X) \otimes Sing_*(X)$ is identified with the cellular chains on this cell complex and hence computes the homology of $X \times X$. The geometric realisation $|Sing(X) \times Sing(X)|$ produces a subdivision of the cell complex structure on $|Sing(X)| \times |Sing(X)|$. Hence there is a map from the cellular chain on $|Sing(X)| \times |Sing(X)|$, which is $Sing_*(X) \otimes Sing_*(X)$, to the cellular chains on $|Sing(X) \times Sing(X)|$. This map assigns to a cell in $|Sing(X)| \times |Sing(X)|$, a linear combination of cells in $|Sing(X) \times Sing(X)|$ that contain an open subset of the original cell with coefficients, where the coefficients are signs determined by the relative orientations. This map induces an isomorphism on cohomology. Thus $\exists \widetilde{\alpha \otimes \beta}$ on $|Sing(X) \times Sing(X)|$ that restricts to a cocycle. The diagonal map $|Sing(X)| \to |Sing(X)| \times |Sing(X)|$ is a cellular map when the range is given the cell structure coming from $|Sing(X) \times Sing(X)|$. It is weakly homotopy equivalent to the diagonal map. The pullback by the Δ_X^* of $\widetilde{\alpha \otimes \beta}$ is a cocycle representing the pullback of the exterior product. Then this pullback by the Whitney map of the restriction of $\widetilde{\alpha \otimes \beta}$ is cohomologous to the pullback by the Whitney map of $\alpha \otimes \beta$, which is $\alpha \cup \beta$. Then by the same acyclic carrier argument applied to the simplicial cochain complex on $|Sing(X) \times Sing(X)|$ shows

that $(\Delta_X)_*$ and Wh are chain homotopic.
$\rightarrow$ for X, Y topological spaces, $H^*(X)\otimes H^*(Y)\rightarrow H^*(X\times Y)$ which is the exterior product of cohomology classes.

3.2.2 The Steenrod Operations

$Wh|\Delta^n| = \sum_{k=0}^{n} P(k,n)$ where $P(k,n)\in Sing_k|\Delta^n|\otimes Sing_{n-k}|\Delta^n|$ is a tensor product of the singular complex that is the inclusion of the face spanned by the vertices $0-k$, with the simplex that is the inclusion of the face spanned by the vertices $k-n$. Denote $P(k,l,n)\in Sing_*|\Delta^n|\otimes Sing_*|\Delta^n|$ a tensor product of the inclusion of the face spanned by vertices $0-k, l-n$ with the inclusion of the face spanned by the vertices $k-l$. Let $H_0 = Wh$ of degree 0 with $\partial H_0 - H_0\partial = 0$, given Wh a chain map. Define $H_1|\Delta^n| = \sum_{0\le k<l\le n}\epsilon(k,l,n)P(k,l,n)\in Sing_*|\Delta^n|\otimes Sing_*|\Delta^n|$, a map of degree 1.

$$\partial H_1 + H_1\partial = T\circ H_0 - H_0$$

pf. $\partial H_1(|\Delta^n|) = \sum\sum_{0\le k<l\le n}\epsilon(k,l,n)P(k,l,n)$ where $P(k,l,n)$ has two types
$\begin{cases}\text{non-degenerate diagram}\\ \text{degenerate diagram}\end{cases}$ where the non-degenerate diagram appears in $H_1\partial|\Delta^n|$.
Thus non-degenerate cases cancel out in pairs, except the extremal diagrams, which splits at H_0. The degenerate diagram occurs twice in the sum except the extremal case which yields $T\circ Wh - Wh$. So the above follows by naturality.

Define $H_k|\Delta^n|\in Sing_*|\Delta^n|\otimes Sing_*|\Delta^n|$ of degree $n+k$. Then $H_k : Sing_*(X)\rightarrow Sing_*(X)\otimes Sing_*(X)$ map of degree k by setting $H_k(\sigma) = (\sigma_*\otimes\sigma_*)H_k|\Delta^n|$. They are defined to satisfy $\partial H_k - (-1)^k H_k\partial = T\circ H_{k-1} - (-1)^{k-1}H_{k-1}$. Then the diagram corresponds to an element of degree $n+k$ in $Sing_*|\Delta^n|\otimes Sing_*|\Delta^n|$ given by the tensor product of the inclusion of the face spanned by the vertices in the front-face with the inclusion of the face spanned by the vertices in the back-face.
$\rightarrow H_k|\Delta^n| = \sum_{0\le a_0<\cdots<a_k\le n}\epsilon(a_0,\ldots,a_k;n)P(a_0,\ldots,a_k;n)$**, which gives a map of degree** k
$H_k : Sing_*(X)\rightarrow Sing_*(X)\otimes Sing_*(X)$ **s.t.**

$$\partial H_k - (-1)^k H_k\partial = T\circ H_{k-1} - (-1)^{k-1}H_{k-1}$$

is a closed element.
pf. $\partial(-1)^k H_k\partial|\Delta^n| = H_k\partial\partial|\Delta^n| + (-1)^k[T\circ H_{k-1}\partial|\Delta^n| - (-1)^{k-1}H_{k-1}\partial|\Delta^n|$
$= (-1)^k[T\circ H_{k-1}\partial|\Delta^n| - (-1)^{k-1}H_{k-1}\partial|\Delta^n|]$. By the inductive hypothesis on H_{k-1},
$(-1)^k T\circ H_{k-1}\partial|\Delta^n| + T\circ\partial H_{k-1}|\Delta^n| = T^2\circ H_{k-2}|\Delta^n| - (-1)^{k-2}T\circ H_{k-2}|\Delta^n|$
$= (-1)^{k-1}(T\circ H_{k-2}|\Delta^n| - (-1)^{k-2}H_{k-2}|\Delta^n|)$. Also $(-1)^k(-1)(-1)^{k-1}H_{k-1}\partial|\Delta^n| - (-1)^{k-1}\partial H_{k-1}|\Delta^n|$
$= (-1)^k(T\circ H_{k-2}|\Delta^n| - (-1)^{k-2}H_{k-2}|\Delta^n|)$. Both of the equations give the claim

above.
→ for any singular n-simplex $\sigma, H_k(\sigma) = (\sigma_*) \otimes (\sigma_*) H_k |\Delta^n|$.

Definition 3.2.5. *For* $\begin{cases} \alpha \in Sing^p(X;\mathbb{Z}/2\mathbb{Z}) \\ \beta \in Sing^q(X;\mathbb{Z}/2\mathbb{Z}) \end{cases}$, $\forall k$, *define* $\alpha \cup_k \beta \in Sing^{p+q-k}(X;\mathbb{Z}/2\mathbb{Z})$ *s.t.*

$$< \alpha \cup_k \beta, \sigma^{p+q-k} > = < \alpha \otimes \beta, H_k(\sigma) >$$

called the higher cup-product.

→ $k = 0$, the above formula gives the Whitney cup-product.

$$\delta(\alpha \cup_k \beta) = (\delta\alpha) \cup_k \beta + \alpha \cup_k (\delta\beta) + \alpha \cup_{k-1} \beta + \beta \cup_{k-1} \alpha$$

pf. $< \alpha \otimes \beta, \partial H_k \sigma + H_k \partial\sigma + T \circ H_{k-1}\sigma + H_{k-1}\sigma > = 0$ by the construction of H_*. Thus

$< \alpha \otimes \beta, \partial H_k \sigma > = < \alpha \cup_k \beta, \partial\sigma > + < \beta \cup_{k-1} \alpha, \sigma > + < \alpha \cup_{k-1} \beta, \sigma >$
$= < \delta(\alpha \cup_k \beta) + \beta \cup_{k-1} \alpha + \alpha \cup_{k-1} \beta, \sigma >$. Additionally,
$< \alpha \otimes \beta, \partial H_k \sigma > = < \delta(\alpha \otimes \beta), H_k \sigma > = < (\delta\alpha) \cup_k \beta + \alpha \cup_k (\delta\beta), \sigma >$, hence the equation follows.

Let $\alpha \in Sing^*(X;\mathbb{Z}/2)$. Then $\alpha \otimes \alpha \in \hom(Sing_*(X) \otimes Sing_*(X), \mathbb{Z}/2)$ is an equivariant cocycle, and $[\alpha \otimes \alpha] \in H^{2p}(X \times X;\mathbb{Z}/2)$.

Definition 3.2.6. *Define the Steenrod square* $Sq^k(\alpha^p) = \alpha \cup_{p-k} \alpha$.

→ $\begin{cases} \text{if } \alpha \text{ is closed, so is } Sq^k(\alpha) \text{ by the above definition.} \\ \text{if } \alpha \text{ vary in coboundaries, so does } Sq^k\alpha \end{cases}$ → Sq^k passes to a map $H^p(X;\mathbb{Z}/2\mathbb{Z}) \to H^{p+k}(X;\mathbb{Z}/2\mathbb{Z})$, which is natural in X.

3.2.3 The Stiefel-Whitney Class

Let $H^i(B;G)$ denote the ith singular cohomology group of B with coefficients in G.

Definition 3.2.7. *To each vector bundle* ξ*, there exists a sequence of cohomology classes*

$$w_i(\xi) \in H^i(B(\xi);\mathbb{Z}/2)$$

called the Stiefel-Whitney classes of ξ*. The class* $w_0(\xi)$ *is equal to the unit element* $1 \in H^0(B(\xi);\mathbb{Z}/2)$ *and* $w_i(\xi)$ *equals zero for* $i > n$ *if* ξ *is an* n*-plane bundle.*

Then the above satisfies

$$\begin{cases} \text{naturality} & \text{if } f : B(\xi) \to B(\eta),\ \xi \to \eta, w_i(\xi) = f^* w_i(\eta) \\ \text{the whitney product theorem} & w_k(\xi \oplus \eta) = \sum_{i=0}^k w_i(\xi) \cup w_{k-i}(\eta) \\ \text{the line bundle } \gamma_1^1 \text{ over } \mathbb{R}P^1 & w_1(\gamma_1^1) \neq 0 \end{cases}.$$

e.g. for the line bundle γ_1^1 over the circle $\mathbb{P}^1$, the Stiefel-Whitney class $w_1(\gamma_1^1)$ is nonzero.

Proposition 3.2.3. *1. if $\xi \sim \eta$, $w_i(\xi) = w_i(\eta)$*

2. *if ϵ is a trivial vector bundle, then $w_i(\epsilon) = 0$, $\forall i > 0$.*
3. *if ϵ is trivial, $w_i(\epsilon \oplus \eta) = w_i(\eta)$*
4. *if ξ is an $\mathbb{R}^n$-bundle with a Euclidean metric, which possesses a nowhere zero cross-section, $w_n(\xi) = 0$. If ξ possesses k cross-sections which are nowhere linearly dependent, then $w_{n-k+1}(\xi) = w_{n-k+2}(\xi) = \cdots = w_n(\xi) = 0$.*

Definition 3.2.8. *$H^{\Pi}(B;\mathbb{Z}/2)$ denote the ring consisting of all formal infinite series $a = a_0 + a_1 + \dots$ with $a_i \in H^i(B;\mathbb{Z}/2)$ with a commutative product operation $(a_0 + a_1 + \dots)(b_0 + b_1 + \dots) = (a_0 b_0) + (a_1 b_0 + a_0 b_1) + \dots$.*

$\rightarrow$the total Stiefel-Whitney class of an n-plane bundle ξ over B is defined to be the element $w(\xi) = 1 + w_1(\xi) + \cdots + w_n(\xi) + 0 + \dots$ of this ring. Then $w(\xi \oplus \eta) = w(\xi) w(\eta)$.

Lemma 3.2.2. *The collection of all infinite series $w = 1 + w_1 + w_2 + \cdots \in H^{\Pi}(B;\mathbb{Z}/2)$ with leading term 1 forms a commutative group under multiplication.*

pf. The inverse $\bar{w} = 1 + \bar{w}_1 + \bar{w}_2 + \dots$ of a given element w can be constructed inductively by the algorithm $\bar{w}_n = w_1 \bar{w}_{n-1} + w_2 \bar{w}_{n-2} + \cdots + w_{n-1} \bar{w}_1 + w_n \rightarrow \begin{cases} \bar{w}_1 = w_1 \\ \bar{w}_2 = w_1^2 + w_2 \\ \bar{w}_3 = w_1^3 + w_3 \\ \vdots \end{cases}$.

Then the lemma follows.

Consider two bector bundles ξ, η over the same base space. Then
$w(\eta \oplus \xi) = w(\eta) w(\xi) \Leftrightarrow w(\eta) = \bar{w}(\xi) w(\eta \oplus \xi)$.

Lemma 3.2.3 (Whitney Duality Theorem)**.** *If τ_M is the tangent bundle of a manifold in Euclidean space and ν is the normal bundle, then*

$$w_i(\nu) = \bar{w}_i(\tau_M)$$

pf. The Whitney sum $\tau_M \oplus \nu$ is the restriction to M of the tangent bundle of the ambient Euclidean space, hence trivial. Then $1 = w(\tau_M \oplus \nu) = w(\tau_M) w(\nu)$ by the Whitney product theorem, so $w(\nu) = \bar{w}(\tau_M)$.

e.g. for the tangent bundle τ of the unit sphere S^n, $w(\tau) = w(S^n)$ is equal to 1, i.e. τ cannot be distinguished from the trivial bundle over S^n by means of the Stiefel-Whitney classes.
pf. The canonical map $f : S^n \to \mathbb{R}P^n$ is locally a diffeomorphism and therefore the induced map $Df : DS^n \to D\mathbb{R}P^n$ of tangent bundles is a bundle map. Then by naturality, $f^* w_n(\mathbb{R}P^n) = w_n(S^n)$ where $f^* : H^n(\mathbb{R}P^n;\mathbb{Z}/2) \to H^n(S^n;\mathbb{Z}/2)$ is zero. Thus $w_n(S^n) = 0$, the statement above follows.

Lemma 3.2.4. *The group $H^i(\mathbb{R}P^n;\mathbb{Z}/2)$ is* $\begin{cases} \text{cyclic of order 2} & 0 \le i \le n \\ 0 & n \ge i \end{cases}$. *If a denote the nonzero element of $H^1(\mathbb{R}P^n;\mathbb{Z}/2)$, each $H^i(\mathbb{R}P^n;\mathbb{Z}/2)$ is generated by the i-fold cup product a^i.*

$\to H^*(\mathbb{R}P^n;\mathbb{Z}/2)$ can be described as the algebra with unit over $\mathbb{Z}/2$ having one generator a and one relation $a^{n+1} = 0$.

e.g.**the total Stiefel-Whitney class of the canonical line bundle γ_n^1 over $\mathbb{R}P^n$**
consider the standard inclusion $j : \mathbb{R}P^1 \to \mathbb{R}P^n$ covered by a bundle map $\gamma_1^1 \to \gamma_n^1$. Then by naturality, $j^* w_1(\gamma_n^1) = w_1(\gamma_1^1) \ne 0$. Thus $w_1(\gamma_1^1) = a$ and the remaining Stiefel-Whitney class of γ_n^1 are determined by the definition, so $w(\gamma_n^1) = 1 + a$. This line bundle γ_n^1 is contained as a subbundle in the trivial bundle ϵ^{n+1} by definition. So if we denote by $\gamma^\perp$ the orthogonal complement of γ_n^1 in ϵ^{n+1}, $(\gamma_n^1 \oplus \gamma^\perp) = \epsilon^{n+1} \Leftrightarrow w(\gamma^\perp) = \bar{w}(\gamma_n^1) = (1+a)^{-1} = 1 + a + \cdots + a^n$. Let L be a line through the origin in $\mathbb{R}^{n+1}$ intersecting S^n in the points $\pm x$, and let $L^\perp \subset \mathbb{R}^{n+1}$ be the complementary n-plane. Then we have the canonical map $\begin{array}{rcl} f : S^n & \to & \mathbb{R}P^n \\ x & \mapsto & \{\pm x\} \end{array}$. Then two tangent vectors $\begin{cases} (x, v) \\ (-x, -v) \end{cases}$ in DS^n have the same image under the map $Df : DS^n \to D\mathbb{R}P^n$ induced by f. Thus the tangent manifold $D\mathbb{R}P^n$ can be identified with the set of all pairs $\{(x,v),(-x,-v)\}$ satisfying $\begin{cases} x \cdot x = 1 \\ x \cdot v = 0 \end{cases}$. Such pair determines and is determined by a linear mapping $\begin{array}{rcl} l : L & \to & L^\perp \\ x & \mapsto & v \end{array}$. Thus the tangent space of $\mathbb{R}P^n$ at $\{\pm x\}$ is canonically isomorphic to the vector space $\hom(L, L^\perp) \cong \hom(\gamma_n^1, \gamma^\perp)$.

Theorem 3.2.2. *The Whitney sum $\tau \oplus \epsilon^1$ is isomorphic to the $n+1$-fold Whitney sum $\gamma_n^1 \oplus \cdots \oplus \gamma_n^1$. Hence the total Stiefel-Whitney class of $\mathbb{R}P^n$ is given by*

$$w(\mathbb{R}P^n) = (1+a)^{n+1} = 1 + \binom{n+1}{1} a + \cdots + \binom{n+1}{n} a^n$$

pf. The bundle $\hom(\gamma_n^1, \gamma_n^1)$ is trivial since it is a line bundle with a canonical nowhere zero cross-section. Thus $\tau \oplus \epsilon^1 \cong \hom(\gamma_n^1, \gamma^\perp) \oplus \hom(\gamma_n^1, \gamma_n^1) \cong \hom(\gamma_n^1, \gamma^\perp \oplus \gamma_n^1) \cong \hom(\gamma_n^1, \epsilon^{n+1}) \cong \hom(\gamma_n^1, \epsilon^1 \oplus \cdots \oplus \epsilon^1) \cong \hom(\gamma_n^1, \epsilon^1) \oplus \cdots \oplus \hom(\gamma_n^1, \epsilon^1) = \gamma_n^1 \oplus \cdots \oplus \gamma_n^1$, given γ_n^1 having a Euclidean metric. By the Whitney product theorem, $w(\tau) = w(\tau \oplus \epsilon^1) \to w(\gamma_n^1) \ldots w(\gamma_n^1) = (1+a)^{n+1}$ and the theorem follows.

Corollary 3.2.2. *The class $w(\mathbb{R}P^n) = 1$ iff $n+1$ is a power of 2. Thus the only projective spaces which can be parallelisable are $\mathbb{R}P^1, \mathbb{R}P^3, \mathbb{R}P^7, \ldots$.*

pf.

- $(a+1)^{2^r} = 1 + a^{2^r} \mod 2$ and therefore if $n+1 = 2^r, w(\mathbb{R}P^n) = (1+a)^{n+1} = 1 + a^{n+1} = 1$
- if $n+1 = 2^r m$ with m odd, $m > 1$, then $w(\mathbb{R}P^n) = (1+a)^{n+1} = (1+a^{2^r})^m = 1 + ma^{2^r} + \frac{m(m-1)}{2} a^{2 \cdot 2^r} + \ldots \neq 1$, given $2^r < n+1$.

Theorem 3.2.3. *Suppose there exists a bilinear product operation $p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ without zero divisors. Then the projective space $\mathbb{R}P^{n-1}$ is parallelisable, hence n must be a power of 2.*

pf. Let $b_1, b_2, \ldots, b_n$ be the standard basis for the vector space $\mathbb{R}^n$. Then the correspondence $y \mapsto p(y, b_1)$ defines an isomorphism of $\mathbb{R}^n$ onto itself. Hence $v_i(p(y, b_1)) = p(y, b_i)$ defines a linear transformation $v_i : \mathbb{R}^n \to \mathbb{R}^n$ where v_is are linearly independent for $x \neq 0$ and $v_1(x) = x$. The functions $v_2, \ldots, v_n$ give rise to $n-1$ linearly independent cross-sections of the vector bundle $\tau_{\mathbb{R}P^{n-1}} \cong \hom(\gamma^1_{n-1}, \gamma^\perp)$. In fact for each line L through the origin, a linear transformation $\bar{v} : L \to L^\perp$ is defined as follows; $\forall x \in L$, let $\bar{v}_i(x)$ denote the image of $v_i(x)$ under the orthogonal projection $\mathbb{R}^n \to L^\perp$. Then $\bar{v}_1 = 0$ but $\bar{v}_i$ are linearly independent for $i \geq 2$. Thus the tangent bundle $\tau_{\mathbb{R}P^{n-1}}$ is a trivial bundle.

Theorem 3.2.4. *If $\mathbb{R}P^{2^r}$ can be immersed in $\mathbb{R}^{2^r+k}$, then k must be at least $2^r - 1$.*

pf. If a manifold M of dimension n can be immersed in $\mathbb{R}^{n+k}$, $w_i(\nu) = \bar{w}_i(M)$ by the Whitney duality theorem. This implies $\bar{w}_i(M) = 0$ for $i > k$. Here $w(\mathbb{R}P^n) = (1+a)^{n+1} = 1 + a + \cdots + a^n \Leftrightarrow \bar{w}(\mathbb{R}P^n) = 1 + a + \cdots + a^{n-1}$ given $n = 2^r$. Thus the theorem follows.

Lemma 3.2.5. *There are no polynomial relations among the $w_i(\gamma^n)$.*

pf. Suppose the contrary. Then $\exists$ a relation of the form $p(w_1(\gamma^n), \ldots, w_n(\gamma^n)) = 0$, where p is a polynomial in n variables with $\mod 2$ coefficients. By the theorem 3.1.6, for any n-plane bundle ξ over a paracompact base space, there exists a bundle map $g : \xi \to \gamma^n$. Hence $w_i(\xi) = \bar{g}^*(w_i(\gamma^n))$ where $\bar{g}$ is the map of base spaces induced by g. then the cohomology classes $w_i(\xi)$ must satisfy $p(w_1(\xi), \ldots, w_n(\xi)) = \bar{g}^* p(w_1(\gamma^n), \ldots, w_n(\gamma^n)) = 0$. Consider the canonical line bundle γ^1 over the infinite projective space $\mathbb{R}P^\infty$. Recall $H^*(\mathbb{R}P^\infty; \mathbb{Z}/2)$ is a polynomial algebra over $\mathbb{Z}/2$ with a single generator a of dimension 1, and $w(\gamma^1) = 1 + a$. Forming the n-fold cartesian product $X = \mathbb{R}P^\infty \times \cdots \times \mathbb{R}P^\infty$, $H^*(X; \mathbb{Z}/2)$ is a polynomial algebra on n generators $a_1, \ldots, a_n$ of dimension 1. Here a_i can be defined as the image $p_i^*(a)$ induced by the projection map $p_i : X \to \mathbb{R}P^\infty$ to the ith factor. Let ξ be the n-fold cartesian product $\xi = \gamma^1 \times \cdots \times \gamma^1 \cong (p_1^* \gamma^1) \oplus \cdots \oplus (p_n^* \gamma^1)$. Then

ξ is an n-plane bundle over $X = \mathbb{R}P^\infty \times \cdots \times \mathbb{R}P^\infty$ and the total Stiefel-Whiteny class $w(\xi) = w(\gamma^1) \times \ldots w(\gamma^1) = p_1^*(w(\gamma^1)) \ldots p_n^*(w(\gamma^1))$ is equal to the n-fold product

$$(1+a) \times \cdots \times (1+a) = (1+a_1)\ldots(1+a_n) \to \begin{cases} w_1(\xi) = & a_1 + a_2 + \cdots + a_n \\ w_2(\xi) = & a_1a_2 + a_1a_3 + \cdots + a_{n-1}a_n \\ \vdots & \vdots \\ w_n(\xi) = & a_1 \ldots a_n \end{cases},$$

i.e. $w_k(\xi)$ is the kth elementary symmetric function of $a_1, \ldots, a_n$. Since the n elementary symmetric function in n indeterminates over a field do not satisfy any polynomial relations, the class $w_1(\xi), \ldots, w_n(\xi)$ are algebraically independent over $\mathbb{Z}/2$ and the lemma follows.

Theorem 3.2.5. *The cohomology ring $H^*(Gr_n; \mathbb{Z}/2)$ is a polynomial algebra over $\mathbb{Z}/2$ freely generated by the Stiefel-Whitney classes $w_1(\gamma^n), \ldots, w_n(\gamma^n)$.*

pf. By the previous lemma, $H^*(Gr_n)$ with $\mod 2$ coefficients contains a polynomial algebra over $\mathbb{Z}/2$ freely generated by $w_1(\gamma^1), \ldots, w_n(\gamma^n)$.
CLAIM : The polynomial algebra over $\mathbb{Z}/2$ above coincides with $H^*(Gr_n)$.
pf. The number of r-cells in the CW complex Gr_n is equal to the number of partitions of r into at most n integers. Hence the rank $H^r(Gr_n)$ over $\mathbb{Z}/2$ is at most equal to this number of partitions. On the other hand, the number of distinct monomials of the form $w_1(\gamma^n)^{r_1} \ldots w_n(\gamma^n)^{r_n}$ in $H^r(Gr_n)$ is also precisely equal to the number of partitions of r into at most n integers. For to each sequence $r_1, \ldots, r_n$ of nonnegative integers with $r_1 + 2r_2 + \cdots + nr_n = r$, we can associate the partition of r which is obtained from the n-tuple $r_n, r_n + r_{n-1}, \ldots, r_n + r_{n-1} + \cdots + r_1$ by deleting any zeros which may occur; and the converse also holds. Since these monomials are known to be linearly independent $\mod 2$, the module $H^r(Gr_n)$ over $\mathbb{Z}/2$ has rank equal to the number of partitions of r into at most n integers, and has a basis consisting of the various monomials $w_1(\gamma^n)^{r_1} \ldots w_n(\gamma^n)^{r_n}$ of total dimension r.
$\bar{g}^* : H^*(Gr_n) \to H^*(\mathbb{R}P^\infty \times \cdots \times \mathbb{R}P^\infty)$ maps $H^*(Gr_n)$ isomorphically onto the algebra consisting of all polynomials in the indeterminates $a_1, \ldots, a_n$ which are invariant under all permutations of these n indeterminates.

Theorem 3.2.6 (Uniqueness Theorem)**.** *There exists at most one correspondence $\xi \mapsto w(\xi)$ which assigns to each vector bundle over a paracompact base space a sequence of cohomology classes satisfying the four axioms for Stiefel-Whitney classes.*

pf. Suppose the contrary; $\exists \begin{cases} \xi \mapsto w(\xi) \\ \xi \mapsto \tilde{w}(\xi) \end{cases}$. For the canonical line bundle γ_1^1 over $\mathbb{R}P^1$, $w(\gamma_1^1) = \tilde{w}(\gamma_1^1) = 1 + a$ by the axiom 1 and 4. Embedding γ_1^1 in the line bundle γ^1 over the infinite projective space $\mathbb{R}P^\infty$, $w(\gamma^1) = \tilde{w}(\gamma^1) = a + 1$ by axioms 1 and 2. Passing to the n-fold cartesian product $\xi = \gamma^1 \times \dot{\times} \gamma^1 \cong p_1^* \gamma^1 \oplus \cdots \oplus p_n^* \gamma^1$. This implies

$w(\xi) = \tilde{w}(\xi) = (1+a_1)\dots(1+a_n)$ by axioms 2 and 3. Since there is a bundle map $\xi \to \gamma^n$ and $H^*(Gr_n)$ injects monomorphically into $H^*(\mathbb{R}P^\infty \times \cdots \times \mathbb{R}P^\infty)$, $w(\gamma^n) = \tilde{w}(\gamma^n)$. For any n-plane bundle η over a paracompact base space, choosing a bundle map $f : \eta \to \gamma^n$, $w(\eta) = \bar{f}^* w(\gamma^n) = \bar{f}^* \tilde{w}(\gamma^n) = \tilde{w}(\eta)$.

For any n-plane bundle ξ with total space E, base space B and projection map p, we denote by E_0 the set of all nonzero elements of E, and by F_0 the set of all nonzero elements of a typical fiber $F = p^{-1}(b)$. Then $F_0 = F \cap E_0$.

Definition 3.2.9. *The Thom isomorphism $\phi : H^k(B) \to H^{k+n}(E,E_0)$ is defined to be the composition of the two isomorphisms* $H^k(B) \xrightarrow{p^*} H^k(E) \xrightarrow{\cup u} H^{k+n}(E,E_0)$

$\to$ the Stiefel-Whitney class $w_i(\xi) \in H^i(B)$ can be considered as Thom's identity $w_i(\xi) = \phi^{-1} Sq^i \phi(1)$, i.e. $w_i(\xi)$ is the unique cohomology class in $H^i(B)$ s.t. $\phi(w_i(\xi)) = p^* w_i(\xi) \cup u$ is equal to $Sq^i \phi(1) = Sq^i(u)$.

Theorem 3.2.7. *The group $H^i(E,E_0)$ is zero for $i < n$ and $H^n(E,E_0)$ contains a unique class u s.t. for each fiber $F = p^{-1}(b)$, the restriction $u|_{F,F_0} \in H^n(F,F_0)$ is the unique nonzero class in $H^n(F,F_0)$. Furthermore, the correspondence $x \mapsto x \cup u$ defines an isomorphism $H^k(E) \to H^{k+n}(E,E_0)$ for every k and u is called the fundamental cohomology class.*

3.3 The Thom Classes

3.3.1 The Euler Class and Thom Isomorphism

Definition 3.3.1. *The Euler class of an oriented n-plane bundle ξ is the cohomology class $e(\xi) \in H^n(B;\mathbb{Z})$ which corresponds to $u|_E$ under the canonical isomorphism $p^* : H^n(B;\mathbb{Z}) \to H^n(E;\mathbb{Z})$.*

1. **If $f : B \to B'$ is covered by an orientation preserving bundle map $\xi \to \xi'$, $e(\xi) = f^* e(\xi')$.**

2. **If the orientation of ξ is reversed, the Euler class of $e(\xi)$ changes sign.**

3. **If the fiber dimension n is odd, $e(\xi) + e(\xi) = 0$.**
   the Thom isomorphism $\phi(x) = p^*(x) \cup u$ equivalently maps $e(\xi)$ to the cohomology class $p^* e(\xi) \cup u = u|_E \cup u = u \cup u \to e(\xi) = \phi^{-1}(u \cup u)$. Since $a \cup b = (-1)^{\dim a \dim b} b \cup a$, the statement follows.

4. **The natural homomorphism $H^n(B;\mathbb{Z}) \to H^n(B;\mathbb{Z}/2)$ carries the Euler class $e(\xi)$ to the top Stiefel-Whitney class $w_n(\xi)$.**
   $e(\xi) = \phi^{-1}(u \cup u) \to \phi^{-1} Sq^n(u)$ by applying the above homomorphism and the statement follows.

5. **The Euler class of a Whitney sum and the Euler class of a cartesian product are given by**

   $$e(\xi \oplus \xi') = e(\xi) \cup e(\xi'), \quad e(\xi \times \xi') = e(\xi) \times e(\xi')$$

   let the fiber dimensions be m,n respectively. The fundamental cohomology class of the cartesian product is given by $u(\xi \times \xi') = (-1)^{mn} u(\xi) \times u(\xi')$. By the restriction homomorphism $H^{m+n}(E \times E', (E \times E')_0) \to H^{m+n}(E \times E') \cong H^{m+n}(B \times B')$, we get $e(\xi \times \xi') = (-1)^{mn} e(\xi) \times e(\xi')$ where the sign can be ignored because the RHS is an element of order 2.

6. **If the oriented vector bundle ξ possesses a nowhere zero cross-section, the Euler class $e(\xi)$ must be zero.**
   let $s : B \to E_0$ be a cross-section, so that $B \xrightarrow{s} E_0 \xrightarrow{i} E \xrightarrow{p} B$ is the identity map of B. Then $H^n(B) \xrightarrow{p^*} H^n(E) \longrightarrow H^n(E_0) \xrightarrow{s^*} H^n(B)$ is the identity map of $H^n(B)$. By the definition, p^* maps $e(\xi)$ to the restriction $u|_E$. Hence the first two homomorphisms map $e(\xi)$ to $(u|_E)|_{E_0}$ which is zero ($H^n(E,E_0) \to H^n(E) \to H^n(E_0)$ exact). Applying s^* gives $s^*(0) = 0$.

Let $\mathbb{R}^n_0$ denote the set of all nonzero vectors in $\mathbb{R}^n$. Then for $n = 1$, $H^1(\mathbb{R}, \mathbb{R}_0)$ with mod 2 coefficients is cyclic of order 2. Let e^1 denote the nonzero element.

Lemma 3.3.1. *For any topological space B and for any $n \geq 1$, a cohomology isomorphism is defined as*

$$\begin{array}{ccc} H^j(B) & \to & H^{j+n}(B \times \mathbb{R}^n, B \times \mathbb{R}^n_0) \\ y & \mapsto & y \times e^n \end{array}$$

pf.

- **CLAIM : For any topological space B, a cohomology isomorphism is defined by** $\begin{array}{ccc} H^j(B) & \to & H^{j+1}(B \times \mathbb{R}, B \times \mathbb{R}_0) \\ y & \mapsto & y \times e^1 \end{array}$ **using the cohomology cross product operation.** pf. Let B' be an open subset of B. Then for each cohomology class $y \in H^j(B,B')$, the cross product $y \times e^1$ is defined with $y \times e^1 \in H^{j+1}(B \times \mathbb{R}, B' \times \mathbb{R} \cup B \times \mathbb{R}_0)$. By the five lemma, $y \mapsto y \times e^1$ defines an isomorphism $H^j(B,B') \to H^{j+1}(B \times \mathbb{R}, B' \times \mathbb{R} \cup B \times \mathbb{R}_0)$.

Thus inductively, we get the n-fold composition $y \mapsto y \times e^1 \mapsto y \times e^1 \times e^1 \mapsto \cdots \mapsto y \times e^1 \times \cdots \times e^1$ is an isomorphism. Setting $e^n = e^1 \times \cdots \times e^1 \in H^n(\mathbb{R}^n, \mathbb{R}^n_0)$, the lemma follows.

Theorem 3.3.1 (Existence of the Stiefel-Whitney Class)**.** *Let ξ be an n-plane bundle with projection $p : E \to B$. There is one and only one cohomology class $u \in H^n(E, E_0)$ with* $\mod 2$ *coefficients whose restriction to $H^n(F, F_0)$ is nonzero for every fiber F. Furthermore, the correspondence $y \mapsto y \cup u$ maps the cohomology group $H^j(E)$ isomorphically onto $H^{n+j}(E, E_0)$,* $\forall j$.

pf.

- ξ is a trivial vector bundle.

  Identify E with the product $B \times \mathbb{R}^n$. Then the cohomology $H^n(E, E_0) = H^n(B \times \mathbb{R}^n, B \times \mathbb{R}^n_0)$ is canonically isomorphic to $H^0(B)$ by the previous lemma. The identity element $1 \in H^0(B)$ is the only class satisfying the condition that each point of B is nonzero. Therefore u exists and is equal to $1 \times e^n$. Since every cohomology class in $H^j(B \times \mathbb{R}^n)$ can be written uniquely as a product $y \times 1$, $y \in H^j(B)$, by the previous lemma, $y \times 1 \mapsto (y \times 1) \cup u = (y \times 1) \cup (1 \times e^n) = y \times e^n$ is an isomorphism.

- B is the union of two open sets B', B''.

  Then for $\xi|_{B'}, \xi|_{B''}, \xi|_{B' \cap B''}$, by the previous statement the theorem follows. By the Mayer-Vietoris sequence, $\cdots \to H^{i-1}(E' \cap E'', E'_0 \cap E''_0) \to H^i(E, E_0) \to H^i(E', E'_0) \oplus H^i(E'', E''_0) \to H^i(E' \cap E'', E'_0 \cap E''_0) \to \ldots$. There exist a unique cohomology classes $\begin{cases} u' \in H^n(E', E'_0) \\ u'' \in H^n(E'', E''_0) \end{cases}$ whose restriction to each fiber are nonzero. By the uniqueness for $\xi|_{B' \cap B''}$, u', u'' have the same image in $H^n(E' \cap E'', E'_0 \cap E''_0)$. Thus they come from a common cohomology class $u \in H^n(E, E_0)$. Since $H^{n-1}(E' \cap E'', E'_0 \cap E''_0) = 0$, u is uniquely defined. Now for the Mayer-Vietoris sequence with $j + n = i$, $\cdots \to H^{j-1}(E' \cap E'') \to H^j(E) \to H^j(E') \oplus H^j(E'') \to H^j(E' \cap E'') \to \ldots$, by the five lemma with the correspondence $y \mapsto y \cup u$, $H^j(E) \to_{\cong} H^{j+n}(E, E_0)$.

- B is covered by finitely many open sets s.t. $\xi|_{B_i}$ is trivial for each B_i.

  Suppose k is the number of open sets that cover B. If $k = 1$, the assertion is true by the previous statement. If $k > 1$, assume $\begin{cases} \xi|_{B_1 \cup \cdots \cup B_{k-1}} \\ \xi|_{(B_1 \cup \cdots \cup B_{k-1}) \cap B_k} \end{cases}$ are true. Then by the previous case, it is true for ξ. By induction the statement follows.

- general case

  Let C be an arbitrary compact subset of the base space B. Then $\xi|_C$ is the case above and therefore the theorem follows. Since the union of any two compact sets is compact, we can form the direct limit $\lim H_j(C)$ of homology groups C that varies over all compact subsets of B. Also we have corresponding inverse limit $\lim_{\leftarrow} H^j(C)$ of cohomology groups. The cohomology group $H^n(E,E_0)$ maps isomorphically to the inverse limit of the groups $H^n(p^{-1}C), p^{-1}(C)_0)$. But each of the latter groups contains one and only one class u_C whose restriction to each fiber is nonzero. Then $H^n(E,E_0)$ contains one and only one class u whose restriction to each fiber is nonzero. Consider $\cup_u : H^j(E) \to H^{j+n}(E,E_0)$. Then for each compact subset C of B,

$$\begin{array}{ccc} H^j(E) & \xrightarrow{\cup_u} & H^{j+n}(E,E_0) \\ \downarrow & & \downarrow \\ H^j(p^{-1}(C)) & \longrightarrow & H^{j+n}(p^{-1}(C),p^{-1}(C)_0) \end{array}$$

  commutes. By passing this diagram to the inverse limit, $\cup_u$ is an isomorphism.

Lemma 3.3.2. *The homology group $H_{n-1}(E,E_0;\mathbb{Z}) = 0$.*

Let Λ be an arbitrary ring. Let ξ be an oriented n-plane bundle, Then for each fiber F of ξ, we are given a preferred generator $u_F \in H^n(F,F_0;\mathbb{Z})$.

Theorem 3.3.2 (Thom Isomorphism)**.** *There is one and only one cohomology class $u \in H^n(E,E_0;\Lambda)$ whose restriction to (F,F_0) is equal to u_F for every fiber F. Furthermore, the correspondence $y \mapsto y\cup u$ maps $H^j(E;\Lambda)$ isomorphically onto $H^{j+n}(E,E_0;\Lambda)$ for every j.*

pf. By the previous lemma, $H^n(E,E_0;\mathbb{Z}) \cong \hom(H_n(E,E_0;\mathbb{Z}),\mathbb{Z})$. Thus $H^n(E,E_0;\mathbb{Z})$ is canonically isomorphic to the inverse limit of the groups $H^n(p^{-1}(C),p^{-1}(C)_0;\mathbb{Z})$ as C varies over all compact subsets of the base space B. Then similar with the case $\Lambda = \mathbb{Z}/2$, the theorem follows, given the fact that under the unique ring homomorphism $\mathbb{Z} \to \Lambda$, the fundamental class in $H^n(E,E_0;\mathbb{Z})$ corresponds to a fundamental class in $H^n(E,E_0;\Lambda)$.

3.3.2 The Thom Isomorphism and the Poincare Duality

Suppose $E \to X$ is an n-dimensional vector bundle. Reduce the structure group of the bundle from $GL_n(\mathbb{R})$ to $O(n)$, i.e. give it a metric. Then

$$\begin{cases} D(E) \to X & \text{the unit disk bundle} \\ S(E) \to X & \text{the unit sphere bundle} \end{cases}.$$

Theorem 3.3.3 (Thom Isomorphism)**.** *Suppose the bundle $p : E \to X$ is orientable n-dimensional vector bundle. Then there is a unique cohomology class $U \in H^n(D(E), S(E))$ whose restriction to any fiber is the relative fundamental class of that fiber. Furthermore, the map*

$$\begin{array}{ccc} H^*(X) & \to & H^{*+n}(D(E), S(E)) \\ a & \mapsto & \pi^* a \cup U \end{array}$$

is an isomorphism.

Definition 3.3.2. *The class U is called the Thom class and the isomorphism is called the Thom isomorphism.*

Suppose M is an n-dimensional manifold and $R \subset M$ is a closed k-dimensional submanifold. Then R has a normal bundle in M $\to \exists \begin{cases} \nu \to R & \text{the vector bundle} \\ \psi_R : D(\nu) \to \text{nhood of } R \text{ in } M & \text{the diffeomorphism} \end{cases}$, which is called a tubular neighbourhood. Suppose M, R are oriented. Then the normal bundle ν inherits the orientation. Let U be the Thom class of this orientation. Then we can view U as a class in $H^{n-k}(M, M \setminus D(\nu))$ by the diffeomorphism and excision theorem.

Proposition 3.3.1. *The image of U in $H^{n-k}(M)$ is Poincare dual to the image of the fundamental class $[X]$ in $H_k(M)$ under the embedding $X \to M$.*

pf. Since $H_k(D(\nu)) = \mathbb{Z}$ is generated by $[X]$ embedded as the zero section, $U \in H^{n-k}(D(\nu), S(\nu))$ is the Poincare dual class to $[X]$ in $(D(\nu), S(\nu))$. Thus $U \cap [D(\nu), S(\nu)] = [X]$. Since the fundamental class of M restricts to the relative fundamental class of $(D(\nu), S(\nu))$, $U \cap [M] = \pm[X]$. The choice of U from the orientation of M and R leads to a sign of $+$.

Appendix A

Poincare Duality

Given a smooth manifold X, a singular cochain $\alpha \in Sing^*(X)$ has compact support if there is a compact subset $K \subset X$ s.t. α vanishes on any singular simplex $\sigma : |\Delta^n| \to X$ whose image is disjoint from K. The set of singular cochains with compact support, denoted $Sing_c^*(X)$ form a subcomplex of the singular cochain complex. The cohomology of this subcomplex is the cohomology of X with compact support, denoted $H_c^*(X)$.

Suppose $U \subset X$ is an open subset. Let α be a cocycle with compact support on U. Denote by $K \subset U$ a compact set containing a support of α. Let $V = X \setminus K$. Then $\{U,V\}$ is an open cover of X. Let $Sing_*^{\{U,V\}}(X) \subset Sing_*(X)$ denote the subcomplex generated by all singular simplexes whose image is contained either in U or V. The inclusion of this subcomplex into $Sing_*(C)$ induces an isomorphism on homology, and therefore the dual map between the algebraic duals induces an isomorphism on cohomology. Define $\tilde{\alpha} \in Sing_*^{\{U,V\}}(X)$ s.t. $< \tilde{\alpha}_K, \sigma > = \begin{cases} < \alpha, \sigma > & \sigma \subset U \\ 0 & otherwise \end{cases}$. Then $\tilde{\alpha}_K$ is a cocycle and its cohomology class $[\tilde{\alpha}_K] \in H^*(Sing_*^{\{U,V\}}(X))$ determines a class $ext[\tilde{\alpha}_K] \in H_c^*(X)$.

$\to \begin{array}{ccc} H_c^*(U) & \to & H_c^*(X) \\ \tilde{\alpha}_K & \mapsto & ext\tilde{\alpha}_K \end{array}$ is well-defined.

pf. Given any two compact subset $K_0, K_1 \subset U$ containing the support of α, $\exists K' \subset U$ s.t. $K_0 \cup K_1 \subset K'$. Let $\begin{cases} V = X \setminus K \\ V' = X \setminus K' \end{cases}$ Then for an arbitrary compact set $K \subset K'$, $Sing_*^{\{U,V'\}}(X) \subset Sing_*^{\{U,V\}}(X) \subset (X) \subset Sing_*(X)$ with all inclusions inducing isomorphisms on cohomology and hence their algebraic duals induces isomorphisms on cohomology. The restriction of $\tilde{\alpha}_K$ to $Sing_*^{\{U,V'\}}(X)$ is $\tilde{\alpha}_{K'}$. Thus the choice of K does not affect the morphism above.

Suppose there is a compactly supported cochain β in U with $d\beta = \alpha$. Let $K \subset U$ be a compact subset containing the support of β. Then $\tilde{\alpha}_K = d(\tilde{\beta}_K)$, and therefore

$ext\tilde{\alpha}_K = 0$. So the choice of representative does not affect the morphism above.

Consider an element $\sum_\sigma \lambda_\sigma \sigma$ s.t. for each compact subset $K \subset X$, only finitely many σ has $\lambda_\sigma \neq 0$ and the image of σ meets K. Then the boundary map ∂ is well-defined and is an expression of the same form with $\partial^2 = 0$. These elements form a chain complex, denoted $Sing_*^{lf}(X)$ and the homology of this is the locally finite homology, denoted $H_*^{lf}(X)$.

The locally finite chain complex is $\lim Sing_*(X, X \setminus K)$ **as** K **ranges over the directed set of compact subsets of** X**.**
pf. Let $\sum_\sigma \lambda_\sigma \sigma$ be a locally finite chain. For any compact subset K, all but finitey many with nonzero coefficient are contained in $X \setminus K$. Thus the image of the formal sum is a finite sum in $Sing_*(X, X \setminus K)$. This gives a chain map $p_K : Sing_*^{lf}(X) \to Sing_*(X, X \setminus K)$. If $c = \sum_\sigma \lambda_c$, $p_K(c) = \sum_{\{\sigma : Im(\sigma) \cap K = \emptyset\}} \lambda_\sigma \sigma$. These maps are compatible as we vary K in the sense that for $K \subset K'$, $\forall c$ a locally finite chain, the image of $p_{K'}(c)$ under the map $Sing_*(X, X \setminus K') \to Sing_*(X, X \setminus K)$ induced by the inclusion of pairs is $p_K(c)$. Then these maps define a map of chain complexes $Sing_*^{lf}(X) \to \lim Sing_*(X, X \setminus K)$. Suppose $\{c_K\} \in \lim Sing_*(X, X \setminus K)$. Represent each c_K uniquely as a finite linear combination of singular simplexes, none of which is disjoint from K. If $K \subset K'$, c_K is obtained from $c_{K'}$ by deleting all the terms in the expression of $c_{K'}$ that are singular simplexes disjoint from K. The formal expression for the localy finite chain is the sum of all the terms that appear in at least one c_K. Then by restricting to $Sing_*(X, X \setminus K)$ we get $c_K \in Sing_*^{lf}(X)$. So the above map is onto. The map is clearly injective and the statement follows.

Any second countable locally compact Hausdorff space X **is a countable union of increasing compact sets.**
pf. Second countable space has a countable basis for the topology $\{U_n\}_{n=1}^{\infty}$.
$\forall x \in X$, $\exists$ a nhood with a compact closure. This nhood contains an element of the countable base U_x, a base element containing x where $\bar{U}_x$ is compact. Thus $X \subset \cup_{x \in X} \bar{U}_x$ where $\forall x \in X$, U_x the open subset is an element of the countable base. Thus X is a countable union of compact sets $\bar{U}_1, \bar{U}_2, \ldots$. By defining $K_k = \cup_{i=1}^{k} \bar{U}_i$, the statement follows.
$\to$ for $K_1 \subset K_2 \subset \ldots$ an increasing union of compact sets, every locally finite chain can be written as a countable sum $\sum_n \lambda_n \sigma_n$ where $\forall k \geq 1$, $\exists n_k \geq 0$ s.t, $\sigma_n \cap K_k = \emptyset$, $\forall n \geq n_k$.

Let $U \subset X$ be an open set. Then there exists a map $\begin{array}{ccc} Sing_*^{lf}(X) & \to & Sing_*^{lf}(X) \\ c = \sum \lambda_\sigma \sigma & \mapsto & c' \end{array}$, where for each σ, we add vertex at the barycenter of each of the faces that is not contained in U and ordered so that the vertices of the original simplex come before all the new vertices and have the same order and the new vertices are ordered by the dimension of the open faces that contain them. Then we form the linear subdivision of σ with this collection of vertices and replace σ by the sum of the newly created simplices with coefficients $\pm\lambda_\sigma$, the sign determined by the relative orientation of the subsimplex to the original simplex. Then remove all simplexes that are not faces of top dimensional simplexes that meet U. Repeat this process ad infinitum and take the union over all the steps of all simplexes whose closure is contained in U. Define such a union as $c' \in Sing_*^{lf}(X)$ (recall the cellular approximation).

Suppose $\alpha \in Sing_c^*(X), A \in Sing_*^{lf}(X)$ with compact support of α is K. Since all but finitely many of the terms giving A are disjoint from K, we can evaluate α on A ; $Sing_c^*(X) \otimes Sing_*^{lf} \to \mathbb{Z}$ with adjoint $Sing_c^*(X) \to \hom(Sing_*^{lf}(X), \mathbb{Z})$, which is not onto (its image is the set of continuous homomorphism $Sing_*^{lf}(X) \to \mathbb{Z}$).

For a cochain α of degree k and a singular chain γ of degree r,

$$< \beta, \alpha \cap \gamma > = < \beta \cup \alpha, \gamma >$$

for any cochain β. Extending linearly gives a pairing called the cap product $\cap : Sing^k(X) \otimes Sing_r(X) \quad \to \quad Sing_{r-k}(X) \quad \text{s.t. } \partial(\alpha \cap \gamma) = \alpha \cap \partial\gamma + (-1)^{r-k}\delta\alpha \cap \gamma.$ Since this map send changes in cobaundaries for α and in boundaries for γ to the changes in boundaries, it induces $\cap : H^k(X) \otimes H_r(X) \to H_{r-k}(X)$ natural w.r.t. the continuous maps. There is a natural map $H_0(X) \to \mathbb{Z}$ which assigns to a zero cycle the sum of its coefficients. Then Analogously we have $\begin{cases} Sing_c^k(X) \otimes Sing_n^{lf}(X) \to Sing_{n-k}(X) \\ H_c^k(X) \otimes H_n^{lf}(X) \to H_{n-k}(X) \end{cases}$

Proposition A.0.1. *Let M be an oriented n-manifold and let $K \subset M$ be a compact set. Then there exists a unique class $\alpha_K \in H_n(M, M \setminus K)$ with the property that $\forall x \in K$, the image $\alpha_K \in H_n(M, M \setminus \{x\})$ is the local orientation of M. Also $H_*(M, M \setminus K) = 0$, $\forall * > n$.*

pf.

- $\begin{cases} M & \text{a round open ball in } \mathbb{R}^n \\ K & \text{a compact convex subset of the open ball} \end{cases}$
  since M is contractible, $H_n(M, M \setminus K)$ is identified with $H_{n-1}(M \setminus K)$. Radial projection from any point $x \in K$ gives a homotopy equivalence between $M \setminus K \sim S^{n-1}$. Thus $M_n(M, M \setminus K) = \mathbb{Z}$ and the higher homology groups vanish. The image of a generator maps to a generator in $H_n(M, M \setminus \{x\})$, so there is a unique class mapping to the local orientation $\forall x \in K$.

- $\begin{cases} M & \text{a round ball} \\ K & \text{a union of } r \text{ compact convex subset} \end{cases}$
  when $r = 1$, by the previous statement trivially the current statement holds when $r > 1$, assume the statement hold for $r-1$. Let $\begin{cases} K = C_1 \cup \cdots \cup C_r \\ C' = C_1 \cup \cdots \cup C_{r-1} \end{cases}$.
  Then by the Meyer-Vietoris sequence, we get $(M, M \setminus K) \to ((M, M \setminus K') \coprod (M, M \setminus C_r)) \to (M, M \setminus K' \cap C_r)$ where the middle term are covered by inductive hypothesis. Since K' is a union of $r-1$compact convex sets, last term is also covered by the inductive hypothesis. In particular, there are classes $\alpha_{K'} \in H_n(M, M \setminus K')$ and $\alpha_{C_r} \in H_n(M, M \setminus C_r)$ as in the proposition. Then by the l.e.s. of the homology, $H_*(M, M \setminus K) = 0,\ \forall * > n$ and $\exists! \alpha_K \in H_n(M, M \setminus K)$ with image $(\alpha_{K'}, \alpha_{C_r})$.

- $\begin{cases} M & \text{a round ball} \\ K & \text{an arbitrary compact set} \end{cases}$
  $\exists B = \cup_i B_i$ of closed round balls contained in M and containing K. By the previous case, $H_*(M, M \setminus \cup_{i=1}^k B_i)$ satisfies the statement. Since taking homology commutes with colimits, the statement holds for $(M, M \setminus B)$ and a fortiori it holds for $(M, M \setminus K)$.

- general case
  consider an arbitrary manifold M and an arbitrary compact set K. Then there exists a set of finitely many open balls that cover K s.t. $K \subset U_1 \cup \cdots \cup U_k \equiv U$. Write $K = K_1 \cup \cdots \cup K_k$ where $K_i = K \cap U_i$ compact. We set $\begin{cases} U' = U_1 \cup \cdots \cup U_{k-1} \\ K' = U_1 \cup \cdots \cup K_{k-1} \end{cases}$. Then by the Meyer-Vietoris sequence, $(M, M \setminus K) \to ((M, M \setminus K') \coprod (M, M \setminus K_r)) \to (M, M \setminus K' \cap K_r)$, which implies $(U, U \setminus K) \to (U', U' \setminus K') \coprod (U_k, U_k \setminus K_k) \to (U_k, U_k \setminus K' \cap K_k)$ by excision. By the inductive hypothesis, the middle term satisfies the statement. The last term also satisfies the statement by the statement for the previous cases. By the l.e.s. of the homology, $H_*(M, M \setminus K) = 0,\ \forall * > n$ and $\exists! \alpha_K \in H_n(M, M \setminus K)$ that maps to $(\alpha_{K'}, \alpha_{K_r})$. For any $x \in K, \alpha_K$ restricts to $H_n(M, M \setminus \{x\})$ to give the local orientation.

Proposition A.0.2. *Let M be an oriented n-manifold and let $K \subset M$ be a compact set. Then there is a unique class $\alpha_K \in H_n(M, M \setminus K)$ with the property that $\forall x \in K$, the image of $\alpha_K \in H_n(M, M \setminus \{x\})$ is the local orientation of M. Also $H_*(M, M \setminus K) = 0,\ \forall * > n$.*

pf.

- $\begin{cases} M & \text{an open ball in } \mathbb{R}^n \\ K & \text{a compact convex subset of the } M \end{cases}$

since M is contractible, $H_n(M,M\setminus K)$ is identified with $H_{n-1}(M\setminus K)$. By the radial projection from any point of $x\in K$ gives a homotopy equivalence between $M\setminus K$ and S^{n-1}. Thus $H_n(M,M\setminus K)=\mathbb{Z}$ and the higher homotopy groups vanish. Then since the generator is mapped to a generator in $H_n(M,M\setminus\{x\})$, there is a unique class mapping to the local orientation $\forall x\in K$.

- $\begin{cases} M & \text{an open ball} \\ K & \text{a union of } r \text{ compact convex subsets} \end{cases}$

  when $r=1$, by the previous statement, it follows.
  when $r>1$, let $\begin{cases} K=C_1\cup\cdots\cup C_r \\ K'=C_1\cup\cdots\cup C_{r-1} \end{cases}$. Then the Mayer-Vietoris sequence $(M,M\setminus K)\to((M,M\setminus K')\coprod(M,M\setminus C_r))\to(M,M\setminus K'\cup C_r)$ exits where middle two terms are covered by the inductive hypotheses. Since K' is a union of $r-1$ compact convex sets, the last term is also covered by the inductive hypothesis. So $\exists\alpha_{K'}\in H_n(M,M\setminus K)$
  $\alpha_{C_r}\in H_n(M,M\setminus C_r)$ as in the proposition. Thus by the homology l.e.s. $\begin{cases} H_*(M,M\setminus K)=0 & *>n \\ \alpha_k=(\alpha_{K'},\alpha_{C_r}) & n \end{cases}$.

- $\begin{cases} M & \text{an open ball} \\ K & \text{arbitrary compact set} \end{cases}$

  $\exists B=\cup_i B_i$ a finite union of closed balls contained in M and containing K. By the previous statement, the proposition holds for $H_*(M,M\setminus\cup_{i=1}^k B_i)$. Since taking homology commutes with colimits, the statement follows for $(M,M\setminus B)$ and $(M,M\setminus K)$.

- general case

  for an arbitrary manifold M and arbitrary compact set K, cover K by finitely many open balls in coordinate patches $\subset U_1\cup\cdots\cup U_k\equiv U$. Write $K=K_1\cup\cdots\cup K_k$ where $K_i\subset U_i$ compact. Set $\begin{cases} U'=U_1\cup\cdots\cup U_{k-1} \\ K'=K_1\cup\ldots K_{k-1} \end{cases}$. Then there exists a Mayer-Vietoris sequence associated to the $(M,M\setminus K)\to((M,M\setminus K')\coprod(M,M\setminus K_k))\to(M,M\setminus K'\cap K_k)$. By excision, it is equivalent to $(U,U\setminus K)\to(U',U'\setminus K')\coprod(U_k,U_k\setminus K_k)\to(U_k,U_k\setminus K'\cap K_k)$. By the inductive hypotheses, each of the middle terms and the last term satisfy the proposition.Then by the homology l.e.s. we get the case.

Theorem A.0.1. *Let M be an oriented topological n-manifold. Then M has a unique locally finite fundamental class, denoted $[M]$ s.t. $H_*^{lf}(M)=0$, $\forall *>n$.*

pf. Let M be an oriented n-manifold. Take an increasing sequence of compact sets $K_1 \subset K_2 \subset \dots$ whose union is M. Then by the previous proposition, there is a unique element $\{\alpha_{K_n}\} \in \lim_{\leftarrow} H_n(M, M \setminus K_k)$ with the property that $\forall x \in K_k$, the image in $H_n(M, M \setminus \{x\})$ of the class α_{K_k} is the local orientation. Note the locally finite chains are the projective limit of the chains on $(M, M \setminus K_k)$.

- **CLAIM:** $H_n^{lf}(X) = H_n(\lim(M, M \setminus K_k)) = \lim H_n(M, M \setminus K_k)$.

  Suppose $\exists\{[\xi_k]\} \in limH_*(M, M \setminus K_k)$. If there is a representative cycles ξ_k; homologous to ξ_k s.t. under the map $(M, M \setminus K_k) \to (M, M \setminus K_{k-1})$ the cycle ξ'_l maps to the cycle ξ'_{k-1}, then the map $H_*^{lf}(M) \to \lim H_*(M, M \setminus K_k)$ above is surjective. Suppose we have fixed $\xi'_1, \dots, \xi'_k$ as required. Then the image of ξ_{k+1} differs from ξ'_k by $\partial\beta_k$ for some $\beta_k \in Sing_*(M, M \setminus K_k)$. Since the inclusion map of relative cochains is onto, $\exists\beta_{k+1} \in Sing_*(M, M \setminus K_{k+1})$ mapping to β_k. Set $\xi'_{k+1} = \xi_{k+1} - \partial\beta_{k+1}$. Then this is the required cycle and by induction, the map above is indeed surjective. Suppose $\{\xi_k\}$ is a cycle in the projective system of chain complexes whose image in the projective system of homology groups is trivial. Then $\forall\xi_k$ is a boundary in $Sing_*(M, M \setminus K_k)$. If given $\xi = \partial\beta_k\ \xi$ compatible, the map above is injective. Assume we have arranged the image of β_i in $Sing_*(M, M \setminus K_{i-1})$ is β_{i-1} for all $i \le k$. Consider the difference of the image of β_{k+1} and β_k. This is a cycle in $(M, M \setminus K_k)$ since the boundaries of the β_i are compatible. But $H_{n-1}(M, M \setminus K_k) = 0$ so this difference is a boundary; i.e. $\exists\gamma_k \in Sing_*(M, M \setminus K_k)$ with $\partial\gamma_k = \beta_{k+1} - \beta_k$. Lift γ_k to an element γ_{k+1} and replace β_{k+1} by $\beta_{k+1} - \partial\gamma_{k+1}$ gives β_{k+1} compatible. Hence by induction the above map is injective.
  For the locally finite chains the projective limits and taking homology commute and therefore the second equality follows.

Let M^n be an oriented n-manifold and choose an increasing sequence $K_1 \subset K_2 \subset \dots$ of compact subsets whose union ins M. By the above claim, $H_n^{lf}(M) = \lim H_n(M, M \setminus K_k)$. The unique classes $\alpha_k \in H_n(M, M \setminus K_k)$ that restricts to the local orientation at any point of K_k are compatible under inclusions and hence form an element in $\lim H_n(M, M \setminus K_k)$. Let $[M] \in H_n^{lf}(M)$ be the corresponding locally finite homology class. $\forall x \in M$, $\exists k$ s.t. $x \in K_k$. Then by construction, the image of α_{K_k} in $H_n(M, M \setminus \{x\})$ is the local orientation, and hence it is also true the image of $[M]$ in $H_n(M, M \setminus \{x\})$ is a local orientation. This implies $[M]$ is a fundamental class for M. The uniqueness follows from the fact that the map from the nth locally finite homology to the inverse limit of $H_n(M, M \setminus K_k)$ is an isomorphism and the uniqueness of α_{K_k}.

Lemma A.0.1. *Let U be an open convex subset of an open unit ball in $\mathbb{R}^n$. Then $H_c^*(U) = \begin{cases} trivial & * \neq n \\ \mathbb{Z} & * = n \end{cases}$. Furthermore, the map $\cap[U] : H_c^n(U) \to H_0(U)$ is an* *isomorphism.*

pf. Assuem wlog $0 \in U$. For any $0 < t \leq 1$, $x \mapsto tx$ defines a diffeomorphism from U to an open subset $U_t \subset U$.

- **the inclusion $U_t \subset U$ induces an isomorphism on compactly supported cohomology.**
  pf. Fix $0 < t < 1$. There is one parameter family of diffeomorphisms $J_s : U \to U$ s.t. J_0 is the identity, and $J_1(U_t)$ contains the support of α. Thus $J_1^*\alpha$ is compactly supported in U_t and $\left\{ \begin{array}{c} J_1^*\alpha \\ J_0^*\alpha = \alpha \end{array} \right.$ represent the same compactly supported cohomology in U. Thus the map is surjective. Suppose α' is a compactly supported cocycle in U_t and $\alpha' = \partial\beta$ for a compactly supported cochain β in U. Then we can assume there is one parameter family J_s s.t. the trivial family on the support of α, i.e. for all s, $J_s|_{supp(\alpha')} = Id$. Then $\delta J_1^*\beta = J_1^*(\alpha') = \alpha'$ and $J_1^*\beta$ is compactly supported in U'. So the injectivity follows.

Then for an open ball B, let $\bar{B}$ be its closure. The extension by zero determines a map $H_c^*(B) \to H^*(\bar{B}, \partial\bar{B})$. Let $B' \subset B$ be a smaller ball. Then we get $H_c^*(B') \to H^*(\bar{B}', \partial\bar{B}') \to H_c^*(B) \to H^*(\bar{B}, \partial\bar{B})$. Then the map $H_c^*(B') \to B_c^*(B)$ is an isomorphism by the above statement and therefore $H^*(\bar{B}', \partial\bar{B}')$ maps onto $H_c^*(B)$. Also the map $H^*(\bar{B}', \partial\bar{B}') \to H^*(\bar{B}, \partial\bar{B})$ is an isomorphism and therefore $H^*(\bar{B}', \partial\bar{B}') \to H_c^*(B)$ is injective. Thus for any U, $H_c^*(U) = \left\{ \begin{array}{cc} 0 & * \neq n \\ \mathbb{Z} & * = n \end{array} \right.$. If we identify $H_0(B)$ with $\mathbb{Z}$ in the natural way, the map $\cap[B] : H_c^n(B) \to \mathbb{Z}$ is the evaluation of $H_c^n(B)$ on $[B]$. The restriction of $[B]$ to $H_n(B, B \setminus \{0\})$ gives the local orientation at the origin and the map $H_c^n(B) \to H^n(B, B \setminus \{x\})$ is an isomorphism. Thus $\cap[B]$ is an isomorphism. Then for any open convex subset U, $\exists B_0 \subset U$ an open ball s.t. $\left\{ \begin{array}{l} H_c^*(B) \to H_c^*(U) \\ H_*(B) \to H_*(U) \end{array} \right.$ are both isomorphism and the restriction of $[U]$ to B_0 is $[B_0]$.

Proposition A.0.3. *For any open subset U of an open ball $\mathbb{R}^n$, $\cap[U] : H_c^*(U) \to H_{n-*}(U)$ is an isomorphism.*

pf. Let U be an open subset s.t. it is a union of r open convex sets.
When $r = 1$, by the previous lemma, the proposition follows.
When $r > 1$, assume the proposition holds for $r - 1$. Suppose $U = B_1 \cup \cdots \cup B_r$, where B_i are open convex balls. Let $\left\{ \begin{array}{c} U' = B_1 \cup \cdots \cup B_{r-1} \\ V = (B_1 \cup \cdots \cup B_{r-1}) \cap B_r \end{array} \right.$. Then by the Meyer-Vietoris sequence,

$$\begin{tikzcd}
H_c^*(V) \arrow[r] \arrow[d] & H_c^*(U') \oplus H_c^*(B_{r+1}) \arrow[r] \arrow[d] & H_c^*(U) \arrow[r] \arrow[d] & H_c^{*+1}(V) \arrow[r] \arrow[d] & \dots \\
H_{n-*}(V) \arrow[r] & H_{n-r}(U') \oplus H_{n-*}(V) \arrow[r] & H_{n-*}(U) \arrow[r] & H_{n-*-1}(V) \arrow[r] & \dots
\end{tikzcd}$$

where the vertical arrows are cap product with the fundamental classes. Since the restriction of the fundamental class of U to $U'$and B_r induces their fundamental classes and the fundamental class of U' and B_r each restrict to V to given its fundamental class, the diagram commutes. Since both U', V are the union of $r-1$ open convex sets, the result follows by the five lemma and the inductive hypothesis. Since any open subset of an ope ball in $\mathbb{R}^n$ is a countable union of open balls. Both homology and cohomology with compact supports commute with taking colimits over such increasing unions. Also the fundamental class of the union restricts to give the fundamental class of any finite union of open balls. Thus the proposition holds for any open set of an open ball in $\mathbb{R}^n$ by taking colimits of the result for finite unions.

Theorem A.0.2 (Poincare Duality)**.** *Suppose M is an oriented n-manifold and let $[M] \in H_n(M)$ be its fundamental class. Then the map*

$$\begin{array}{rcl} \cap[M] : H_c^*(M) & \to & H_{n-*}(M) \\ \alpha & \mapsto & \alpha \cap [M] \end{array}$$

is an isomorphism for all $*$.

pf. Let M be an oriented n-manifold that is a union of a finite number of open n-balls. Denote r as the number of such balls.
When $r = 1$, $\cap[M] : H_c^*(M) \to H_{n-*}(M)$ is an isomorphism by the previous proposition.
When $r > 1$, write $\begin{cases} M' & \text{the union of the first } r-1 \text{ balls} \\ B & r\text{th ball} \\ V & = M' \cap B \end{cases}$. Then by the Meyer-Vietoris sequence for the compactly supported cohomology and for ordinary homology for $V \to M' \coprod B \to U$ with the l.e.s. given by cap product with the fundamental classes. Since the fundamental classes restict to the fundamental classes on open sets, the diagram commutes. Then by the inductive hypothesis, the cap product with the fundamental class is an isomorphism for M', B. Since V is an open subset of an open ball, so is for V. Then the five lemma gives the case for r.
When $r = \infty$, taking colimits over a countable increasing union of the result for the finite unions of pen balls give the theorem for this case.

Corollary A.0.1 (Lefschetz Duality)**.** *Let M be a compact oriented manifold with boundary. Then $\forall k$, the map*

$$\cap[M, \partial M] : H^k(M, \partial M) \to H_{n-k}(M)$$

is an isomorphism.

pf. Let $\partial M \times I \subset M$ be a collar nhood of the boundary, with $\partial M \times \{0\}$ being the boundary of M. $\forall 0 < t \leq 1$, let M_t be the open sub-manifold that is the complement of $\partial M \times [0,t]$. Then the inclusion $M_t \subset \int M$ induces an isomorphism on compactly supported cohomology by the previous lemma. For any t, $\bar{M}_t \subset M$ induces isomorphism $H^*(\bar{M}_t, \partial\bar{M}_t) \cong H^*(M, \partial M \times [0,t]) \cong H^*(M, \partial M)$. Then $H_c^*(M_1) \to H^*(\bar{M}_1, \partial\bar{M}_1) \to H_c^*(\int M) \to H^*(M, \partial M)$. Since $H(\bar{M}_1, \partial\bar{M}_1) \to H^*(M, \partial M)$ is an isomorphism, $H_c^*(\int M) \to H^*(M, \partial M)$ is surjective. Also since $H_c^*(M_1) \to H_c^*(\int M)$ is an isomorphism, $H_c^*(M_1) \to H_c^*(\bar{M}_1, \partial\bar{M}_1)$ is injective. Then given a diffeomorphism $\bar{M}_1 \to M$, $H_c^*(\int M) \to H^*(M, \partial M)$ is an isomorphism.

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